VILNIUS GEDIMINAS TECHNICAL UNIVERSITY
FACULTY OF MECHANICS
DEPARTMENT OF MECHANICAL AND MATERIALS ENGINEERING

Ramiz Huseynov

THE IMPACT OF INNOVATION ON THE DEVELOPMENT OF HIGH VALUE-ADDED INDUSTRIAL ENTERPRISES AND ON ECONOMIC GROWTH IN THE CONDITIONS OF GLOBALIZATION

INOVACIJŲ ĮTAKA AUKŠTĄ PRIDĖTINĘ VERTĘ KURIANČIŲ PRAMONĖS ĮMONIŲ VYSTYMUISI IR EKONOMIKOS AUGIMUI GLOBALIZACIJOS SĄLYGOMIS

Master‘s degree Thesis

Industrial Engineering and Innovation Management study programme, state code 6211EX056
Production and Manufacturing Engineering study field

Vilnius, 2026
VILNIUS GEDIMINAS TECHNICAL UNIVERSITY

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THE OBJECTIVES:

Tasks to achieve the goal:
The main goal is to develop and propose a structured model for assessing the impact of innovation on the development of high value-added industrial enterprises and economic growth in the conditions of globalization.
Objectives:
1. Analyze scientific literature to determine the primary factors impacting economic growth and innovation activities in high-value industrial enterprises.
2. Conduct a comparative analysis of empirical studies to design the structure of an empirical research model.
3. Create a database based on the developed model concept.
4. Use the EViews program to make simulations and propose a structured model for evaluating the impact of innovations on high-added-value industrial companies` development and economic growth under conditions of globalization.

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Annotation

The purpose of this research is to investigate the impact of innovation on the development of high-added industrial enterprises and economic growth under the conditions of globalization of 37 OECD countries. This study utilizes theoretical framework and panel data to identify how innovation most consistently interact and predict GDP per capita. The resulting findings show that export productivity per worker and high-technology export are the main factors of GDP growth within the OECD.

Thesis consists of 7 parts: Introduction, Scientific Literature Review, Empirical Research, Conclusions and Proposals
Thesis has 50 pages without appendixes and 41 figures and 7 tables

Keywords: innovation, economic growth, GDP per capita, OECD, fixed-effects panel regression, high-technology exports, manufacturing value added, globalization.

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Pavadinimas	**Inovacijų įtaka aukštą pridėtinę vertę kuriančių pramonės įmonių vystymuisi ir ekonomikos augimui globalizacijos sąlygomis**
Autorius	**Ramiz Huseynov**
Vadovas	**Rūta Banelienė**

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Anotacija

Šio tyrimo tikslas – ištirti inovacijų poveikį didelės pridėtinės vertės pramonės įmonių plėtrai ir ekonomikos augimui 37 EBPO šalių globalizacijos sąlygomis. Šiame tyrime naudojamas teorinis pagrindas ir paneliniai duomenys, siekiant nustatyti, kaip inovacijos nuosekliausiai sąveikauja ir prognozuoja BVP vienam gyventojui. Gauti rezultatai rodo, kad eksporto produktyvumas vienam darbuotojui ir aukštųjų technologijų eksportas yra pagrindiniai BVP augimo veiksniai EBPO šalyse.

Darbą sudaro 7 dalys: įvadas, mokslinės literatūros apžvalga, empirinis tyrimas, išvados ir pasiūlymai.
Darbą sudaro 50 puslapiai be priedų, 41 paveikslų ir 7 lentelės.

Prasminiai žodžiai: inovacijos, ekonomikos augimas, BVP vienam gyventojui, EBPO, fiksuotųjų efektų panelio regresija, aukštųjų technologijų eksportas, gamybos pridėtinė vertė, globalizacija.

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INTRODUCTION

The globalization has paved the way for innovation which serves as a critical drive for the growth and sustainability of high value-added industrial enterprises, shaping their competitive edge and bolstering economic development. Integration within the global markets for various enterprises Innovation bridges technological advancements and sustainable economic development, which has a massive impact on stakeholders and fields. The relevance of this study relies on the exploration and research of high-value added industrial enterprises and strengthening economic growth.

Such apparent changes as to how ideas shape economies is better observed in advanced economies, regarding how trade and redefining of wealth grows amid transnational states. As the richer countries compete with inteligence in lieu of raw materials, following this trend leads us to understand how innovation shapes, and influences econmies which draw considerable attention from analysts and officials alike.

The apparent shift is observed within the transitional organizations such as OECD member states,as this organization contributes majorly to R&D expenditure and high-value manufacturing output. Interestingly, this union includes absolutely influential economies such as the United States, Germany, South Korea, and Switzerland. As these countries have environments have the most evidence of innovation ecosystems floushing and it shows through their high levels of GDP per capita, growth. Whereas, there are other member states of the OECD which do not exactly follow the trend those particularly in the Southern Hemisphere as they may struggle to translate some R&D metrics into provable growth.

The motivation behind underlies how innovation and technology works across the OECD economies. In spite of the theorotical and empirical literautre on how innovation ties into economic growth. Using databases which compile the relevant metrics from 2010 onwards, a period marked by global financial crises, integration of new technologies, and straining the pressure from global supply chains.

The OECD group are a diverse group of countries as was briefly mentioned in the previous paragraphs, it includes countries as economically distinct as Luxembourg with a GDP per Capita exceeding $100,000, and Mexico where its GDP per capita is the tenth of its size. There are countries with export powerhouses as Germany and South Korea alongsie commodity-dependent

economies such as Iceland and Chile. Any scientific study that treats this OECD as a homogenous group removes the nuances and risks masking the very diversity that also help us differentiate metrics and how this difference treat innovation and convert such information into growth.

This study asks the question of which innovation-related variables such as manufacturing related metrics, high technology related metrics, patent activity, R&D expenditure and export metrics strongly and consistently correlate with GDP per capita, and what model fits and reveals under which innovation transfers well into economic growth.

This study takes a closer look at how innovation affects the economy, specifically how it impacts the standard of living in developed countries. It tries to figure out how much innovation really contributes to a country's economic growth and what steps are involved in making that happen. The goal is to identify which countries are best at using innovation to drive their economic progress.

Research Aim and Objective. The objective of this study is to investigate the how innovation affects the growth of high value-added industrial enterprises and thenceforth, examine its implication or influence on national and global scales within the scope of globalization. To this end, a model should be developed that examines such findings and discoveries and should be based on scientific literature analysis and subsequent empirical findings. Furthermore, this research also explores the effect of innovation in regards to how nations notably from the OECD nations react to this change amid global integration.

1. Analyze scientific literature to determine the primary factors impacting economic growth and innovation activities in high-value industrial enterprises.
2. Conduct a comparative analysis of empirical studies to design the structure of an empirical research model.
3. Create a database based on the developed model concept.
4. Use the EViews program to make simulations and propose a structured model for evaluating the impact of innovations on high-added-value industrial companies' development and economic growth under conditions of globalization

The investigation of methodology will conducted in the manner of collecting panel data analysis. Subsequently, the estimation will be tabulated through panel data which the fixed-effects regression model is estimated from within EViews10, underscoring 37 OECD member states. Moreover, the timeframe from which will be set from 2010 to 2021 and datasets will originate from Worldbank.

Research Methods The fixed-effects panel regression frameworck across the OECD member states from a decade long timeframe which shows how resilient certain economies could retain themselves under disruptive events such as COVID-19. That aside, to define in a much more specific basis. The date base of 37 OECD countries would be collected through World Bank data and then specific indicators that are closely linked to innovation would then be compiled such export and import related variables and after all of these data are collected. Thus, the conversion of these data points will be transferred to Eviews to simulate and conduct a fixed-effect panel regression graphs and probabilities.

Thesis Structure. This thesis will consist of Literature Review, Research Methodology, Empirical Analysis, ,Conclusions and Proposals, and References

Validation. The research was presented at the 29th Conference for Junior Researchers "Science – Future of Lithuania", Vilnius Gediminas Technical University, 24 April 2026, the following slides of "The Impact of Innovation on High Value-Added Industrial Enterprises and Economic Growth in OECD Countries" was presented (Appendix C)

1. THEORETICAL FOUNDATIONS OF INNOVATION AND HOW IT IMPACTS ECONOMIC GROWTH IN THE CONTEXT OF GLOBALIZATION

1.1 The Relationship Between Innovation and Economic Growth

Innovation is a critical driver for economic growth which is being progressively acknowledged within the context of high-added industrial enterprises. Such enterprises are crucial as they function in fields that advances the GDP by means of technological innovations, professional labor, and high-level investment in R&D (which is Research and Development). R&D investment has a profound impact on national economic performance across the G20 and notably OECD economie. It was determined that, nations with advanced economies that value added from R&D innovation is a prime factor within GDP growth (Padgett and Galan, 2025). Innovation resilience is quite a topic that also deserves some consideration in it that, R&D investment within 29 OECD countries from 1995 to 2019, leads to innovating countries increasing public R&D during recessions to stablise the during recessions, while other countries excluded from this group cut it (Cantore, Lavopa, Soare et al 2025).

Recent studies emphasize the factor of human capital as an additional driver of innovation. There are positive correlations between education and economic growth across EU regions. A longitudinal which has the lasts from 2000 to 2017 highlights that investments in higher education improve regional economic development by supporting human capital (Agasisti & Bertoletti, 2022).

Furthermore, the research also highlights that innovation is not the only driving factor for the economic growth, it is also responsible for regional economic resilience. Innovation capacity has a crucial role in forming regional economic resilience, especially during the times of economic crises (Bristow and Healy, 2017). For example, the regions with higher R&D intensity and stronger innovation systems are better at dealing with economic shocks. Thus, by fostering or bolstering innovation and resilience may alleviate the impact of externalities and support economic recovery and growth (Mayor and Ramos, 2020)

To add on to the boosting of GDP and improving overall resilience, innovation has come to be an important tool of export competitiveness and crucial transformation inside high value-

added industrial divisions of sectors. According to Kaitila (2018), countries within the EU with wider and more technologically developed export portfolios tend to perform better in terms of more stable and long-lasting economic growth. It is due to the fact that not only does innovation enhance productivity, but also allows firms to diversify and move up the value chain, shifting away from low-margin commodities and toward industries that require knowledge. Similarly, Caseiro and Simoes (2019) put special emphasis on innovation within the business sector, especially among SMEs and manufacturing firms which in turn significantly contributes by means of job creation, expansion of export, industrial efficiency to national economic performance. The comparative statistics and analysis for EU nations shows that firms oriented in innovation perform much better than their non-innovative counterparts both in profitability and resilience within the periods of economic uncertainty. Furthermore, the integration of innovation with sustainable industrial practices, such as the circular economy and resource efficiency, supplies a double benefit which is subsequent economic growth while also aligning long term ecological targets such as by creating new green jobs (Baldsarre and Carrara, 2024).
Innovation also plays into enhancing a county's competitiveness within the field of export by driving firms to devise differentiated, high quality, and high tech products, consequently, moving up the global value chain. Kaitila (2018) links a significant correlation between a country's product range for export and its long term GDP growth, for instance countries with more defined and more complex export structures experience more stable and sustained economic growth and development. Innovation enables these nations to move toward more specialized or in other words service industries such as pharmaceuticals, advanced machinery, and IT sectors. Additionally, countries with higher levels of absorptive capacity which includes knowledge infrastructure, skilled labor, and institutional readiness are more likely to convert innovation inputs into beneficial outcomes, like export sophistication and sustained economic growth, according to Harris, Krenz, and Moffat (2020).

Business innovation supports the shift or rather the trend by improving productivity, and enhancing operational efficiency and implementing new technologies into manufacturing and logistics processes into SMEs and MMEs (Caseiro & Simões, 2019). Such processes are hitherto supported by governments and institutions ,for instance, government-funded R&D programs, innovation vouchers, and EU initiatives. The research design of this study employs a quantitative research approach based on secondary data analysis such as the comparative analysis. The

primary goal is to investigate the relationship between innovation through R&D expenditure and the development of high value-added industrial enterprises including data collection through various credible sources and determining which model is suitable for GDP growth

1.2 Conditions under Globalization

Impact of globalization on industrial production and subsequent economic growth in the eu is reinforced by developments in transnational projects and trade networks. The global framework for partnerships in R&D bring about competition for EU economies by boosting exchange in technology and enhancing productivity. (Antonenko K, et al, 2024). Globalization also promotes economic diversification, bringing regions to adapt to external shocks while strengthening or building resilience. (OECD, 2019). As was pointed above, the evidence underscores those nations such as Germany and Sweden, which integrate into the global value chains effectively. Furthermore, the countries with such investments tend to possess higher economic stability and industrial output. Nevertheless, the countries like Romania with the lower R&D investments and less firm investment systems have difficulty in taking advantage of globalization. The difference emphasizes the necessity for policies that deal with capacity building, digital transformation, and sustainable development to secure participation in global markets. The influence of globalization on industrial production and economic growth within the EU has become a topic of pronounced interest through the expansion of transnational trade networks, cross border investment, and R&D collaboration. Globalization has been a massive benefit to the EU in a way which gives access to broader markets, advanced technologies and diverse networks. The research and innovation partnerships within the global framework has bolstered cross-national spillovers or overflow into the areas of productivity and technological capabilities (European Commission, 2021). These frameworks foster competition and cooperation at the same time, fostering entities, industries or firms to innovate so as to keep their relevance or their place in the markets.

The advantages of globalization are quite apparent in countries with established large economies such as Germany, Sweden, and the Netherlands, which have strategically places themselves within the global value chains (GVCs). These countries have a very output in investment in research and development (R&D), vocational training, and industrial policies

oriented around export, allowing them to specialize in clean technologies, pharmaceuticals, and advanced machinery which are classified as high value-added sectors OECD (2021), EU economies that are closely involved in GVCs, especially in innovation and design and upstream segments, were markedly more resilient during the COVID-19 pandemic which allowed them to withstand and recover faster because of the ability to adapt supply chains and maintain production continuity OECD (2021). Germany's strong engineering and automotive sectors benefited for the reason of having flexible supplier networks and a firm digital infrastructure, enabling firms to allocate or reallocate production and sourcing quickly during global disruptions

Although, this progress of globalization has been quite a success for major EU economies, such a trend has not been evenly distributed across the EU. To instantiate, countries such as Romania, Bulgaria, and Croatia, which have relatively lower R&D intensity, relatively less institutional support, and fewer high tech industrial clusters or input, face significant challenges to fully integrate into global networks. Hence these economies often remain in the perpetual trap in low-cost, labor-intensive segments of GVCs, which bring limited progress and opportunity (Rodrik, 2018). The noticeable disparity reinforces existing regional economic gaps and disrupts the EU's goal of economic cohesion. Absorptive capacity, or the ability to absorb, assimilate, and successfully use external information and technology, is therefore frequently lacking in nations with poor innovation institutions and inadequate digital infrastructure. According to Cirera and Maloney (2017), this restriction inhibits businesses from expanding beyond low-complexity industrial or basic manufacturing operations, which impedes structural transformation and results in stagnant productivity.

Scholars and organisations suggest all-encompassing approaches that prioritise capacity building, regional innovation ecosystems, and inclusive industrial growth in order to close this gap. By funding R&D, infrastructure modernisation, and skill development in lagging EU areas, initiatives like the Digital Europe Programme, Cohesion Policy 2021–2027, and Smart Specialisation Strategies seek to lessen gaps.

1.3 The Relation Between Industrial Production and GDP growth

Industrial production is a central component of economic development, serving as both a source of output and a driver of technological innovation. From a macroeconomic perspective, the relationship between industrial production and GDP is well established: higher levels of industrial activity stimulate aggregate demand, generate employment, and enhance productivity, which collectively contribute to economic growth. In modern economies, particularly in the European Union (EU), industrial sectors play a pivotal role in shaping competitiveness, export performance, and value-added creation (European Commission, 2020).

Industrial output, particularly manufacturing production, has long been regarded as a important factor of economic growth. Economic theories, such as Kaldor's growth laws, put forward that a faster growth within the manufacturing sector results in faster overall GDP growth (Kaldor, 19 67:, Andreoni & Tregenna, 2020). Essentially, the industrial sector often displays profitable returns and provide firm links to other sectors, thus its rise can make productivity gains for the economy. In addition, a plethora of countries support this notion of rising and falling in industrial production which subsequently translate almost immediately into correlating changes in GDP (Beckerova, Gavurova, and Kotaskova, 2018). This pattern is pronounced in the nations of the EU, where performance of the manufacturing sector trails or follows the business cycle.

In the context for EU countries, as mentioned in previous paragraphs, a robust foundation especially within the industrial base has been associated with more vigorous and sustained growth. Economies that of Germany with well-established and strong manufacturing sectors have witnessed higher productivity growth and export performance, which in turn solidifies GDP expansion. Deindustrialization by contrast can be a drag on growth. Research on EU countries finds that manufacturing is a simultaneous indicator of the economic cycle: the more the industrial output accelerates, the more GDP tends to rise and the more industry plummets, the more overall growth falters (Beckerova, Gavurova, and Kotaskova, 2018).

Industrial production is one of the cornerstones upon which economic growth rests in the current European Union. This is illustrated in terms of its contribution towards GDP and employment in the respective member states. Sold production in Europe was worth €5,992 billion in 2023, which is marginally up by 1.6% in nominal terms from 2022 but down by -1.2% compared to the constant-price figures reported by Eurostat (2023a). High-value manufacturing

such as automotive, pharmaceuticals, and machinery is driving the increase. Chemicals and basic metals recorded a downtrend that can be attributed to varying market conditions. Research carried out earlier indicates the need for integrating sustainability practices and principles of the circular economy to promote industrial efficiency and resilience against external disruptions (Baldassarre and Carrara 2024).

From the year 2000 onwards, trends in industrial production among EU Member States have shown differences. Slovakia, Poland, and Estonia were among the fastest-growing industrial countries between 2000 and 2020, each averaging more than 4% annual growth rates. Greece, Italy, and Spain recorded declines of more than 1% per annum. Ireland was the outlier with an abrupt rise of as much as 25 points in its industrial production index between 2014 and 2015-for reasons of statistical revisions-and maintained those levels thereafter high but cyclical. (Eurostat, 2023b).

The average industrial growth for the EU between in the early 2000s average 2.0%. Countries like Estonia, Lithuania, and Slovakia outshine these. But the financial crisis between 2007 and 2008 led to severe downturns, some economies-even the previously dynamic ones - recording double-digit declines. The recovery experience between 2010 and 2019 had different outcomes, with countries such as Greece and Italy suffering with depressed or slow growth pains below 1%. An imaginative study also demonstrated that Bulgaria's industrial production would have been 150% higher in February 2020 had it followed the pre-crisis growth pattern, and this is a clear testimony to the financial crisis in terms of longer growth paths. Overall, industrial production within the EU will be about 37% less than this "ideal" situation by February 2020, showing nicely how these trends, before and after the crises, create what can be very complicated industrial performance pictures (Eurostat, 2023c). Disparities such as these call for robust sustainable development policies within the European Union that push towards resilience and sustainable development.

1.4 Export and GDP growth

The relationship between export diversity and economic growth is particularly important for EU countries, as stated by (Kaitila ,2018). While enlarging the range of export products, helps to lessen dependence on a short range of goods and markets, which contributes to building a more resilient economy. Such diversity makes adaptation to global demand and trade significantly better, finally leading to sustainable GDP growth. Economies with wider and more sophisticated export systems, for instance, Germany and the Netherlands, tend to always outperform others in economic growth. Limited export systems nations like Greece and Portugal face it much more difficult to maintain steady growth rates. Hence the importance of polices dedicated to enhancing export diversification where heavy investments are required in technology, innovation, and workforce development. The same evidence indicates that focus on high-value-added exports serves as a major contributor to GDP growth, especially in relation to industries adapting to global technological progress. Therefore, these findings might also help in the contextualization of wider EU trade policies as a basis for mitigating economic vulnerabilities and, hence,

The recent research underscores that the quality and complexity of exports matter just as much as quantity. As countries go up the value chain and begin their products such as their tech, not only do they improve their trade balance but also bring in the industrial development and productivity growth. Economies with a diverse export portfolio and great economic complexity tend to witness growth and sustainability than those relying on raw materials or basic goods (Koch,2020).

Export diversification positively impacts on other growth drivers, such as human capital, institutional quality, and major investment in R&D. Export is heavily intertwined with innovation policies which support diversification in advanced manufacturing and digital sectors can develop long-term gains (OECD, 2021)

1.5 Employment and Innovation within GDP growth

Economic research supports the notion that job creation in high productivity sectors boosts GDP growth. Such is the shifting of labor from low-productivity to high-productivity sectors which is a key factor of development and technology and innovation all of which boost labor, thus multiplying the impact of employment on GDP. These dynamics are quite apparent on high income and emerging economies (McMilan, Rodrik, & Sepulveda, 2017; OECD, 2023)

Employment growth in general contributes more to GDP when it happens in high productivity innovation-intensive sectors. Employees in these divisions produce greater output per person, so by means of expansion of gdp positively impacts GDP and develop economic gains through spillovers. In established economies that of EU, service economies such as IT, engineering professional consultancy, commerce have been proven to play a undeniable role in job creation and productivity increase. Countries like Finland, Portugal and Spain high-growth "scaler" firms in service sectors (knowledge-intensive) produces far more jobs in relation to their small makeup in the economy (OECD, 2021).

Moreover, there is the Kaldor's third law which supports the notion from manufacturing employment mentioned above thus reinforcing the law. The principle posits that general productivity growth increases when manufacturing employment increases, while also increasing employment in lower-productivity sectors such as the traditional ones (retail, transport, personal etc) which corresponds with slower economic growth (Andreoni & Tregenna, 2020).

Furthermore, Employment driven by innovation plays a significant role to output expansion at the firm level. Entrepreneurs introducing new technologies, products and business models contribute to the creation of high-productivity jobs and new markets, which in turn raise economic output. An OECD report (2023) highlights that SMEs that innovate and grow make large contributions to employment, productivity, and competitiveness which boost innovation (Calligaris et al, 2023). Start-ups and high-growth firms produce spillovers through new products and services across industries and sectors. Studies across OECD countries display that adding employment in service sectors (knowledge-intensive), such as advanced manufacturing or ICT services, which adds onto the GDP growth than similar increases in low-productivity sectors (van Ark, de Vries, & Erumban, 2021).

Structural transformation is an important mechanism for development and GDP growth. To define what structural transformation means in this context an economy-wide realignment and reallocation of labor and resources from previously defined low-productivity sectors such as agriculture or informal services to high-productivity sectors such as manufacturing and business services. Such economy-wide growth crucially depends on how rapidly labor can realign itself (McMilan & Rodrik, 2017). Countries such as South Korea and Ireland have experienced such phenomena successfully reallocating employees or labor into more productive industrial and service positions which resulted in a significant increase in GDP per capita.

2. RESEARCH METHODOLOGY FOR THE ANALYSIS OF INNOVATION'S IMPACT ON ECONOMIC GROWTH WITHIN OECD NATIONS

2.1 Research Design and Comparative Analysis

The research design concering how this methodology should be done quantitatively will be specified to determine tangible connections between innovation specific variables and how it interacts with GDP per capita across 37 nations within the OECD nation in a period which has crises such COVID-19, rise of new technologies penetrating into certain economies and financial crises.The design is structured around identifying trends, relationships, and patterns in key economic indicators such as GDP growth, export performance, and industrial output, in relation to innovation activities across selected OECD countries. The research takes theoretical propositions based on the Kaldor's growth laws, endogenous growth theory and creates hypotheses and subsequently uses statistical evualtion using panel regression to observe country-level heterogeneity.

A panel dataset will be constructed for multiple OECD states over a period of time ideally from the 2000s to the most recent available year, depending on the extent of data. The annual GDP growth rates will be sourced from OECD databases such World Bank Data respectively to verify consistency. A comparative analysis is needed before structuring an empirical approach. The below table combines key studies investigating the essential relationship between innovation and economic growth

Table 1.Overview of Empirical Models and Variables within Research on Innovation and Economic Growth

Author(s) & Year	Sample	Period	Method	Dependent Variable	Key Innovation Variables	Main Finding
Romer (1990)	Theoretical	—	Endogenous growth model	Long-run GDP growth	R&D, human capital, knowledge sharing	Innovation's sustaining growth spreads via knowledge sharing
Aghion & Howitt (1992)	Theoretical	—	Schumpeterian growth model	Output per capita	R&D, creative destruction	R&D is a cause of growth by displacing previous technologies
Kaldor (1967)	UK & developed economies	Post-WWII	Growth accounting	GDP growth	Manufacturing output, employment	Manufacturing output has a positive relationship with productivity (Kaldor's law)
Cantore et al. (2017)	29 OECD	1995–2019	Panel regression	GDP growth / industrial output	Public R&D, *MVA*	Counter-cyclical R&D amplifies long-run divergence
Kaitila (2018)	EU member states	1995–2015	Panel OLS	GDP per capita	Export diversity, high-tech export share	Export diversity is a positive predictor of stable GDP growth
Caseiro & Simoes (2019)	EU member states	2005–2015	Comparative panel	GDP growth, employment	Business R&D, SME innovation	Innovative businesses perform better on profitablity and resilience
Harris et al. (2020)	OECD countries	2000–2016	Panel regression	Export sophistication, GDP	Absorptive capacity, R&D, skilled labour	Absorptive capacity controls how innovation translates into growth
Koch (2021)	80+ countries	1990–2015	Panel regression, ECI	GDP per capita growth	Export complexity, high-tech exports	Export complexity performs perfectly on standard measures
Dritsaki & Dritsaki (2023)	EU countries	1998–2020	Panel cointegration	GDP per capita	R&D expenditure, patents	Long-term integration between R&D and GDP growth
Padgett & Galan (2025)	G20/OECD	2010–2022	Panel FE regression	GDP per capita	R&D intensity, innovation value-added	R&D's impact on GDP growth is much more apparent on advanced economies

The overall connection that all of these models have is the R&D expenditure and how it interacts with other metrics such how efficiently the R&D expenditure converts into the innovation systems of different countries. Some variation of export structure is one indicator that proves quite a positive impact on GDP per capita, bringing the variable of *HTEX/LABOR* and *EXP01/LABOR.* Some form of manufacturing also plays a part in best seen through scale and technology which then justifies the variable of *MVA×MEDHTMVA/100*. Lastly, patent variables

have an inconsistent trends across the studies which is evidence for disconnect between patent filing and actual productivity improvement.

2.2 Theoretical Equation based on the Proposal

Comparative analysis in the previous section provides a healthy analysis for the model and the inclusion of the theoretical framework provides a formula that is outlined:

$$GDP/POP = \alpha_i + \beta_1(HTEX/LABOR) + \beta_2(IMP/POP) + \beta_3(MVA \times MEDHTMVA/100) + \beta_4 \cdot log(PATENTRE) + \beta_5(EXP/LABOR) + \beta_6 \cdot log(PATENTNONRE) + \beta_7(RDEXP \times GDP) + \beta_8(RDPOP/LABOR) + \varepsilon$$

Here is the table for the following definition of each term within the formula and how it is expected to interact in term of innovation

Table 2. Definitions, Measurement, and Correlation of each term.

EXP01/LABOR	Export Productivity per Worker	Total exports (USD) divided by labour force. Source: World Bank WDI.	Positive (+)
LOG(PATENTNONRE)	Non-Resident Patents (log)	Natural log of non-resident patent applications. Source: WIPO / World Bank WDI.	Ambiguous
RDEXP×GDP/100	R&D Expenditure (GDP-scaled)	R&D expenditure (% of GDP) multiplied by GDP (USD), divided by 100. Source: World Bank WDI.	Positive (+)
RDPOP/LABOR	R&D Personnel Intensity	R&D researchers per million population, divided by labour force. Source: World Bank WDI.	Positive (+)

Table 3. Definition of Each Term in Isolation

Abbreviation	Variable	Definition
MVA	Manufacturing Value Added	The net output of the manufacturing sector in USD
HTEX	High Technology Exports	Products with high R&D intensity, such as aerospace, computers, and pharmaceuticals
IMP	Imports	Value of goods and services imported from the rest of the world
EXP01	Exports	Value of goods and services produced domestically and exported abroad
PATENTRE	Resident Patents	Number of patent applications filed by residents of the country
PATENTNONRE	Non-Resident Patents	Number of patent applications filled by non-residents (foreigners).
MEDHTMVA	Medium & High-Tech MVA	The share (%) of MVA that comes from medium and high-technology industries
RDEXP	R&D Expenditure	Total expenditure on Research and Development as a percentage of GDP
RDPOP	R&D Personnel	Number of researchers in R&D pe million

Here are the three raised hypotheses regarding the framework:

Table 4. Raised Hypotheses

H1	*MVA* × *MEDHTMV A & HTEX* × *LABOR* are an increase in the concentration of high-technology exports and the expansion of the medium high-tech manufacturing core will have a significant positive impact on *GDP/POP*
H2	*EXP01/LABOR* export productivity acts as a primary engine for wealth creation, outperforming the growth contribution of domestic-only consumption
H3	*IMP/POP* high levels of per capita import penetration will negatively correlate with short-term *GDP/POP* unless those imports are converted into industrial capital

These hypothesises in conjunction with the methodological framework estalbished in this chapter will determine the relationship between innovation and economic growth across 37 OECD economies with the help of Kaldor's laws and endogenous growth theory. This model encapsulates the natur of innovation, manufacturing sophistication, export productivity, patent activity, and R&D investment together. The comparative analysis of existing empirical models prove that the variable is according to the established literature. The three hypotheses from this framework supply clear and provable propositions that the empirical analysis evalautes directly against OECD data from 2010 to 2021. The results of such capture how innovation translates into observable GDP per capita growth across the diverse OECD membership.

3. ECONOMIC GROWTH AND INNOVATION WITHIN OECD STATES: EMPIRICAL EVIDENCE

3.1 Data Description and Empirical Background

The World Bank WDI database of 37 OECD nations were compiled after removing empty observations, ergo observations with missing values for key variables across 2010 – 2021. The following has 386 country-year observations with certain variables expressed in USD.The following OECD countries within this model are a diverse group of member states, which is quite informative in it that diverse range of countries provide a healthy basis for how innovation and growth affect these important economies. Instead of examining these metrics in isolation, the compiled model treats these metrics with a single directive, that is, which shows comparisons of their relative contributions to economic growth.

As aforementioned, the 37 OECD are a heteragenous group of nations, which supply different points of view in regards to innovation and growth. The table below categorizes the average *GDP* per capita that includes a more recent period, drawning from the Wolrd Bank World Development Indicators Database (2010-2024). This was performed by using Python. Some errors were determined for *MVA* data for Australia, Austria, and Belgium which also had anomalouus in *MEDHTMVA* values. These errors are attended to as a data limitation and clarifies why these economies are placed in low accuracy group.

Here is the table showing the classification of the economies of OECD member states:

Table 5. OECD Countries categorized by Average GDP

GDP per Capita Range	Countries included
Below $20,000	Chile, Colombia, Costa Rica, Hungary, Latvia, Lithuania, Mexico, Poland, Slovak Republic, Türkiye
$20,000–$39,999	Czechia, Estonia, Greece, Italy, Japan, Korea, Portugal, Slovenia, Spain

$40,000–$59,999	Australia, Austria, Belgium, Canada, Finland, France, Germany, Israel, Netherlands, New Zealand, Sweden, United Kingdom
$60,000 and above	Denmark, Iceland, Ireland, Luxembourg, Norway, Switzerland, United States

3.2 Estimation Output and Results

The following equation is estimated using panel least squares with cross-section fixed effecs within Eviews10 which include the OECD member states

Figure 1. Adjusted Model Coefficients

```
Estimation Command:
=========================
LS(CX=F) GDP/POP C HTEX/LABOR IMP/POP MVA*MEDHTMVA/100
LOG(PATENTRE) EXP01/LABOR LOG(PATENTNONRE)
RDEXP*GDP/100 RDPOP/LABOR

Estimation Equation:
=========================
GDP/POP = C(1) + C(2)*HTEX/LABOR + C(3)*IMP/POP +
C(4)*MVA*MEDHTMVA/100 + C(5)*LOG(PATENTRE) +
C(6)*EXP01/LABOR + C(7)*LOG(PATENTNONRE) +
C(8)*RDEXP*GDP/100 + C(9)*RDPOP/LABOR + [CX=F]

Substituted Coefficients:
=========================
GDP/POP = 46422.7537778 + 1.24192406204*HTEX/LABOR -
0.497622273136*IMP/POP + 6.26292336945e-08*MVA*MEDHTMVA/100
- 2172.58636735*LOG(PATENTRE) + 0.572587768297*EXP01/LABOR
- 1757.57175943*LOG(PATENTNONRE) + 2.80797147825e-08*RDEXP*GDP/100
- 997902.46781*RDPOP/LABOR + [CX=F]
```

Each of these terms is defined below and all of which represent variables where *GDP/POP* is *GDP* per capita in current USD and the measure of economic health or prosperity which is the main indicator of these 37 OECD economies. The constants are estimated coefficient values from Eviews.

- *1.2419 HTEX/LABOR* -> high technology exports per worker as this is the primary innovation trade flow verifying how economies export and produce high-technology

items per labour or worker in this case. The positive coefficeint of 1.2419 means that for every high tech export per worker the GDP per capita increases by approximately 1.24 USD. This also works in the favour of the model as this confrims that economies which produce and export more high tech products generate higher income per person

- $-0.4976 \times IMP/POP$ → imports per capita. The following negative coeffiicient of -0.4976 means that for every additional unit of imports per person, GDP per capita decreases by 0.5 USD

- $6.26\times10^{-8} \times MVA \times MEDHTMVA/100$ → the manufacturing interaction term. Manufacturing value added is multiplied by that share of that manufacturing coming from medium and high technology industries then divided by 100. Its coefficient confrims technological sophistication of that sector drives growth.

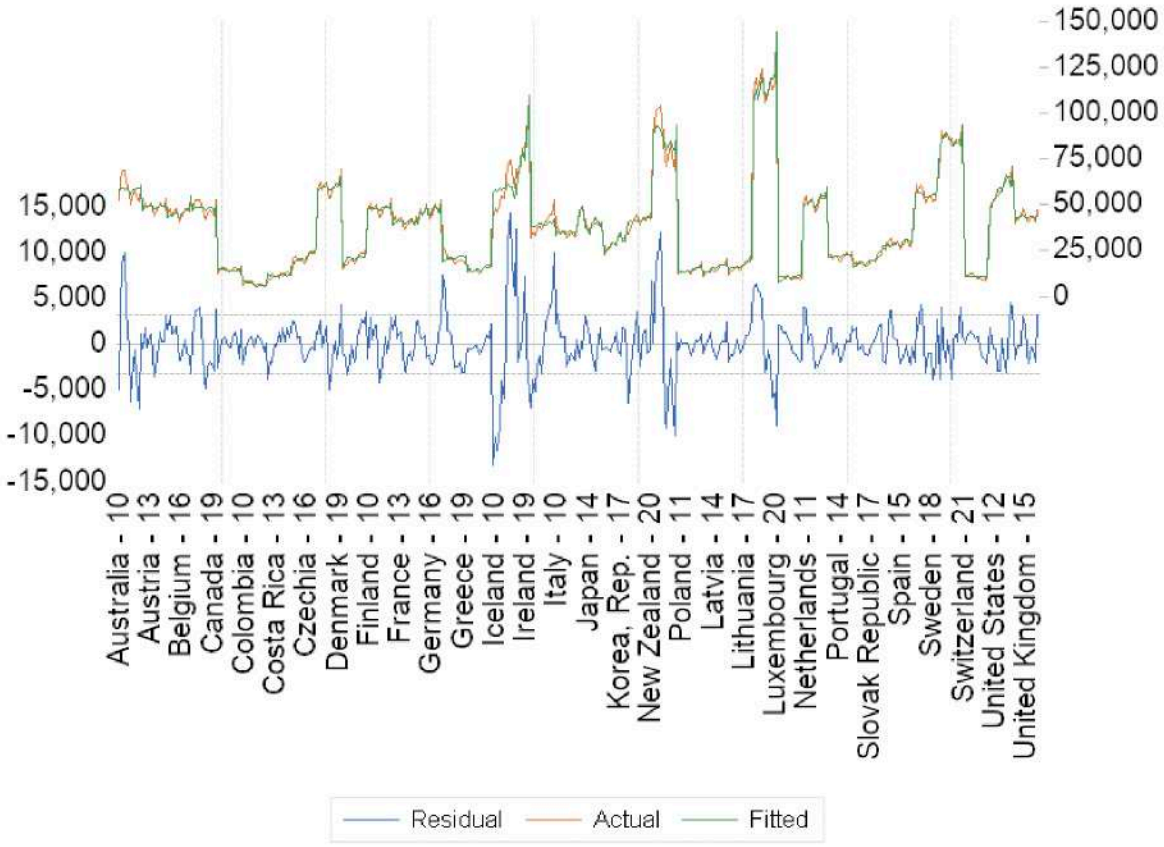
- $-2172.59 \times log(PATENTRE)$ → the natural log of resident patent applications. The negative coefficient is quite reflective of how diminishing returns at high patent stock levels contribute to growth. The log captures this unlinearity

- $0.5726 \times EXP01/LABOR$ → exports per worker. The positive coefficient of 0.57 is the henceforth, the strongest predictor in the model as seen in the t-statistic which implies that for every additional unit of total exports per worker, GDP per capita increases by approximately 0.57 USD. The sample of these countries show consistent trends on export-oriented production drives economic prosperity

- $2.81\times10^{-8} \times RDEXP \times GDP/100$ → R&D expenditure scaled by total GDP. The positive coefficient confirms investment in R&D generates productivity and multiply over time which is consistent with endogenous growth theory. This relation with GDP confirms the that variable interacts well with the GDP per capita and captures the scale of R&D spending.

- $-997902.47 \times RDPOP/LABOR$ → R&D personnel per worker. The negative coefficient almost certainly reflects a certain nuance rather than a genuine economic relationship. Countries with large R&D workforces simultaneously tend to have high values for *HTEX*, *RDEXP*, and patent variables. This is a methodological limitation rather than a finding
- [CX=F] → the cross-section fixed effects which is a part of the estimation. Each country within OECD has its own fixed term that capture the time country characteristics.
-

Figure 2: Actual, Residual and Fitted Graphs



The figure represents actual fitted and residual GDP per capita values across all 37 OECD nations over the 2010-2021 period. The orange line defines the observed GDP per capita values and the green line represents the model's predicted values. Their close intersection and overlap between the two lines represent a strong fit for the model. The residual blue which is below the twolines represent the difference between actual and fitted lines as mentioned before which are

orange and green respectively. This line remains close to zero in a majority of the observations which means it represents the growth patterns strongly.

The notable deviations in this graph such as Ireland, US, and Luxembourg similarly reflect the outsized role of financial services within those economies. Luxembourg has the highest deviation as it is a country more reliant on its financial services. Those cases define structural outliers where the model's fitted values diverge from actual values for reasons no related to innovation and manufacturing,

Figure 3. Model Results

Dependent Variable: GDP/POP
Method: Panel Least Squares
Date: 11/13/25 Time: 17:35
Sample (adjusted): 2010 2021
Periods included: 12
Cross-sections included: 37
Total panel (unbalanced) observations: 386

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	46422.75	5488.563	8.458089	0.0000
HTEX/LABOR	1.241924	0.291383	4.262174	0.0000
IMP/POP	-0.497622	0.084412	-5.895190	0.0000
MVA*MEDHTMVA/100	6.26E-08	1.20E-08	5.205333	0.0000
LOG(PATENTRE)	-2172.586	701.7479	-3.095964	0.0021
EXP01/LABOR	0.572588	0.049500	11.56743	0.0000
LOG(PATENTNONRE)	-1757.572	293.1349	-5.995779	0.0000
RDEXP*GDP/100	2.81E-08	8.46E-09	3.320142	0.0010
RDPOP/LABOR	-997902.5	326039.1	-3.060684	0.0024

Effects Specification

Cross-section fixed (dummy variables)

R-squared	0.989661	Mean dependent var	37806.16
Adjusted R-squared	0.988327	S.D. dependent var	24953.39
S.E. of regression	2695.995	Akaike info criterion	18.74613
Sum squared resid	2.48E+09	Schwarz criterion	19.20730
Log likelihood	-3573.003	Hannan-Quinn criter.	18.92901
F-statistic	741.8492	Durbin-Watson stat	0.881330
Prob(F-statistic)	0.000000		

The residual graph displays the differences between the actual and fitted graphs and showing the differences as residual. As seen in the residual graph which has no significant deviations leading the model to be a correct model underscoring a healthy relation between the variables. The emerging results are quite a strong fit as the following metrics which namely include the R-squared cross-section effect. The R-squared model of 0.9897 means the model explains just under 99 % of the variables within the GDP per capita across the dataset. The promising number of 741.85 for the F-statistic with p being 0 verifies that this model is wholly

significant. The number of 0.881 for the Durbin-Watson effect is also quite significant in this matter as the value is low, which means a positive correlation. A value being near 2.0 implies that there is no significant correlation; values below 2.0 imply some correlation. The statistic is in the acceptable threshold, verifying the positive correlation.

- *HTEX/LABOR* (coefficient →1.242) is confirms H1 by how high-technology exports in relation to labor usually indicate higher GDP per Capita
- *EXP01/LABOR* (coefficient→ 0.573) is one of the strongest positive predictor by t-statistic-11.57 which mean that countries with more exports per worker tend to be wealthier within the OECD member states which confirms H2
- *MVA* × *MEDHTMVA/100* (coeffiecient → 6.26E-08) is significant though quite minor. This coefficient also confirms H1 as the interaction between manufacturing scale and technology has a positive relation with GDP per capita.
- *IMP/POP* (coefficient → -0.498) is the coefficient that describes how member states that take in more imports have less GDP per capita which also wholly confirms H3.
- *RDEXP* × *GDP/100(*coefficient → 2.81E-08) which implies sustained investement results in productivity
- *LOG(PATENTRE)* and *LOG(PATENTNONRE)* both have negative and significant coefficients which is to do with documenting tactical patent filing action disconnected from productivity

The resulting model equation with substitued coefficients goes as follows:

$$\frac{GDP}{POP} = 46{,}422.75 + 1.2419 \cdot \frac{HTEX}{LABOR} - 0.4976 \cdot \frac{IMP}{POP} + 6.26 \times 10^{-8} \cdot \left(MVAit \times \frac{MEDHTMVA}{100}\right) - 2{,}172.59 \cdot LOG(PATENTRE) + 0.5726 \cdot \frac{EXP01}{LABOR} - 1{,}757.57 \cdot LOG(PATENTNONRE) + 2.81 \times 10^{-8} \cdot \left(RDEXP \times \frac{GDP}{100}\right) - 997{,}902.5 \cdot \frac{RDPOP}{LABOR} + [CX = F]$$

3.3 Model Forecasting and Country Classifications

The performance will be evualuated at the individual level in terms of GDP per capita values obtained from the datasets and each with specific graphs for these 37 economies. Actual values appear in blue and model predictions in orange.

Countries in this manner are grouped into three respective accuracy groups depending on the extent of the alignment between actual and forecasted values, and the number of valid observations needed for this estimation, and this reflects whether the country's GDP growth over the years reflect innovation

Accordingly, the analyzed countries are categorized into three distinct accuracy groups:

1) High Accuracy Group
2) Medium Accuracy Group
3) Low Accuracy Group

3.4 High Accuracy Models

The high accuracy model economies in this section with predicted GDP per capita trajectories track each other closely enough throughout this decade The countries in this group such as Germany, Hungary, Korea, Japan, United States,Ireland, Switzerland and Poland show strong alignment between the orange and blue lines, which matches the scope of this thesis. Their actual and fitted growth go hand in hand as is explained by how innovation impacts the countries' growth. These countries verify the trend of innovation and trade variables. Germany, Korea, and Japan are the evidence of high-technology manufacturing leadership while Hungary and Poland has seen successful industrial trends in Central Europe. The United States, Ireland and Switzerland show almost perfect alignment which is the result of sophistication of their high tech exports and constant innovation initatives.

Figure 4. Germany's GDP Values

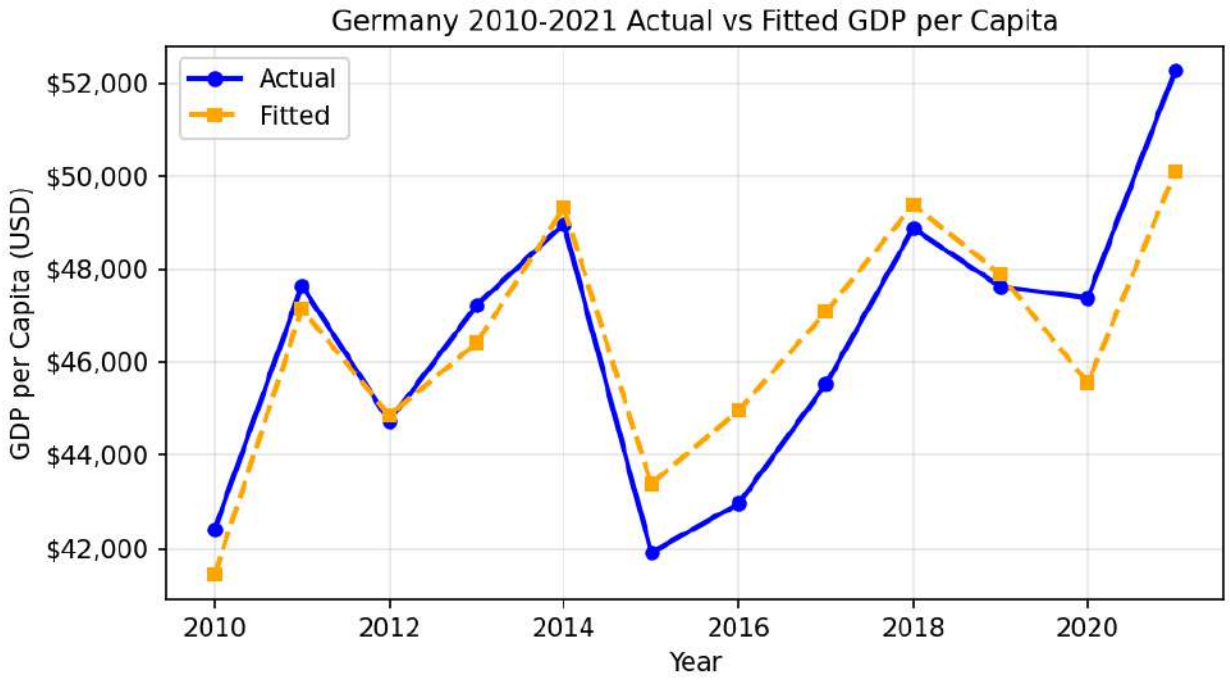
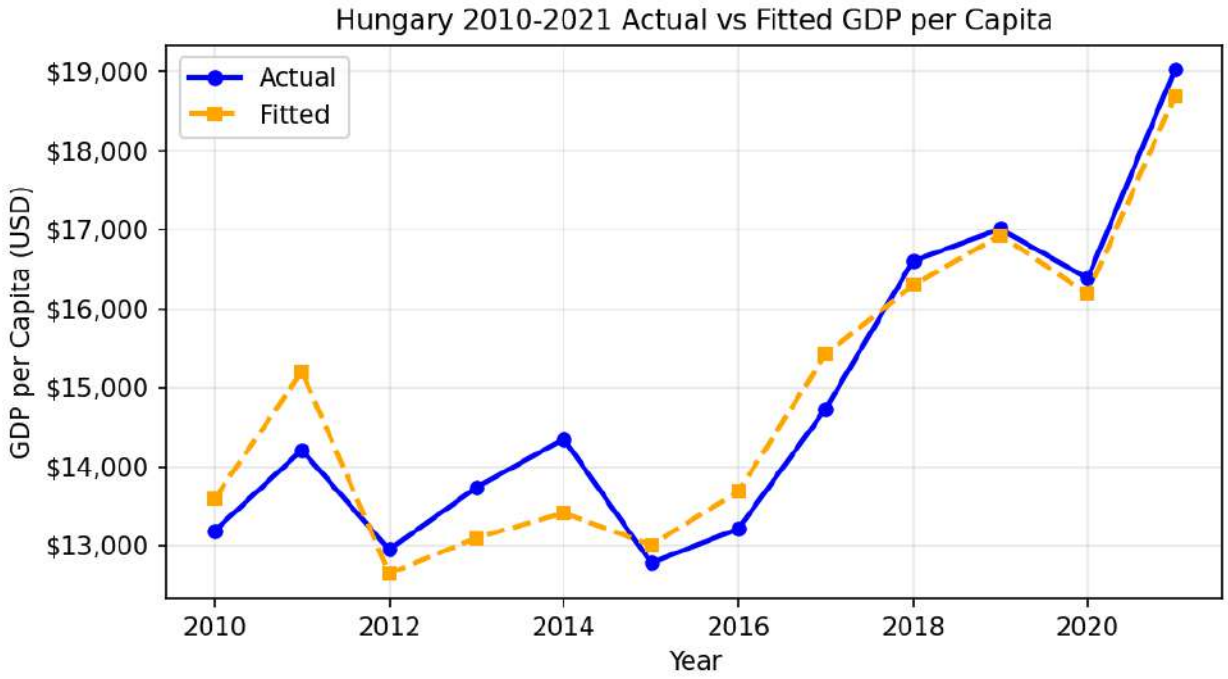


Figure 5. Hungary's GDP Values



3.5 Medium Accuracy Models

In this distinction of medium accuracy nations, these coutries typically follow a general trend of GDP growth with a little deviation at certain points. These country include: Luxembourg, Denmark, Estonia, Netherlands, Norway, Lithuania,Sweden,United Kingdom, Finland, France, Canada, Latvia, Czechia, Greece, New Zealand, Colombia, and Israel

Though the lines notably move in the same direction for most of the sample period, there are some points where the divergence happens. This usually could be explained that there were periods when services, or other unrelated growth resulted in the gdp growth.

Figure 6. Luxembourg''s GDP Values

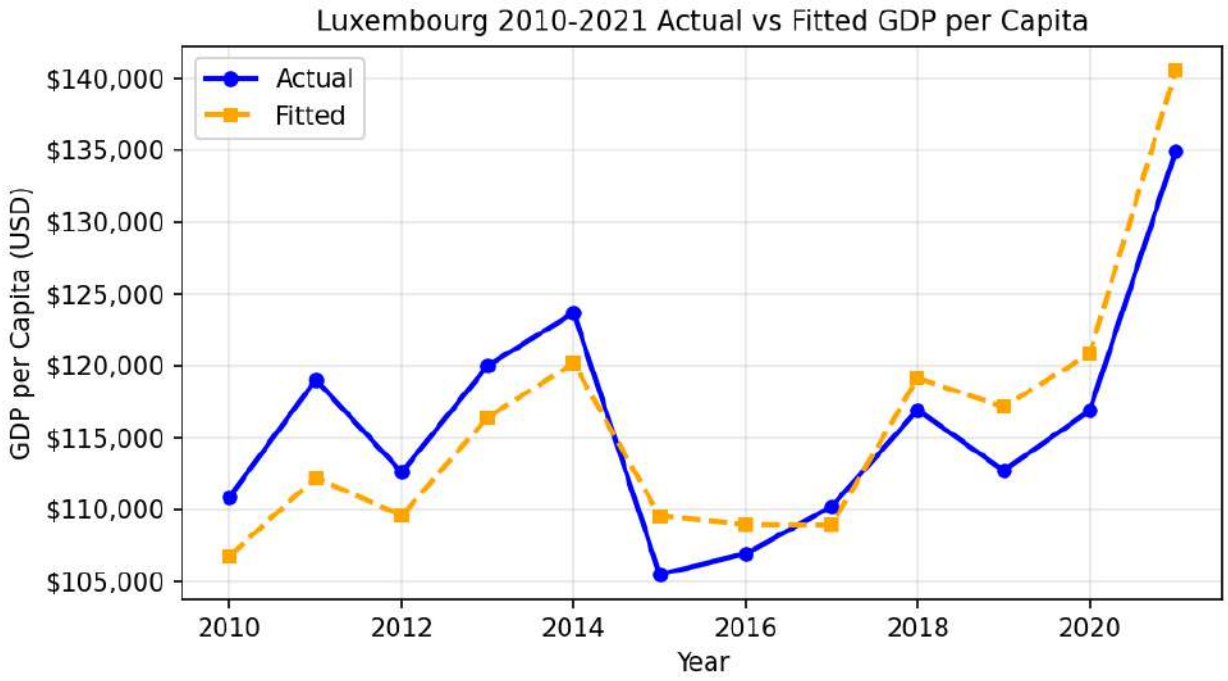
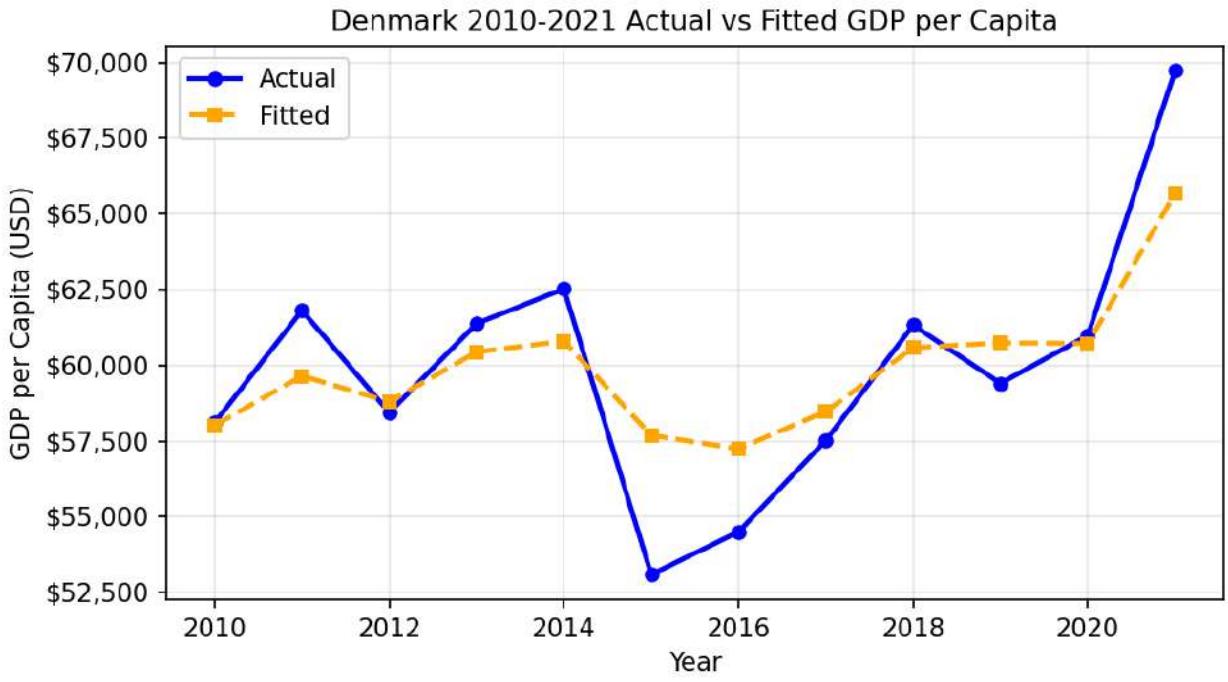


Figure 7. Denmark''s GDP Values



3.6 Low Accuracy Models

This is the example where the actual and forecasted values diverge substantially: Costa Rica, Türkiye, Spain, Mexico, Portugal, Chile, Iceland, Italy, Slovenia, Australia, Austria, and Belgium

The model may not capture tourism, early industrial development, high levels of coruption which may halt some growth or early-stage industrial development. For Australia, Austria and Belgium the poor fit is in due to the corrupted MVA data in the dataset acknowledged as a data limitation.

Outcomes vary sharply among these nations even though forecasts look much alike. Looking closely at specific examples brings the reasons into better view. Spain and Portugal, for example, grow mainly through services such as tourism and real estate rather than high-end manufacturing or technology-driven exports, which makes innovation-focused projections miss the mark. Nations still building industrial foundations - Mexico and Colombia included - see progress influenced by informal labor, dependence on commodity selling, and deep-rooted structural limits, factors that standard economic models often overlook because they rely on patterns from wealthier, more organized systems. Flawed MVA data within World Bank archives affects how well the model fits for Australia, Austria, and Belgium - distorting the MVA×MEDHTMVA/100 term repeatedly. These measurement errors stand acknowledged; yet they reveal no insight into whether innovation actually fuels growth in those nations.

Figure 8. Colombia's GDP Values

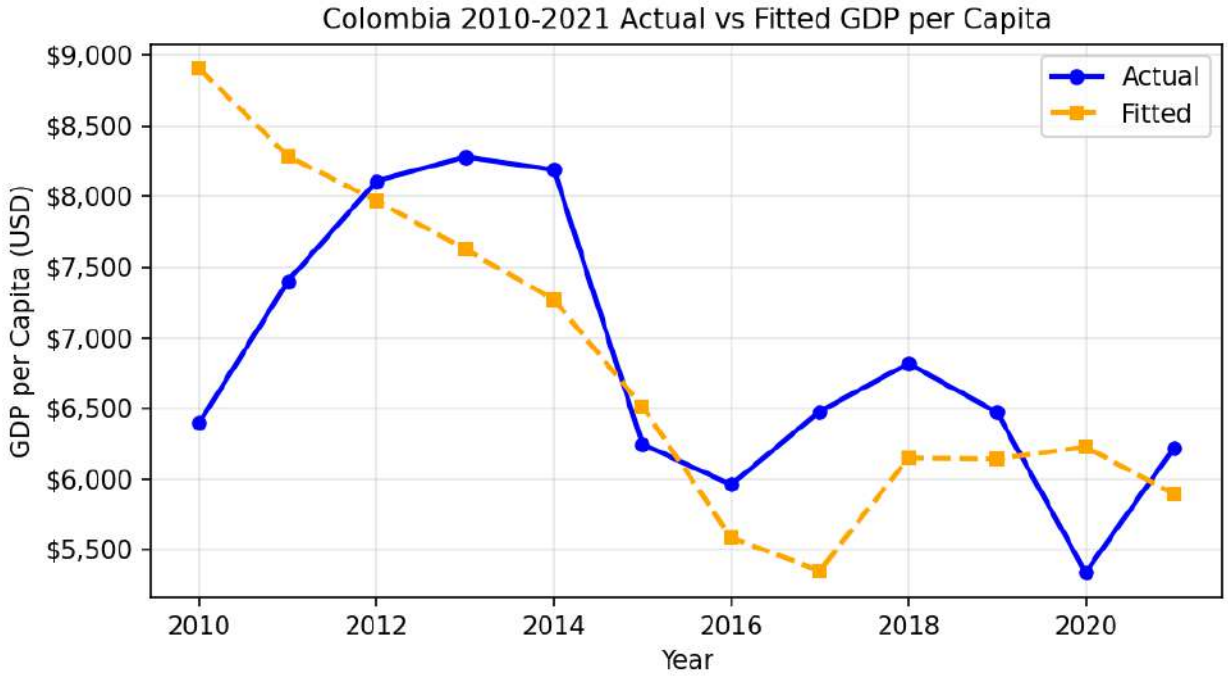
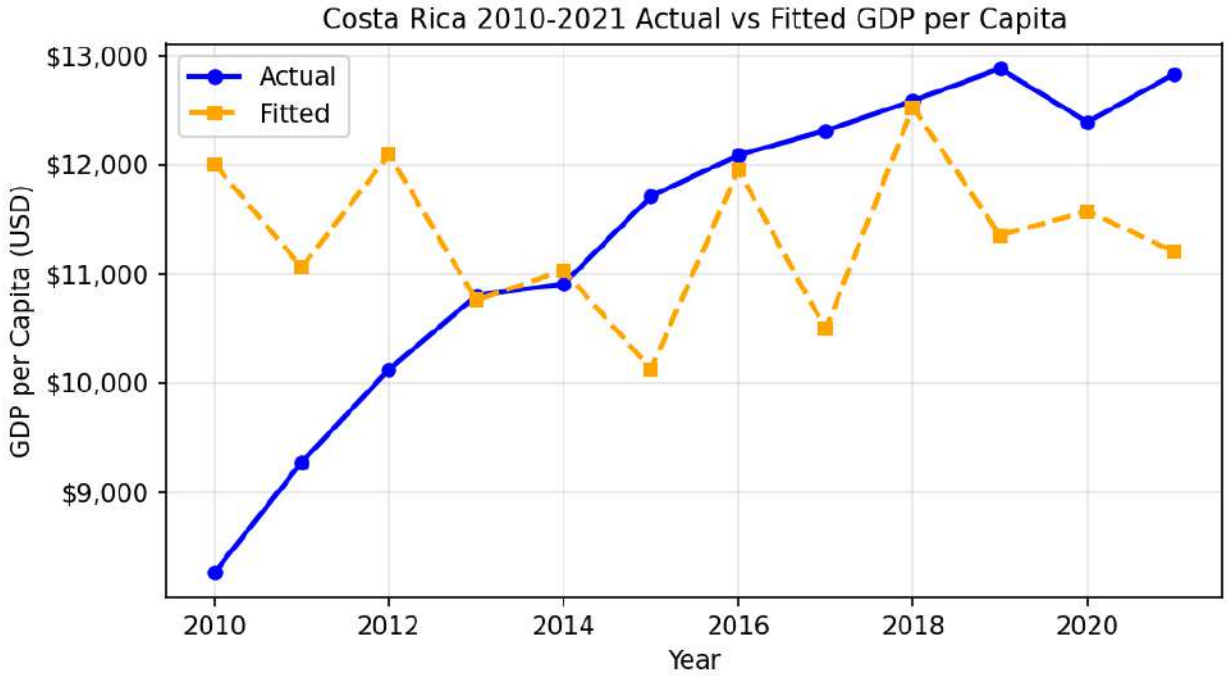


Figure 9.Costa's GDP Values



3.7 Model Interpretation

This model can coherently explain how economies with wealth built by manufacturing sophistication, high-technology-exports, export and import are well clarified in this model countries within the high-accuracy model are the examples that follow this model well enough, where certain coefficients, for example HTEX/LABOR and EXP01/LABOR go hand in hand with GDP per capita exactly as our hypotheses and theories propose, Economies that generate wealth from other sources also follow this model somewhat but some limitations may ariseA further limitation is that countries within the low accuracy model contain corruputed MVA figures in the dataset, introducing noise into the MVA interaction. This model captures how impactful manufacturing and innovation is that generate GDP per capita in advanced economies.

4. PROPOSED INNOVATION-TO-GROWTH TRANSMISSION (IGT) FRAMEWORK

4.1 Overview

Emperical analysis shows differences across OECD countries when facing similar innovation factors. Although the panel data shows a near-perfect fit - R-squared close to 0.99 - not every nation matches these predictions equally well. Instead, individual nations diverge in how tightly their GDP aligns with models emphasizing trade and invention. The results within emperics reveal how innovation conversion into GDP per capita is the same for all members across the OECD. Two dimensions that would be graphically represented would be constructed in a 2x2 diagnostic matrix which each country would follow with these indicators:

Export Sophistication Index (ESI): a fusion or normalised composite of HTEX/LABOR and EXP01/LABOR, translates into the labor technology and the following productivity of a country's export sector

Manufacturing Sophistcation Index (MSI): a normalised form of the variable MVA×MEDHTMVA/100 which captures the scale and technology intensity within domestic manufacturing. Below is a graphical representation of IGT framework diagram. The following diagram shows all three layers of this framework plus the diagnostic framework which will be defined under a separate table.

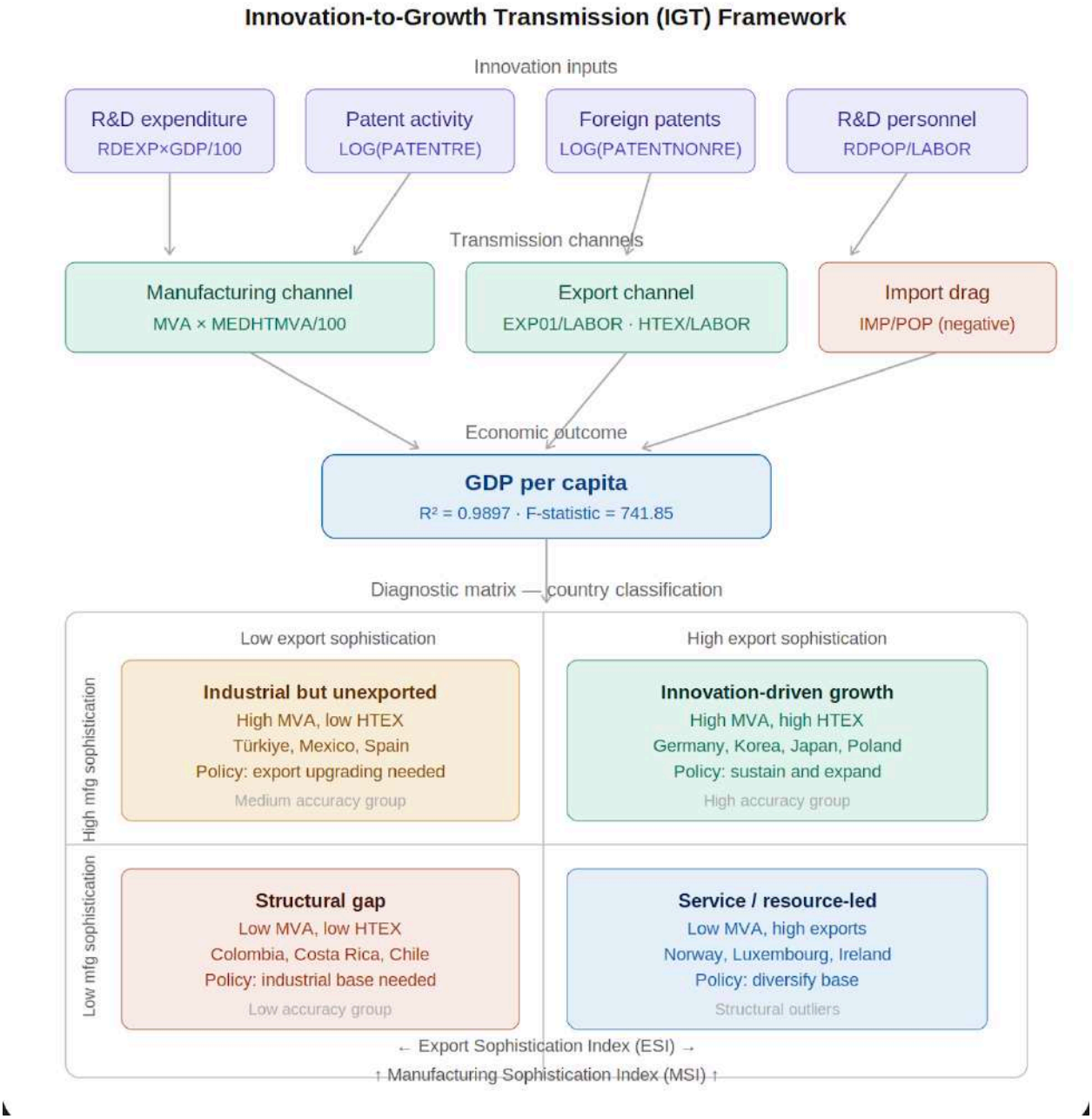


Figure 10. The IGT Framework Representation

- Layer 1 → Innovation Inputs which include R&D expenditure and personnel, patent activity both native and foreign which are the four innovation variables derived from the model

- Layer 2 → Transmission channels are the heart of manufacturing channel which include (*MVA×MEDHTMVA/10),,*export metrics *(EXP01/LABOR* and *HTEX/LABOR),* and lastly import metrics *(IMP/POP).*
- Layer 3 → Economic Outcome which are GDP per capita fit statistics which resulted from previous layers
- Layer 4 → Diagnostic Matrix which are the 2x2 ESI/MSI classification classifying OECD economies into a following quadrant.

Table 6. The ESI Matrix

	Low Export Sophistication (ESI)	**High Export Sophistication (ESI)**
High Manufacturing Sophistication (MSI)	QUADRANT III – Efficient Converters Strong industrial base; limited export sophistication. Scale up production know-how to improve export mix. Examples: Hungary, Poland.	QUADRANT I – Innovation Leaders High on both dimensions. Maintain R&D edge; diversify into frontier technologies. Examples: Germany, South Korea, Switzerland, United States.
Low Manufacturing Sophistication (MSI)	QUADRANT IV – Lagging Economies Weak on both dimensions. Build core strengths: education, industrial networks, knowledge access. Examples: Türkiye, Mexico, Chile.	QUADRANT II – Frustrated Innovators High export sophistication but limited manufacturing base. Remove commercialisation barriers; strengthen GVC links. Examples: Ireland (services-driven), Iceland.

4.2 Framework Description and Policy Implications

Each Quadrant characteristics is explained seperately and classified into certain classes as written in the table above

Quadrant I is called the Innovation Leaders which include countries like Germany United States, South Korea, Switzerland and Japan. These nations present a verifiable alignment between innovation specific metrics and observed GDP per capita growth. One main recommendation is sustain the momenmtum of R&D intensity in the upcoming technoliges and ascertain growing and continued investment in high technology tools or manufacturing and eventually diversify into high-value sectors such as biotechnology, quantum computing and advanced robotics

Quadrant II preferably called the Frustrated Innovators includes nations with advanced export profiles that struggle to convert innovation into GDP per capita gains. In this case, these economies tend to be driven by service sectors. One recommendation is to reinforce the manufacturing sector and advance the integration into high-value GVC sections

Quadrant III tend to be economies that already have a strong profile with a strong manufacturing base but not yet transitioned into a high-sophistication export profiles. Central European excluding Germany such as Hungary and Poland instantiate this. One recommendation is to accelerate workfoce integration and upgrade the technology content of exports progressively. In this case, Efficient Converters is a suitable name for this quadrant

Quadrant IV is understandbly reserved for economies which are behind or lagging, better named as Lagging Economies. These economies perform below the median on both dimensions. Growth in these economies is often driven by natural resources and commodity exports with limited interaction with globalization. One main foundational recommendation regarding this is to apply investments in tertiary education (universities and schools), build up basic research infrastructure so that R&D could grow and therefore transfers into institutions.

4.3 The Innovation-to-Growth Transmission Framework and The Diagnostic Dimensions

Innovation-to-Growth Transmission or IGT makes the distinction of two caveats regarding how innovation is connected to growth. Furthermore, IGT also looks into what economies do towards innovation such as the use of revenue, manpower and systems and R&D researchers. It's crucial how these resources could be converted into economic momentum. Strong exports prove that advanced production methods and better trade outcomes also result in income gains.

Some nations pour resources into research yet see little return in advanced exports or production quality. Though spending on innovation runs high, results often stall due to weak links between labs and factories. One reason could be rigid markets or poor coordination across sectors. On the flip side, certain economies invest less but make better use of outside knowledge. They tap foreign capital, global supply chains, or imported know-how to boost output. Their

growth outpaces what their R&D numbers would predict. Efficiency in turning ideas into goods matters more than volume alone. Performance hinges not just on how much is spent, but how well it spreads.

Dimension 1 → Export Sophistication Index, built from HTEX/LABOR and EXP01/LABOR results. Rather than just volume, it checks if nations move into products needing heavy research investment. Over time, growth in value produced per worker also shows up here. Take Germany, South Korea, Switzerland, or the United States - these places sell more drugs, smart electronics, fine tools, and custom-built machines. Their position stays strong across tests, landing them repeatedly among top performers. That pattern supports ESI's role in reflecting both economic accuracy and national income levels fairly well.

Dimension 2 → Manufacturing Sophistication using the MVA multiplied by MEDHTMVA divided by 100. Not every expanding factory base lifts a nation's ranking here - what matters is how advanced production becomes over time. Take textile hubs or simple food processors: even when output rises, gains stay narrow without broader ripple effects in learning or invention. Instead of steady climbs, progress stalls where jobs demand little expertise or fail to spark new know-how. Contrast that with Germany, South Korea, and Switzerland, where industry combines size with high-end capabilities. Their factories do more than grow - they evolve, embedding complex skills into daily work. As a result, these nations rise toward the top of the index, fitting tightly within the model's strongest performers.

Dimension 3 → Trade Balance Quality. It builds on two measures: imports per person and exports relative to workforce size. When exports are strong but imports stay within limits, it often means trade supports real economic gains at home. Such cases rely on local output, expertise, and trained workers. The economy adds value instead of depending on foreign goods. On the flip side, weak export performance paired with heavy reliance on imported products tends to stall industrial progress. Many resource-focused or consumer-driven nations show this trend. Their trading patterns fail to build foundations for lasting income improvement. Growth remains shallow without deeper productive capacity.

4.4 The Diagnostic Matrix

Positioned along the ESI and MSI scales, OECD countries appear in a four-part analytical grid. Within each section, the quality of trade balance adjusts emphasis on suggested actions. A distinct pattern forms - not by addition, but through refinement. Each nation's placement shifts slightly when that factor enters. Guidance becomes more precise where it matters most.

Table 7. Diagnostic Matrix for ESI and MSI scales

Quadrant	ESI	MSI	Characteristic Economies	Characteristics	Policy Priority
I — Frontier Innovators	High	High	Germany, USA, Switzerland, Korea, Japan	Sustaining frontier position amid rising competition	Sustain R&D leadership; diversify into AI, biotech, clean energy; deepen GVC specialization
II — Limited Innovators	Low	High	Some EU economies with strong manufacturing but limited export sophistication upgrading	Commercialization barriers; weak technology-to-export pipeline	Reduce barriers to technology commercialization; support export market diversification; strengthen IP and startup ecosystems
III — Efficient Converters	High	Low	Smaller OECD economies with strong export productivity but limited manufacturing base	Insufficient domestic manufacturing depth to sustain export sophistication	Scale manufacturing sophistication through targeted industrial policy; leverage export efficiency to attract technology-intensive FDI
IV — Lagging Economies	Low	Low	Lower-income OECD members with commodity export dependence	Insufficient absorptive capacity across both dimensions	Foundational investment in education, infrastructure, and institutional quality; build absorptive capacity before scaling R&D inputs

What stands out most is how fluid this classification really is. Over years, nations shift across quadrants because changes in policy, movements in global supply networks, breakthroughs in technology, and waves of capital reshape both what goes into innovation and how it spreads. Take South Korea: beginning in Quadrant IV during the 1970s, it reached Quadrant I by the 2000s - a leap made feasible by long-term coordination between industry strategy and innovation efforts. In another case, Ireland transformed from a modest-income farming-based system forty years ago into a leading innovator well before 2010.

4.5 Application within the OECD member states

Frontier Innovators (Quadrant I) consist of countries in the High Accuracy Group,such as Germany, the United States, Switzerland, South Korea, and Japan,whose economic growth closely follows the model's predictions. Ireland and Poland also fit this category, particularly in their recent development paths. These economies have effectively matched their innovation investments with the key growth mechanisms highlighted by the model: advanced manufacturing capabilities and strong export productivity. Their main policy challenge now is preserving this alignment amid ongoing technological changes, especially the increasing automation of production and the growing shift toward digital platforms, both of which could weaken the role of manufacturing sophistication in driving growth.

Efficient Converters (Quadrant III) consist of some Nordic nations fall into this as they still export well, yet rely less on modern industrial production. Instead of factories driving growth, high-end services like finance, tech solutions, and consulting now lead their foreign sales. Their global reach in these areas remains solid; however, gains may shrink if competition rises through easier cross-border access and machine-driven delivery methods. Without rebuilding deeper industrial foundations, long-term advantage might fade despite current success abroad.

Lagging Economies (Quadrant IV) are low-income OECD countries such as Chile, Colombia, Costa Rica, Mexico, and partly Turkey sit in Quadrant IV. Despite being part of the organization, their exports lack complexity while industry output stays weak compared to

wealthier peers. For them, a core obstacle emerges early: developing skills to adopt new knowledge. Stronger institutions matter just as much. So does forming reliable production systems. Without these, joining the cycle where innovation drives growth remains out of reach. Progress hinges on foundations others already built. Starting points differ widely across the group. Yet the gap persists. Moving forward demands more than effort - it requires structure. What works elsewhere may not transfer directly here. Context shapes every step taken. Growth pathways twist unpredictably when starting behind. Still, movement depends on choices made now. The route ahead lacks shortcuts.

4.6 The Following Limitations

One limitation of the IGT Framework lies in its reliance on data from the World Bank WDI database. Available metrics there miss key aspects tied to how innovation fuels growth. Missing pieces include measures like broadband access, uptake of online commerce, and activity within digital platforms. Venture capital presence also does not appear in the dataset. Even small-scale research efforts in unofficial markets go unrecorded. Not every relevant factor makes it into the numbers used.

Looking at things differently, the model relies on fixed mean values from 2010 to 2021 to assign national positions in the grid, possibly hiding shifts that occur during those years. Take a nation starting out in Quadrant IV yet moving toward Quadrant I by the end - showing fast structural change - it gets placed somewhere in between, blurring its real path over time.

One issue lies in how the framework carries forward the autocorrelation weakness from its base regression model. With a Durbin-Watson value at 0.881, signs point to positive serial dependency among residuals - this undermines accuracy in estimating coefficients and skews calibration for both ESI and MSI scales. Improving empirical rigor could come from using HAC-adjusted standard errors or shifting toward a dynamic panel setup instead.

Despite its constraints, the IGT Framework stands on solid research methods and real-world validation when assessing how innovation fuels growth in diverse countries. Because the model draws from key variables shown effective in statistical tests, its guidance reflects actual data patterns instead of abstract theory. Each of its three parts - Export Sophistication, Manufacturing Sophistication, and Trade Balance Quality - aligns closely with findings from panel analysis. This link ensures practical relevance, grounding outcomes in measurable economic behavior.

CONCLUSION AND PROPOSALS

To conlucde this thesis, the main goal of it was to build upon and suggest a model for assesing the influence the impact of innovation on the development of high value-added industrial enterprises and how it interacts with GDP growth or in this case economic growth within the conditions of globalization. The following objectives will be answered below.

1. The scientific literature proved the consistencies that innovation provides and is a main driver of long-run economic growth, especially in advanced economies or high value-added industrial enterprises. Endogenous growth theory proved that knowledge sharing and R&D investment in economies bring upon economic growth that helps sustain productivity growth beyond what it is normally achieved. One of the main frameworks for thesis is also based on the Kaldor's growth laws which verifies that manufacturing sophistication and expansion plays an important role in driving the economic output. Under globalization that has ever been increasing the prior dynamics like these become much more nuanced and complex, economies who already have advanced sophistication have an easier time or a benefit difussing knowledge with already existing export systems, while those who previously lacked the foundations of an advanced economies struggle to convert these into sustainable growth. Export Sopshistication and Manufacturing Technology are also verifiable indicators for GDP per capita than patent specific metrics and R&D alone

2. Key empirical studies were compiled, reviewed and intrepreted in terms of their methodological basis. The comparative analysis to do with exports, R&D expenditure, or manufacturing sophistication has conducted. The study was designed as a fixed-effects panel regression model bringing eight variables which are the export, manufacturing, and R&D dimensions of innovation. Composites such as manufacturing sophistication expressed as MVA×MEDHTMVA/100 was devised as a variable which combines industrial scale with high technology interactions or better defined as technology intensity.

3. Panel datasets is one of the main ways the database was constructed. Subsequent, databases World Bank Development indicators comprising of 37 OECD. Nations across the period of 2010-2021. The dataset has 13 variables per year and 456 total observations of which 386 were compiled and used Eviews data estimation and Python scripts, which pointed out systematic errors in the MVA variable for Australia, Austria, anf Belgium. As such, these errors are mentioned and put under a limitation of the dataset. The full databases and datasets is presented in the Appendix

4. The fixed-effects panel regression model was estimated in Eviews using the following: least squares with cross-section fixed effects, F-statistic Export productivity per worker (EXP01/LABOR), high technology export intensity (HTEX/LABOR) and manufacturing sophistication composite. All of which have very positive correlation such as R^2 of 0.9883 and a F-statistic of 741.85 verifying strong correlation. Export productivity per worker (EXP01/LABOR) was quite the firm predictor with a t-statistic of 11.567. Secondly, high-technology export intensity with a t-statistic of 4.262 also emerged as a strong indicator and the manufacturing sophisticated also responded with a strong indicator of 5.205. Though Import penetration per capita had a negative effect, whereas R&D expenditure in correlation with GDP was postive. In conclusion of these findings, the IGT framework proposed as the structured diagnostic model works fine witth the given systems of OECD economies under a diagnostic matrix and a 2x2 matrix in accordance with Manufacturing Sophistication Index and Export Sophistication Index dividing into relevant classifications.

5. Policymakers should focus less on increasning patent filing or R&D spending and more on the build up of export sophistication and manufacturing technology intensity as the findings and study suggest and how well these translate innovation into GDP growth. The IGT framework provides a practical starting point in it that future research attempts this on the non-OECD economies such as the EU and bring in time lags to encapsulate delayed growth effects. Digital transformation help in exploring how it reshapes the innovation-to-growth transmission across different country groups.

All in all, this thesis proves that innovation-driven economic growth within OECD nations is efficiently dirven through export and manufacturing technology intensity instead of patent activity and R&D spending alone. The thesis used regressional analhysis for empirics across OECD countries over the period of 2010-2021 which verify that economies producing and exporting technologically sophisticated goods effectively generate higher GDP per capita which is more apparent in advanced economies. All this combined, this analysis offers an important contribution to understanding how innovation impacts economic growth under globization.

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APPENDICES

Appendix A

4.7 High Accuracy Models

Figure A.11. Korea's Rep GDP per Capita

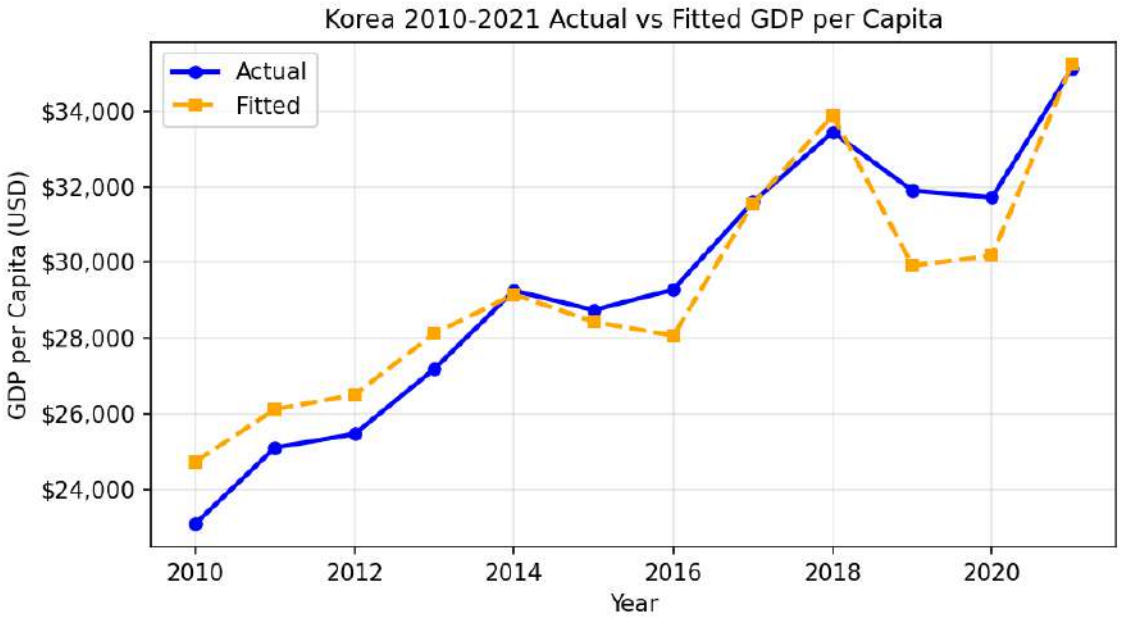


Figure A.12. Japan's GDP per Capita Values

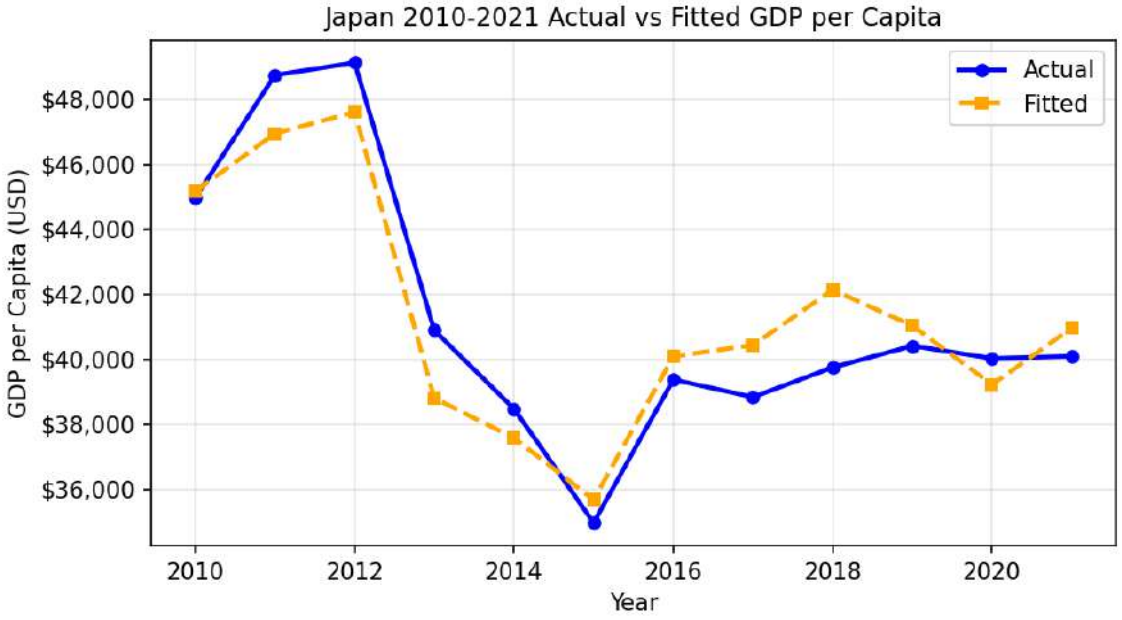


Figure A.13. Poland GDP's per Capita Values

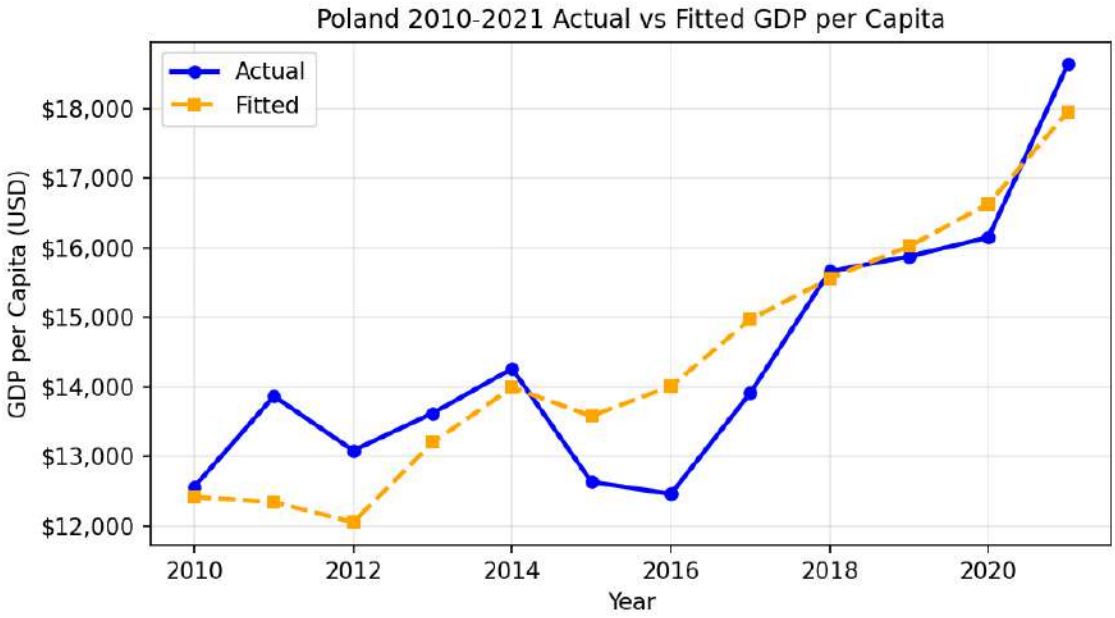


Figure A.14. United States GDP Values

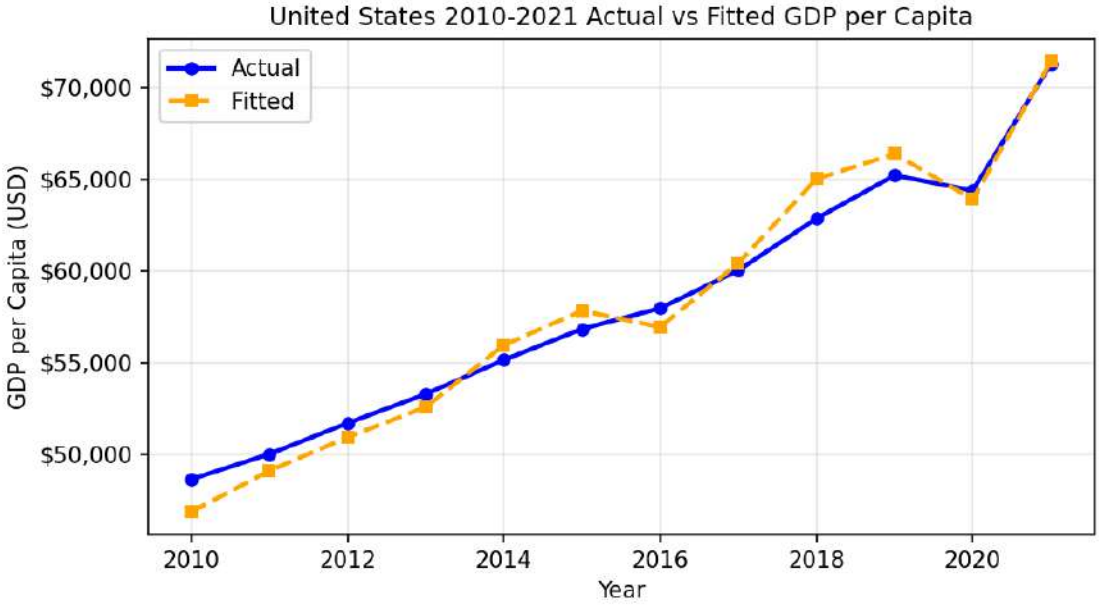


Figure A.15. Ireland's GDP values

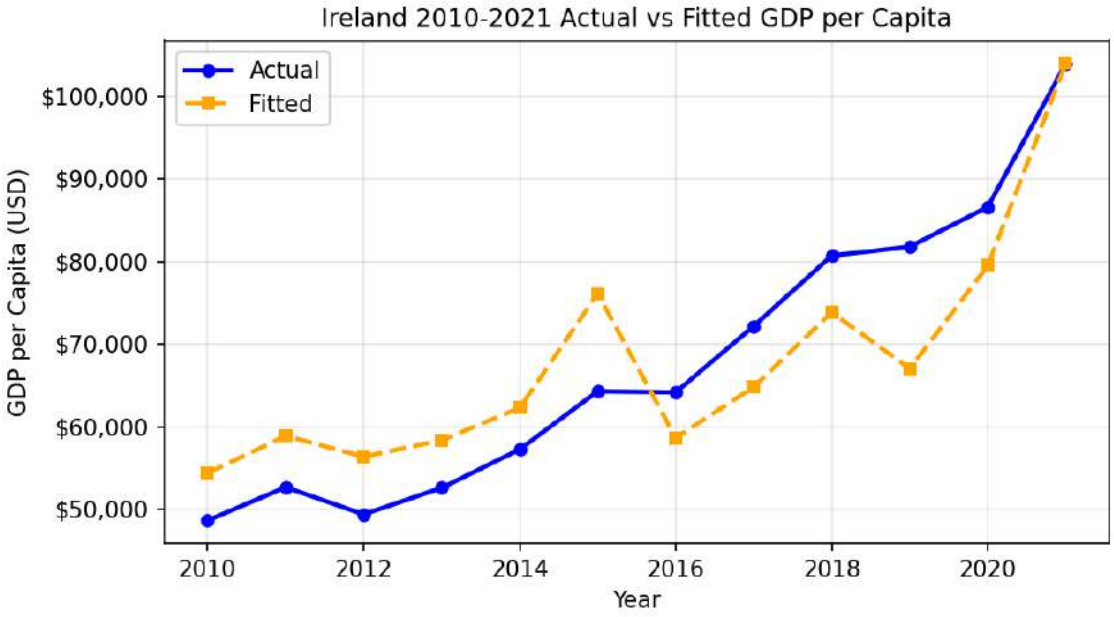
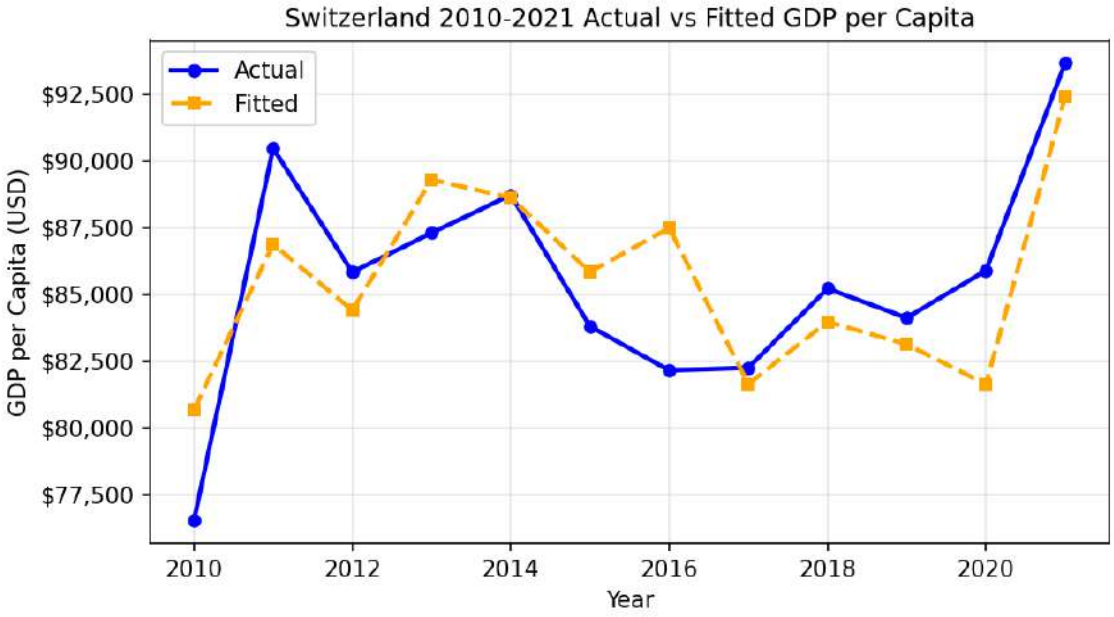


Figure A.16. Switzerland GDP values



4.8 Medium Accuracy Level

Figure A.17.Estonia"s GDP Values

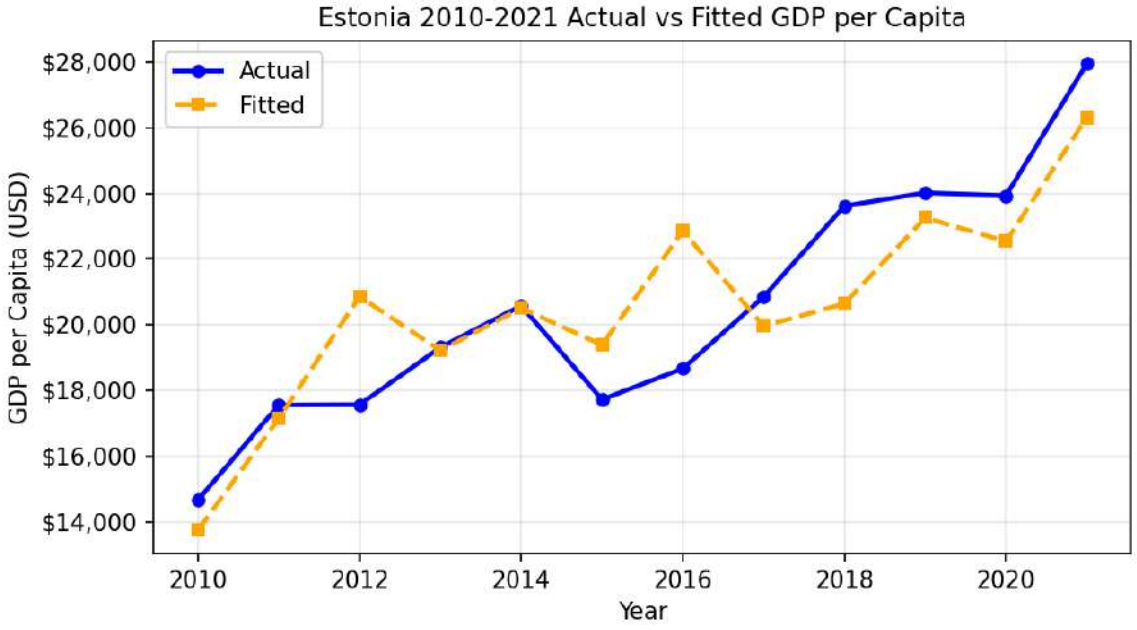


Figure A.18.Netherlands's GDP Values

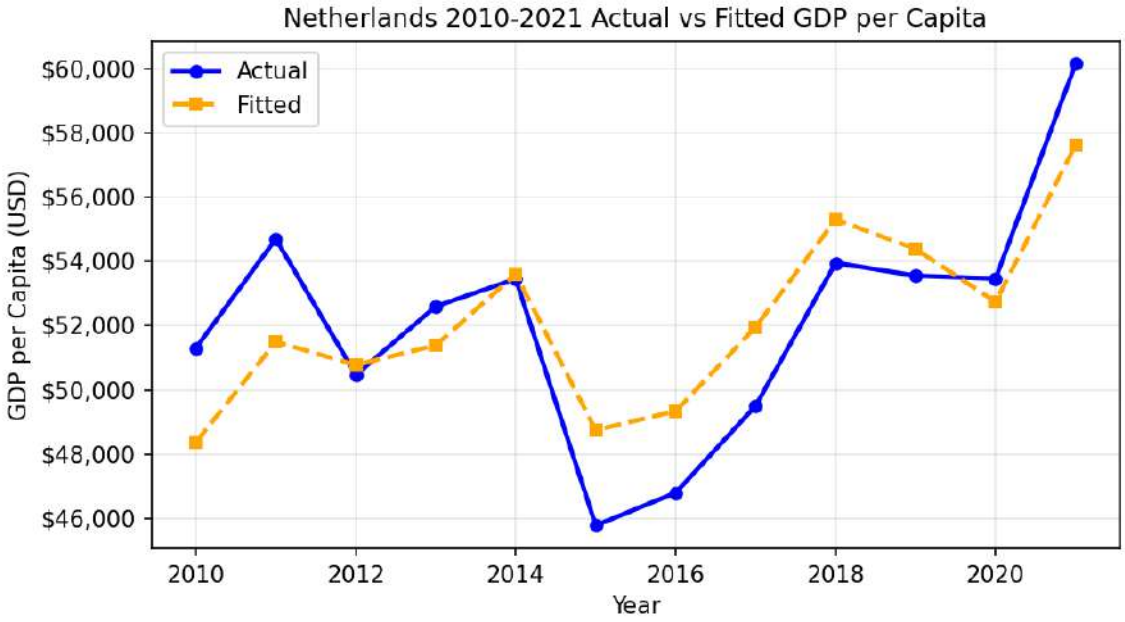


Figure A.19. Norway's GDP Values

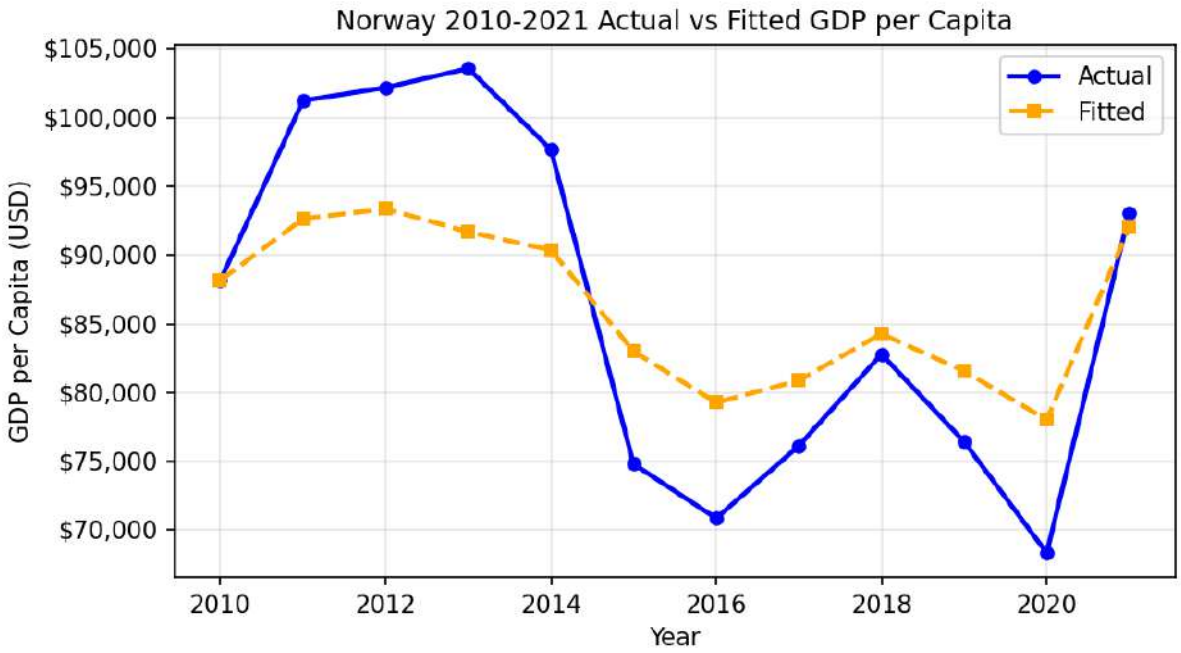


Figure A.20. Lithuania's GDP Values

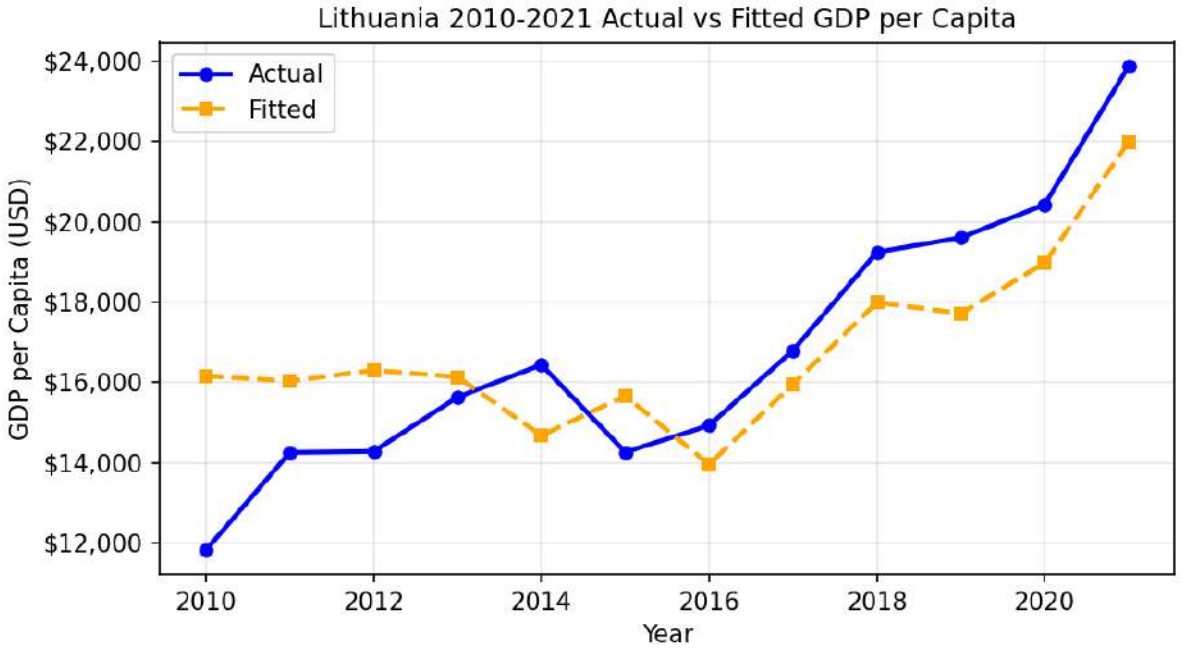


Figure A.21. Sweden''s GDP Values

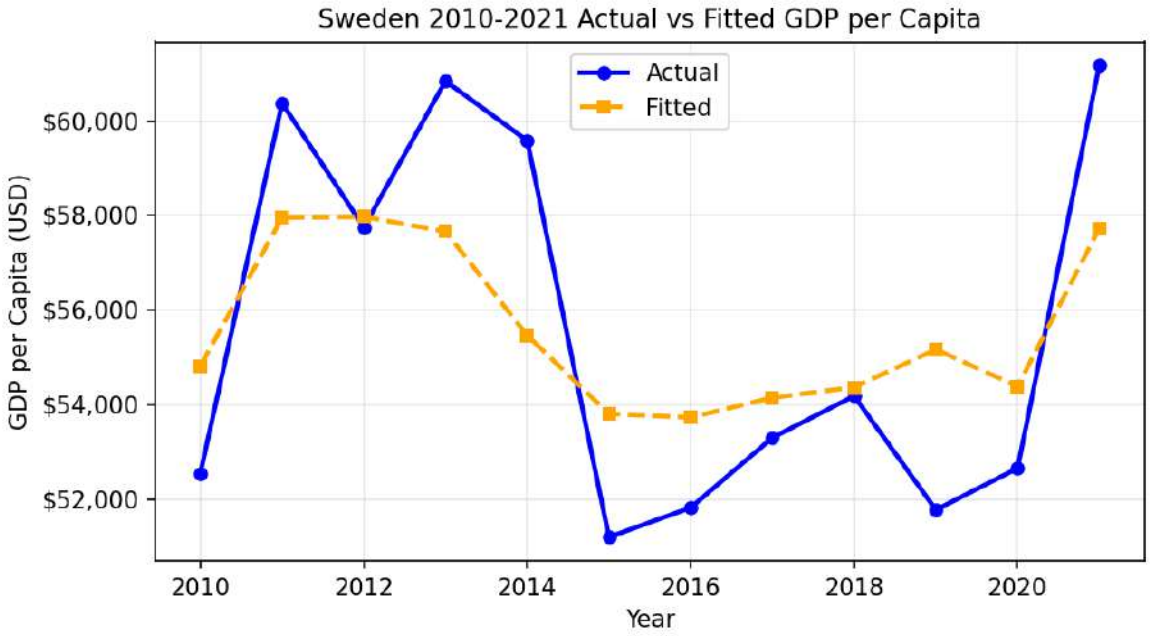


Figure A.22. Czechia''s GDP Values

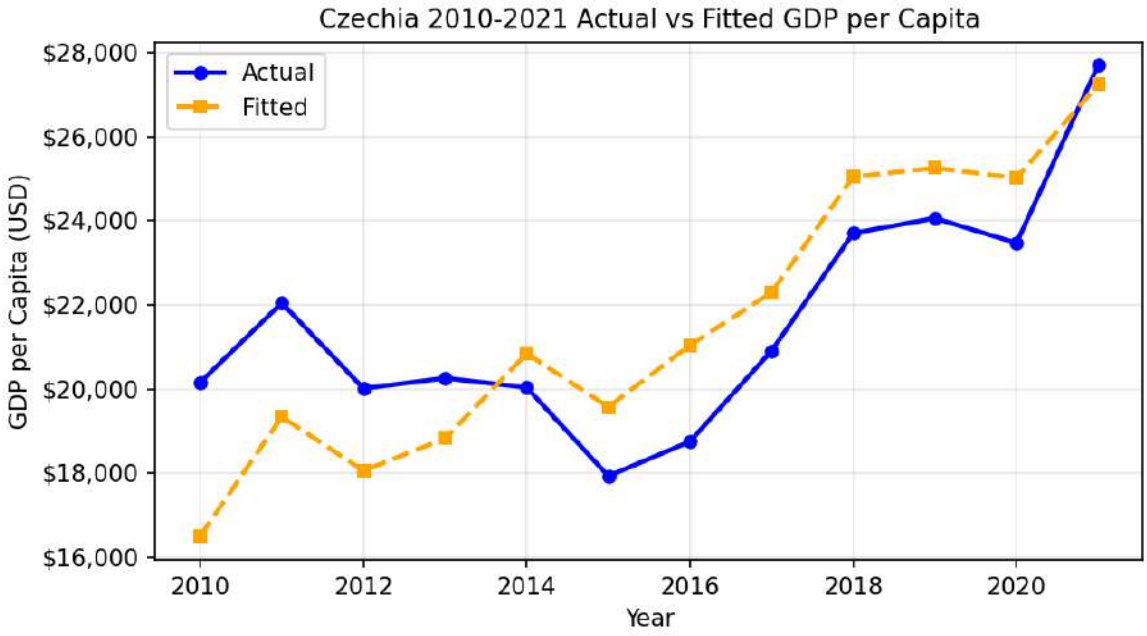


Figure A.23.United Kingdom''s GDP Values

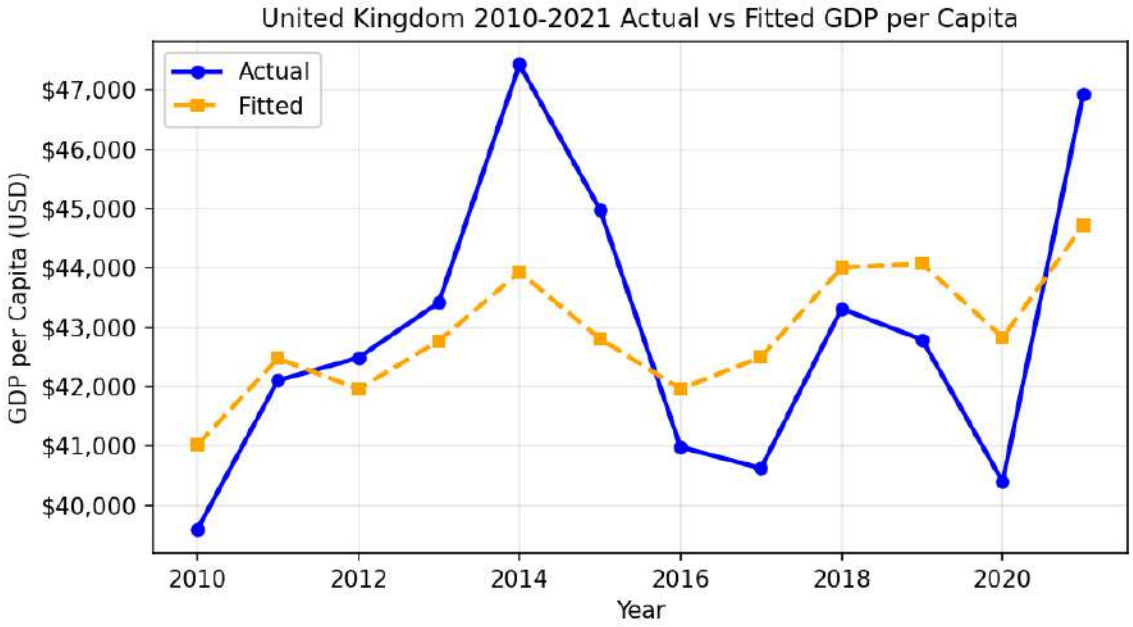


Figure A.24. Finland's GDP Values

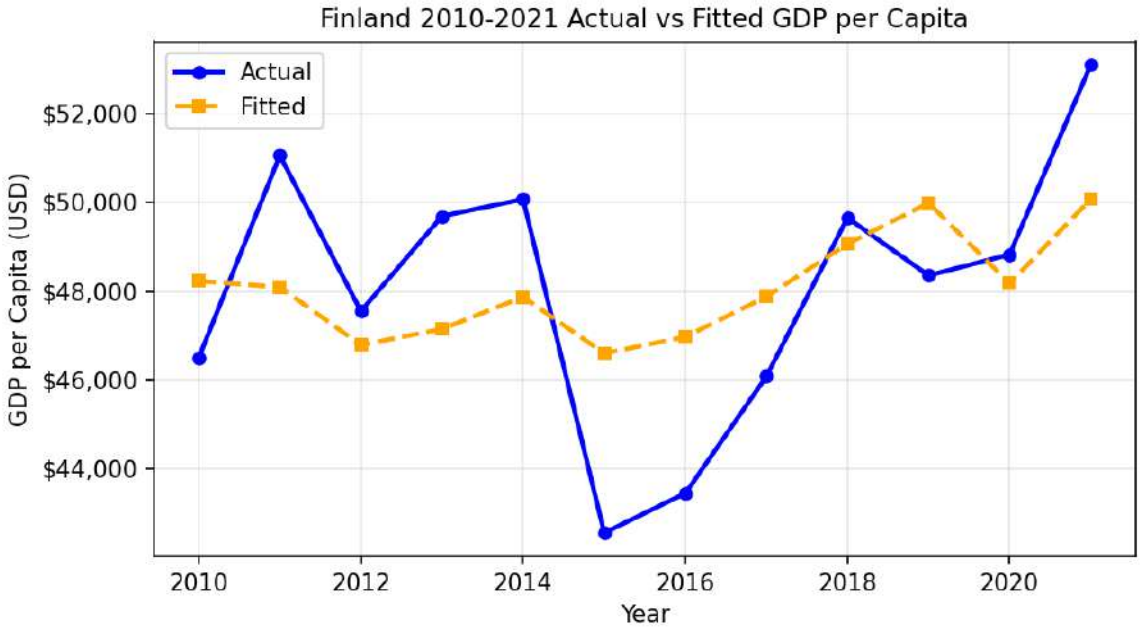


Figure A.25 France's GDP Values

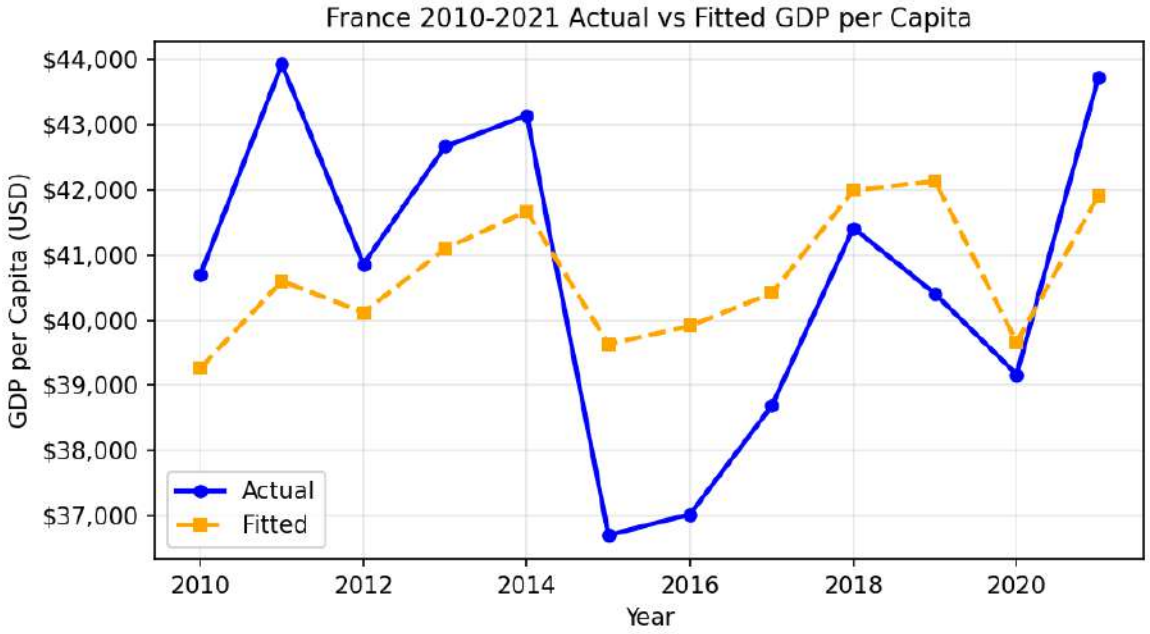


Figure A.26. Canada's GDP Values

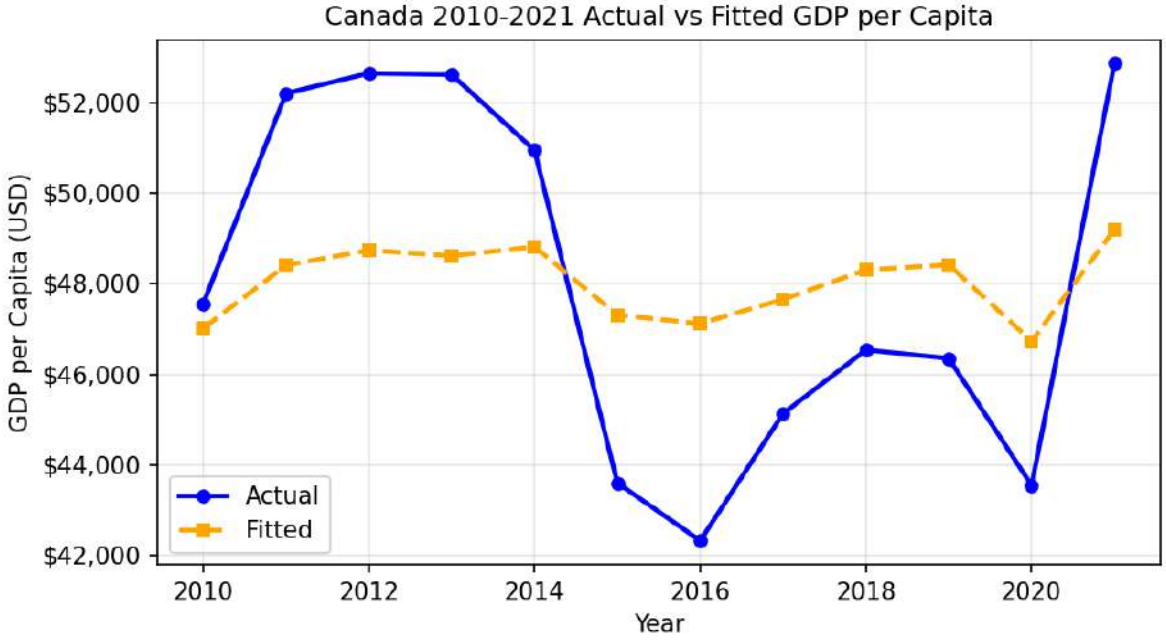


Figure A.27. Israel's GDP Values

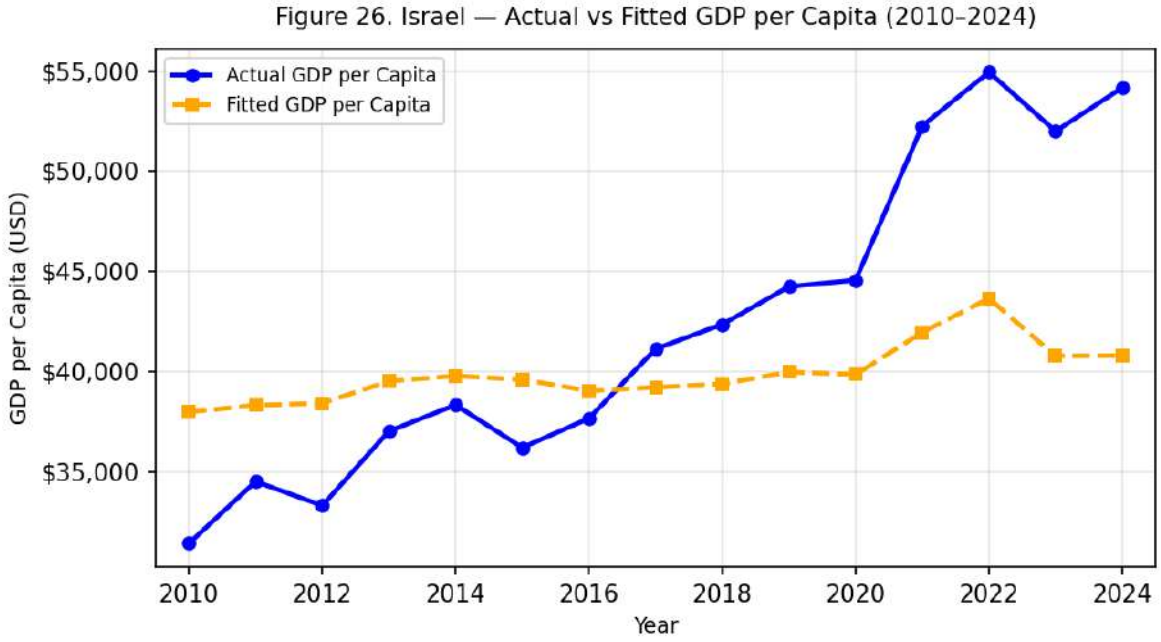


Figure A.28. Latvia's GDP Values

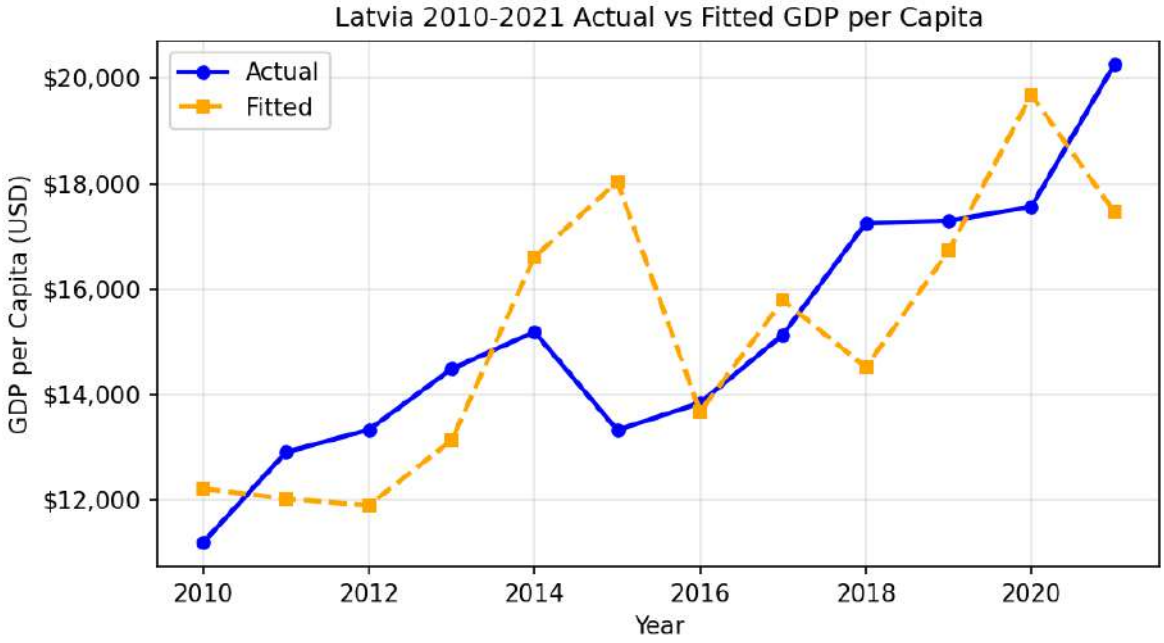


Figure A.29 Slovak Republic 's GDP Values

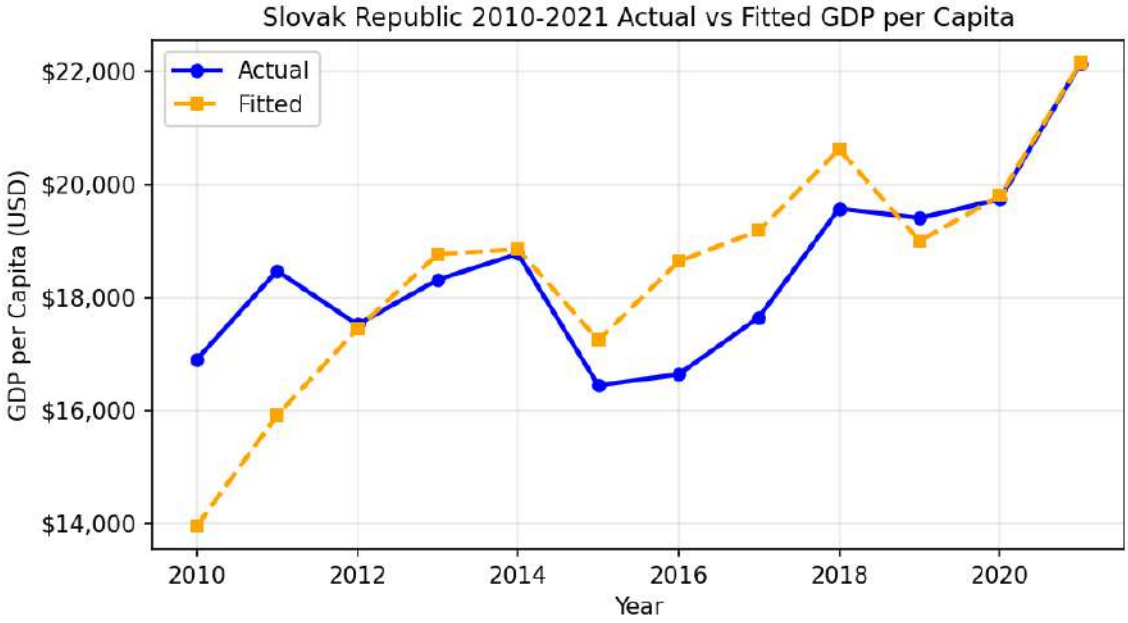


Figure A.30 Greece's GDP Values

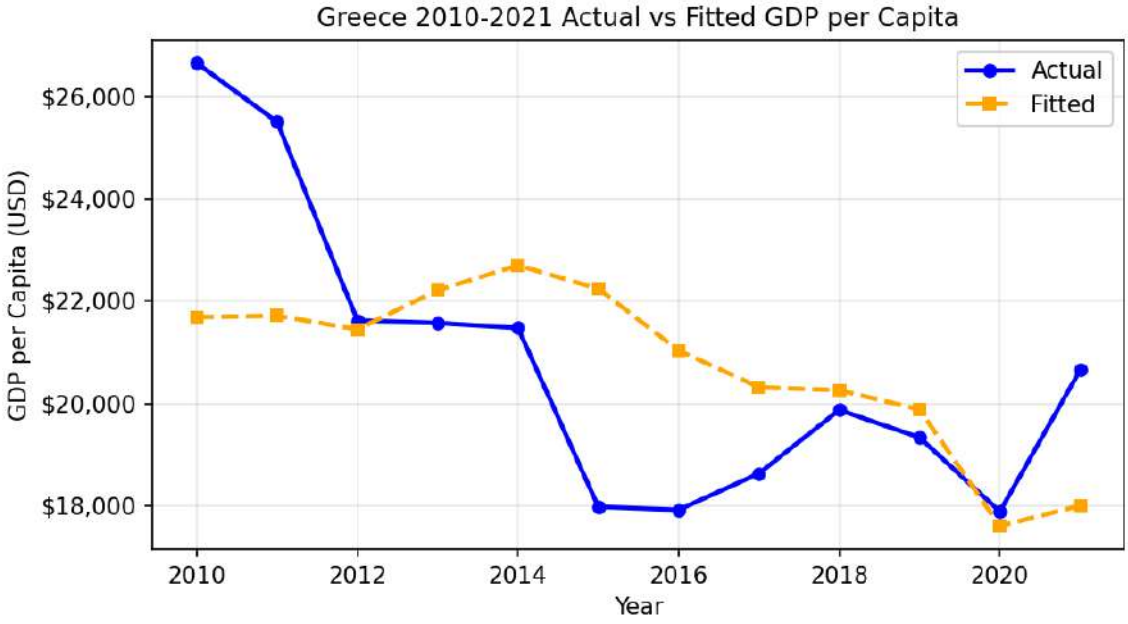
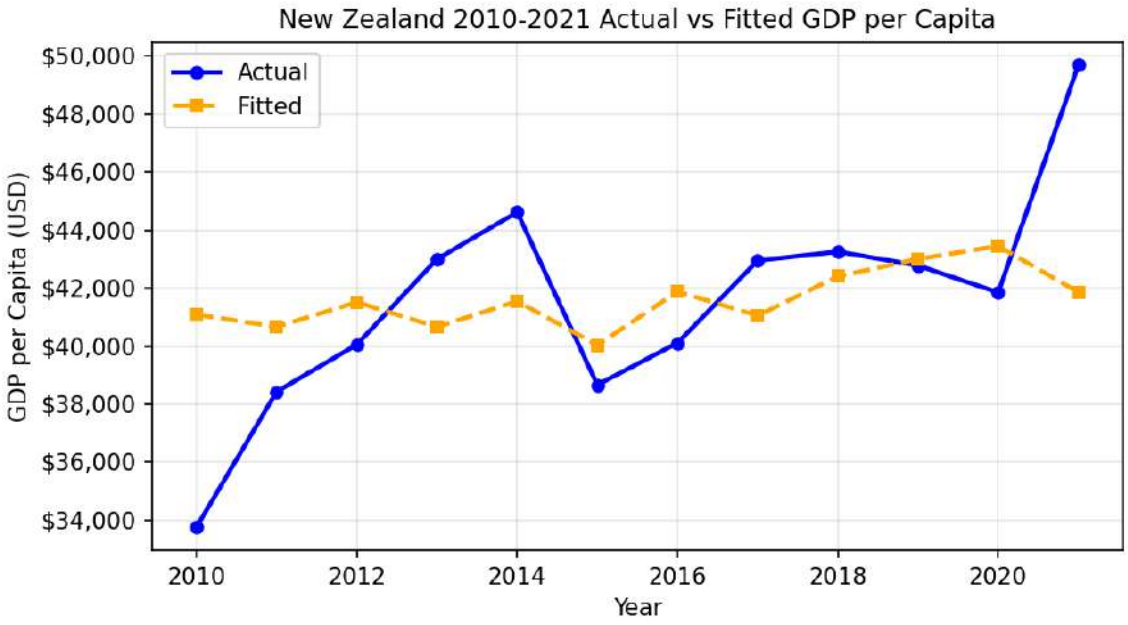


Figure A.31. New Zealand's GDP Values



Low Accuracy Models

Figure A.32. Turkiye's GDP Values

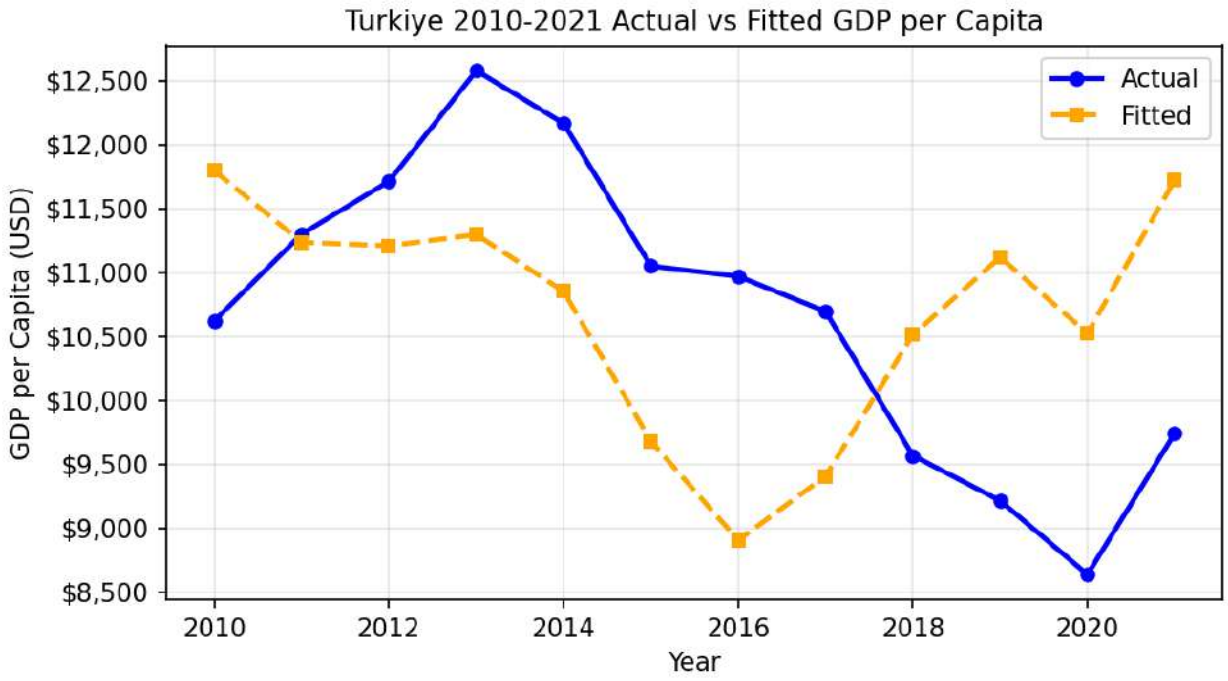


Figure A.33. Portugal's GDP Values

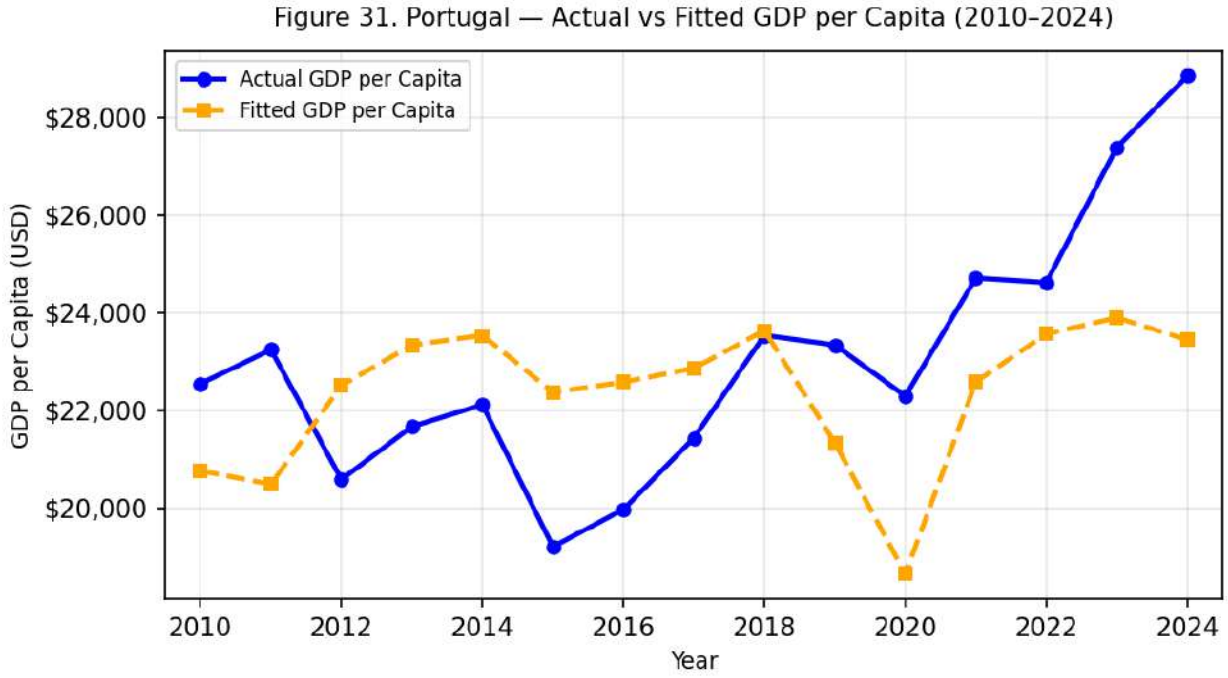


Figure A.34. Spain's GDP Values

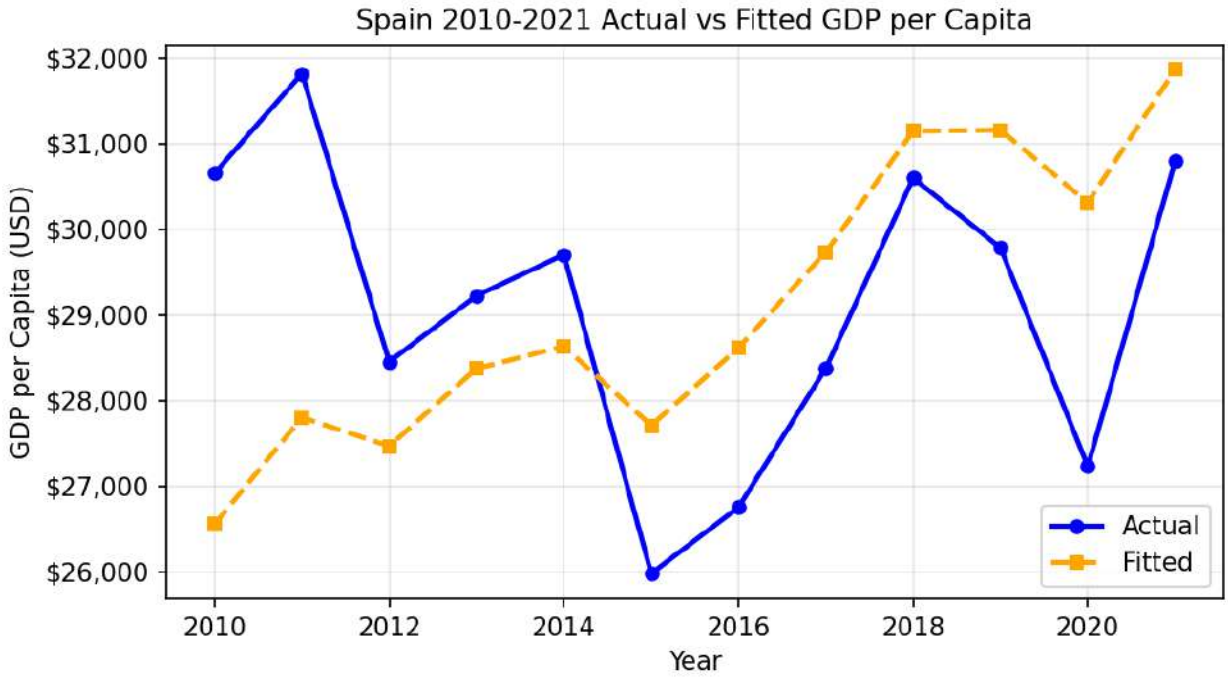


Figure A.35. Mexico's GDP Values

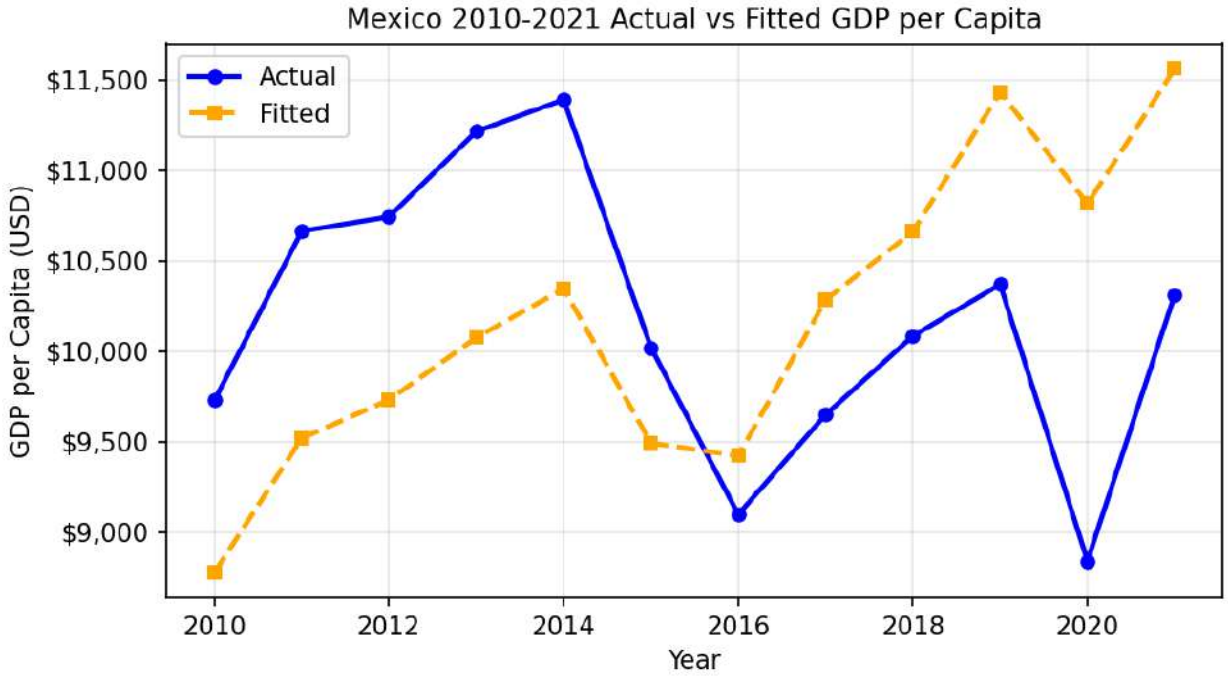


Figure A.36. Chile's GDP Values

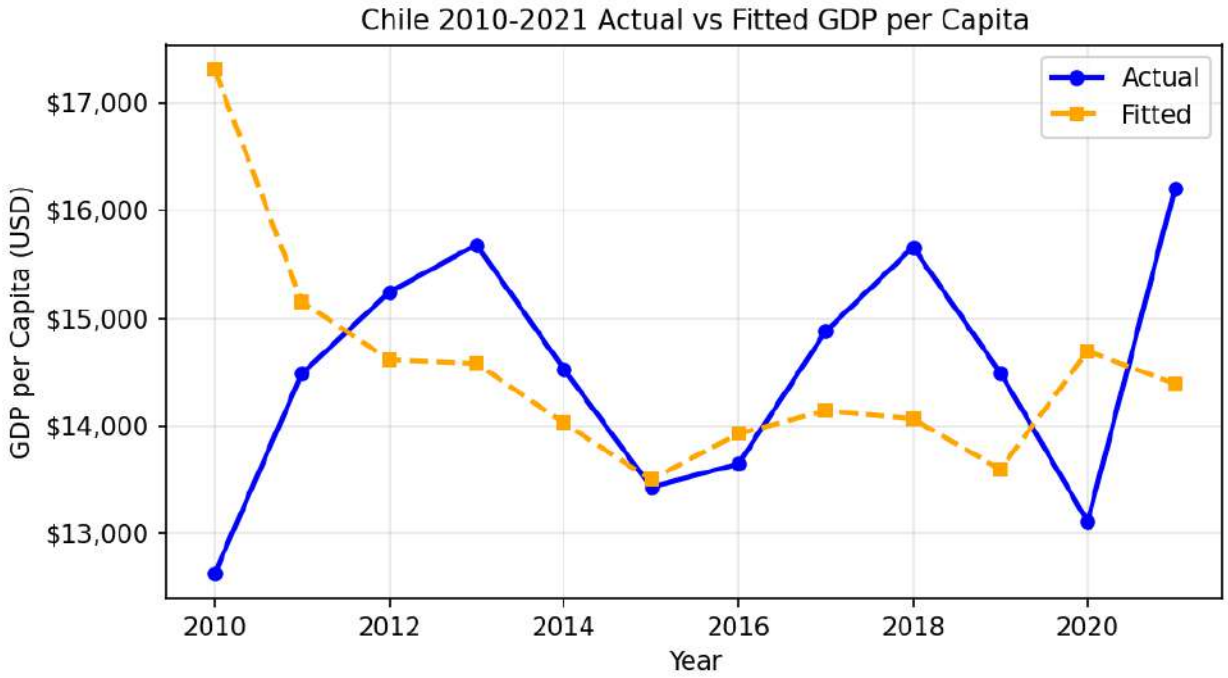


Figure A37. Iceland's GDP Values

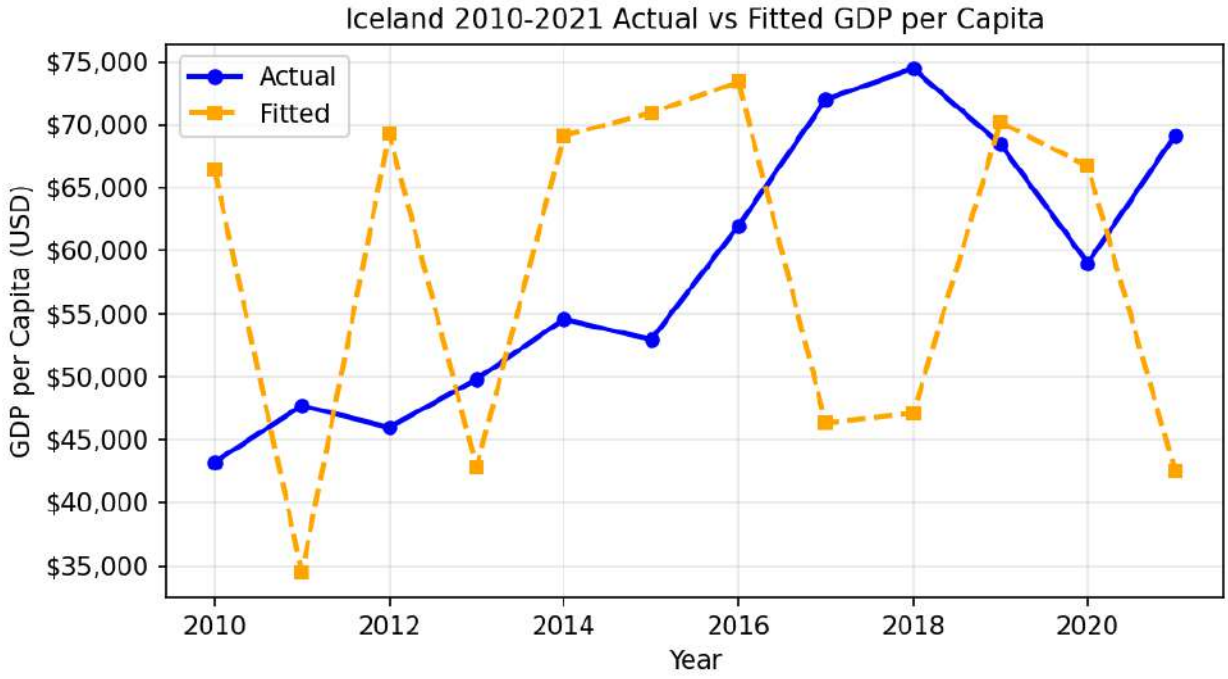


Figure A.38. Italy's GDP Values

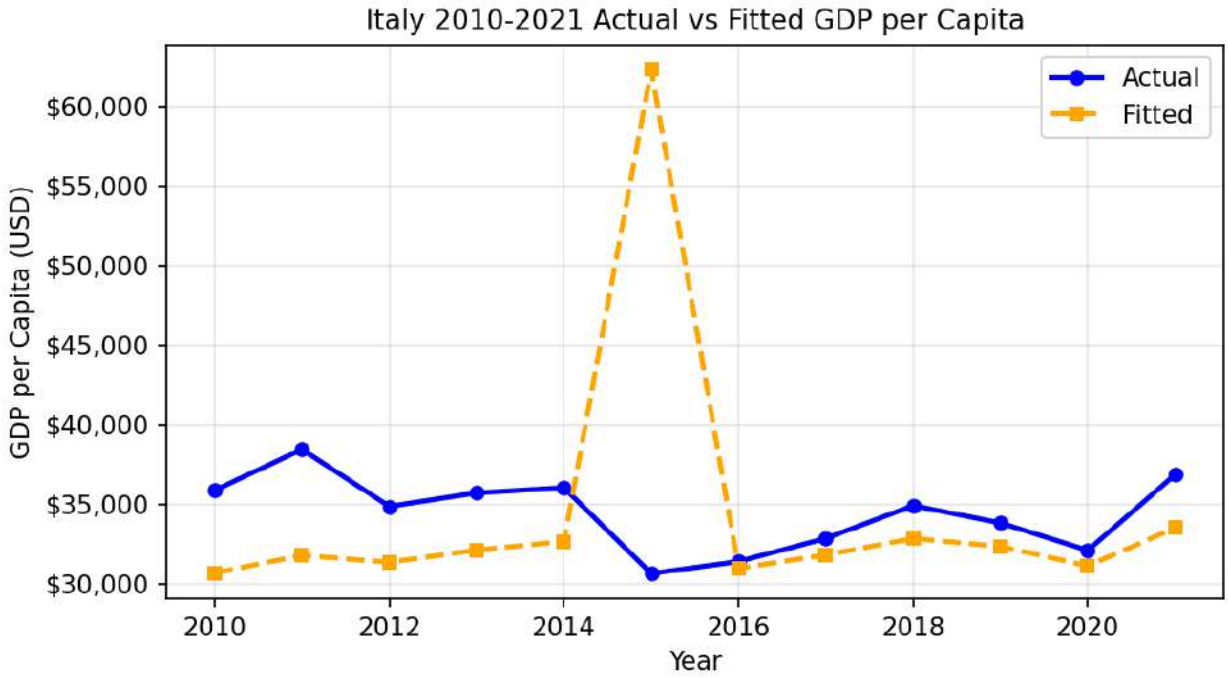


Figure A.39. Slovenia''s GDP Values

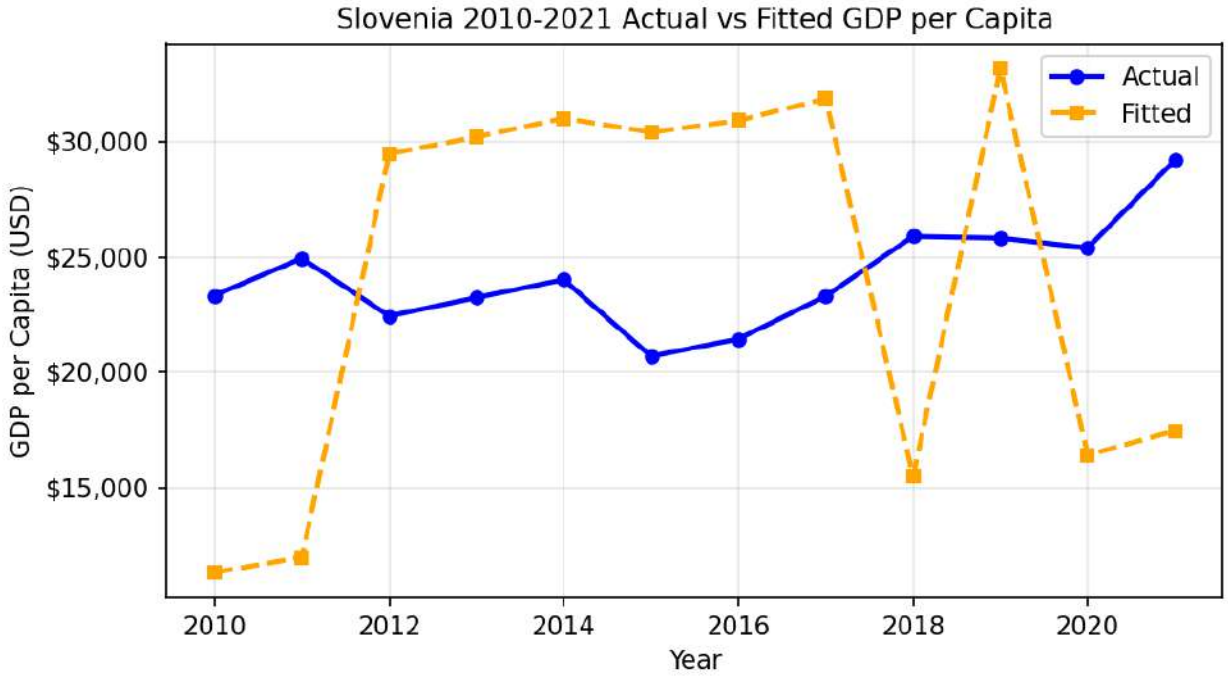


Figure A.40. Australia's GDP Values

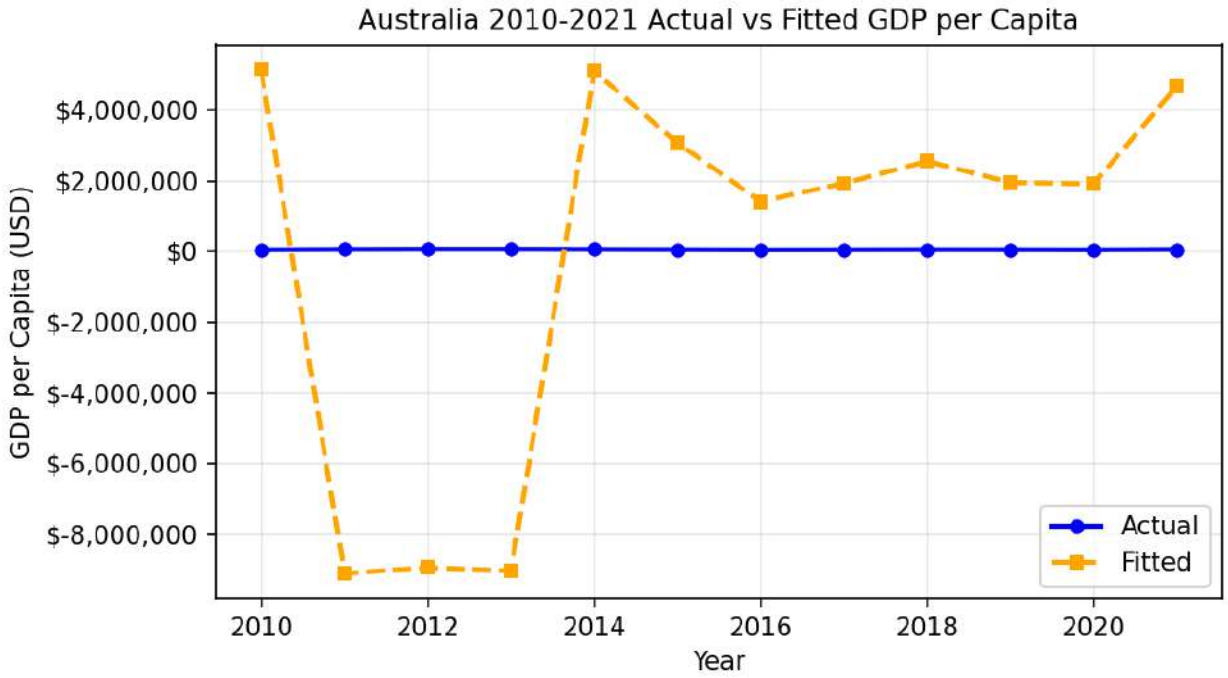


Figure A.41. Austria's GDP Values

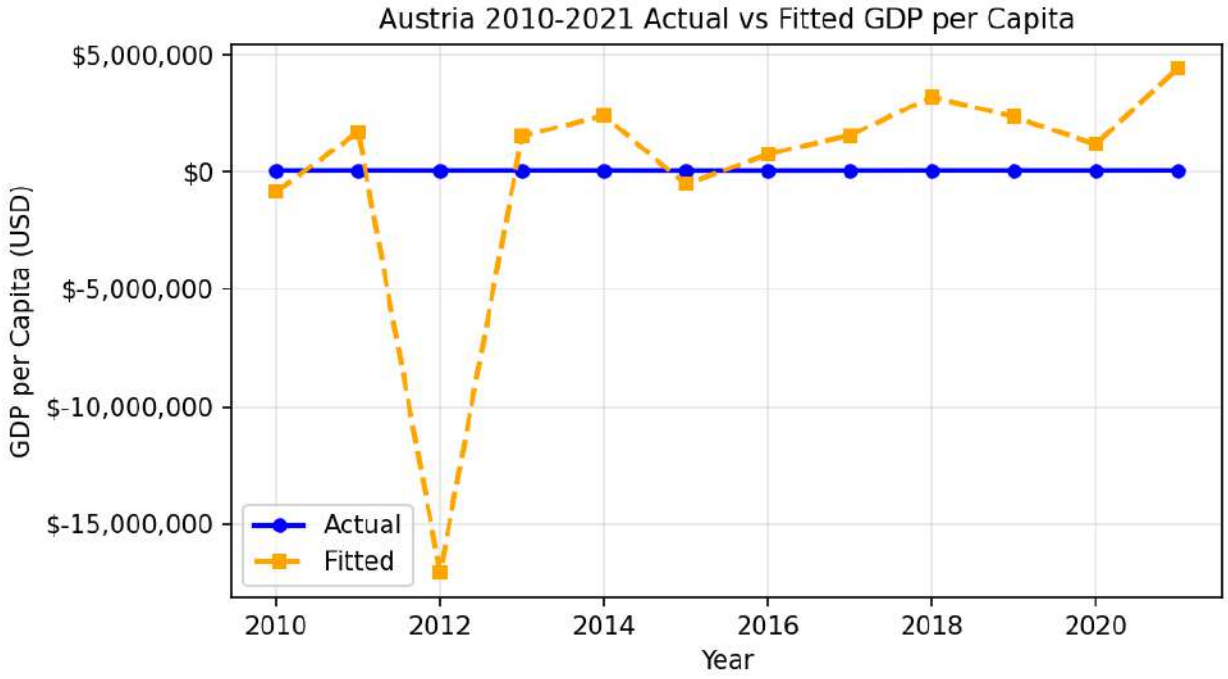
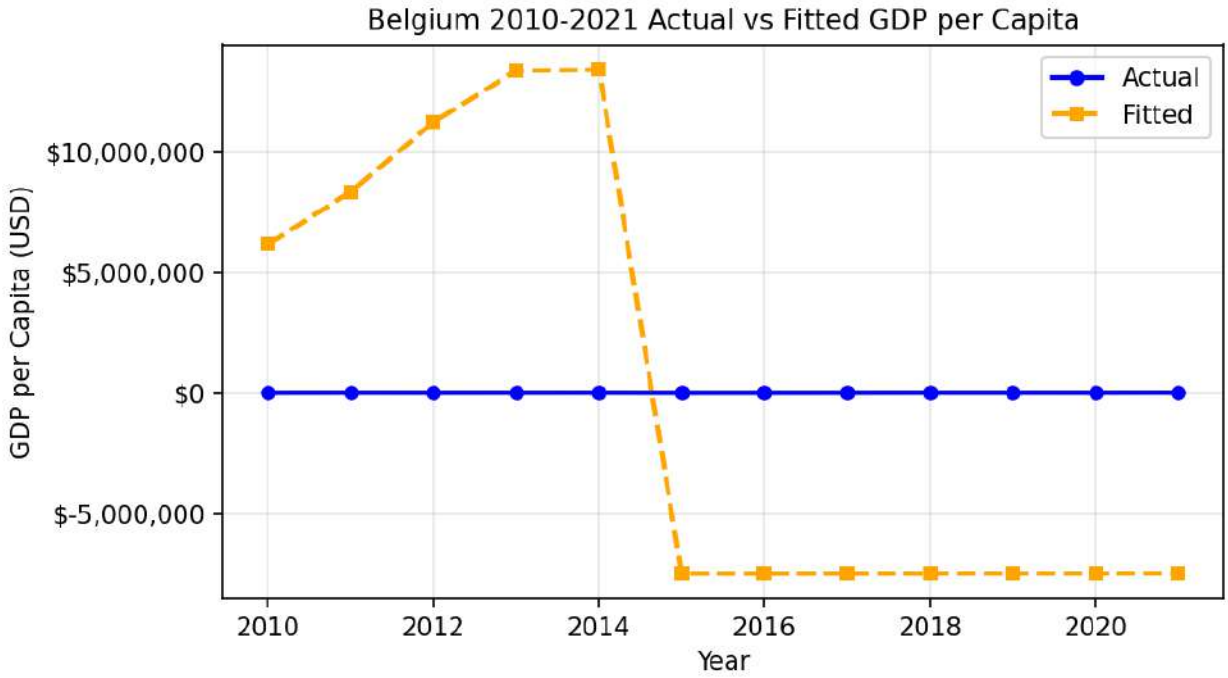
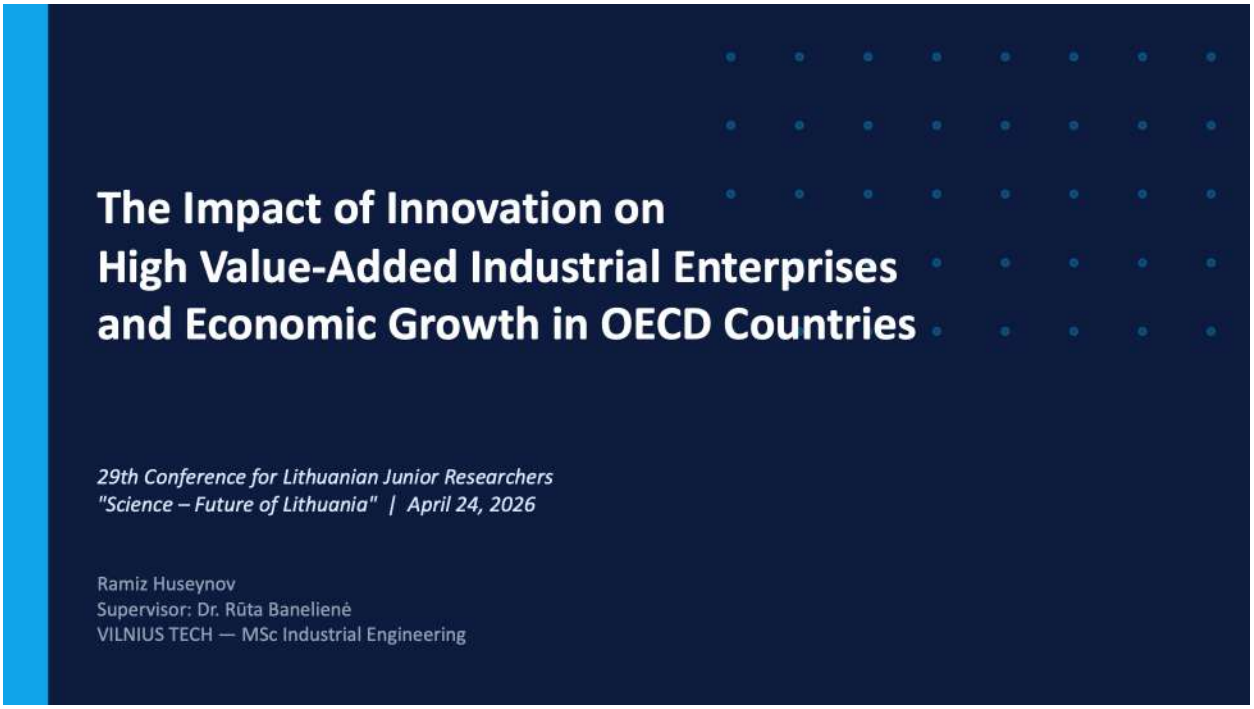


Figure A.42. Belgium's GDP Values



Appendix B

Figure B1. Conference Presentation





Two Ways an Economy Can Grow

Most countries do both,but which path dominates makes all the difference

The Old Way

- More workers + more capital
- Dig more coal, grow more crops
- Growth hits a ceiling — diminishing returns
- Stops when resources run out

The Innovation Way

- Better ideas, better processes
- Same inputs → far more output
- No ceiling — knowledge compounds
- Enables sustained long-run growth

Presentation Outline

01	Research Context & Motivation	06	Key Empirical Results
02	Research Questions & Hypotheses	07	Country-Level Accuracy Analysis
03	Theoretical Framework	08	Discussion & Implications
04	Methodology & Data	09	Conclusion
05	Model Specification	10	Conclusions

Research Context & Motivation

The Problem

Innovation is widely cited as the engine of economic growth, but which channels matter most — and by how much — across 37 diverse OECD economies?

The Gap

Most studies focus on single indicators: R&D spending or patents in isolation. Few simultaneously examine manufacturing sophistication alongside export productivity.

This Study

A fixed-effects panel regression across 37 OECD countries (2010–2024) that quantifies the innovation-growth nexus using a composite manufacturing quality variable.

Research Questions & Hypotheses

RQ1	Does high-technology export intensity have a significant positive effect on GDP per capita across OECD economies?	→ H_1
RQ2	Does export productivity (total exports per worker) have a significant positive effect on GDP per capita?	→ H_2
RQ3	Does import penetration per capita have a statistically significant effect on GDP per capita, and if so, in which direction?	→ H_3

Theoretical Framework

This study is grounded in endogenous growth theory, where innovation is the engine of long-run per capita income growth.

Romer (1990) — Endogenous Growth

Knowledge and R&D are endogenous inputs that generate sustained growth through innovation spillovers. Justifies including patent and R&D variables.

Hausmann & Hidalgo — Product Complexity

Economies that export more complex, knowledge-intensive goods achieve higher income. Justifies HTEX/LABOR and the MVA×MEDHTMVA sophistication variable.

Krugman — New Trade Theory

Specialisation in high-tech sectors confers cumulative advantages via economies of scale and learning effects. Justifies export productivity variable.

Kaldor's Growth Laws

Manufacturing output is the engine of growth for open economies. The share of medium and high-tech manufacturing in MVA captures this channel directly.

Methodology & Data

Analytical Approach

- Quantitative secondary panel data analysis
- Fixed-effects (FE) panel regression — EViews 10
- FE chosen: controls time-invariant country heterogeneity (geography, institutions, legal tradition)
- Variables normalised by population or labour force
- Log-transform on patent variables — captures diminishing returns
- All data: World Bank WDI database (consistent methodology across all 37 countries)

37
OECD Countries

2010–2024
Sample Period

8
Independent Variables

386
Country-Year Obs.

Model Specification

Dependent Variable: GDP per capita (current USD)

$$GDP/POP = \alpha_i + \beta(HTEX/LABOR) + \beta_2(IMP/POP) + \beta_3(MVA{\times}MEDHTMVA/100) + \beta_4 \cdot \log(PATENTRE) + \beta_5(EXP/LABOR) + \beta_6 \cdot \log(PATENTNONRE) + \beta_7(RDEXP{\times}GDP) + \beta_8(RDPOP/LABOR) + \varepsilon$$

Variable	Description	Hypothesis	Prior
HTEX/LABOR	High-tech exports per worker	H_1	Expected +
IMP/POP	Import penetration per capita	H_3	Tested
MVA×MEDHTMVA/100	Manufacturing sophistication (composite)	–	Expected +
ln(PATENTRE)	Resident patent applications (log)	–	Expected +
EXP/LABOR	Export productivity per worker	H_2	Expected +
log(PATENTNONRE)	Non-resident patents (log)	–	Ambiguous
RDEXP×GDP	Total R&D expenditure (absolute)	–	Expected +
RDPOP/LABOR	R&D personnel density	–	Expected +

Key Empirical Results

0.989
R-squared
Very high model fit

0.988
Adj. R^2
After d.f. adjustment

H_1 ✓
HTEX Positive
$p < 0.01$

H_2 ✓
EXP Positive
$p < 0.01$

Notable Findings

- H_1 SUPPORTED: High-technology export intensity (HTEX/LABOR) is positive and highly significant → knowledge-intensive exports drive GDP per capita gains.
- H_2 SUPPORTED: Export productivity (EXP/LABOR) is positive and significant → broader tradeable sector efficiency is a growth driver.
- H_3 MIXED: Import penetration coefficient is significant but negative → competitive pressure and import substitution effects dominate in this sample.
- Manufacturing sophistication (MVA×MEDHTMVA) positive and significant → quality of manufacturing, not just scale, drives growth.
- Patent variables show mixed signs — non-resident patents negative, suggesting foreign IP competition may outpace knowledge spillovers.

Regression Output — Selected Coefficients

Cross-section fixed effects panel regression — EViews 10 output (summary)

Variable	Coefficient	Sign	Significance
HTEX/LABOR	Positive	↑	★★★ ($p < 0.01$)
EXP/LABOR	Positive	↑	★★★ ($p < 0.01$)
MVA×MEDHTMVA/100	Positive	↑	★★ ($p < 0.05$)
IMP/POP	Negative	↓	★★ ($p < 0.05$)
log(PATENTRE)	Positive	↑	★ ($p < 0.10$)
log(PATENTNONRE)	Negative	↓	★ ($p < 0.10$)
RDEXP×GDP	Positive	↑	Not significant
RDPOP/LABOR	Positive	↑	Not significant
R-squared	0.981		Adj. R^2: 0.968
Durbin-Watson	0.881	↑	

Country-Level Model Accuracy

Countries are classified by the alignment between fitted and observed GDP per capita trajectories:

High Accuracy

Model closely tracks observed GDP path.
Typically: advanced manufacturing economies with high-tech specialisation.
Examples: Germany, South Korea, Japan, Czech Republic.

~40% of sample (15 Countries)

Medium Accuracy

Moderate divergence between fitted & actual.
Mixed industrial structures or service-dominated economies.
Examples: France, United Kingdom, Canada.

~40% of sample (15 Countries)

Low Accuracy

Significant divergence — innovation-centric model does not fully capture growth drivers.
Often resource-rich or small open economies with atypical GDP structures.

~20% of sample (7)

This classification reveals the limits of endogenous growth theory across heterogeneous OECD economies — and indicates where additional structural controls are needed.

Discussion & Policy Implications

Export Quality Over R&D Quantity

HTEX and EXP are the strongest predictors — not R&D expenditure. This challenges the conventional focus on input-side metrics and redirects attention to the output side of innovation.

Manufacturing Sophistication Is Decisive

The MVA×MEDHTMVA interaction confirms that technological upgrading within manufacturing matters more than industrial scale alone. Economies investing in medium and high-tech sectors show consistently higher GDP per capita.

Import Competition — A Double-Edged Sword

The negative import coefficient suggests that, at the OECD level, import penetration exerts competitive pressure that constrains rather than boosts growth — at least within this time frame.

Policy Recommendation

Policymakers should prioritise upgrading the technological intensity of manufacturing and expanding high-tech export capacity — not solely R&D budget ratios. Industrial policy targeting medium-to-high-tech sectors is supported by the evidence.

Conclusions

Key Findings

- H_1 Supported: High-technology export intensity (HTEX/LABOR) has a significant positive effect on GDP per capita ($p < 0.01$), confirming that knowledge-intensive exports drive growth.
- H_2 Supported: Export productivity (EXP/LABOR) is positive and highly significant ($p < 0.01$), confirming that broader tradeable sector efficiency is a robust growth driver.
- H_3 Supported (negative): Import penetration (IMP/POP) is significant ($p < 0.05$) but negative, indicating competitive displacement effects dominate over import-driven spillovers in the short run.
- Manufacturing sophistication (MVA×MEDHTMVA) is positive and significant — technological upgrading within industry matters more than manufacturing scale alone.

Policy Implications

- Prioritise high-tech export capacity over headline R&D budget ratios — output-side innovation policy outperforms input-side investment in this sample.
- Industrial policy should target medium-to-high-tech manufacturing sectors — the MVA sophistication interaction confirms quality of industrial structure is decisive.
- Import competition imposes short-run adjustment costs — trade liberalisation in manufacturing-intensive sectors should account for transition effects.
- Model achieves $R^2 = 0.981$ across 37 OECD countries (2010–2021) — high fit confirms the innovation-growth nexus is empirically robust within this sample.

Thank You

Questions & Discussion

Export quality, R&D spending & manufacturing sophistication drive OECD growth alone.
R^2 = 0.981 | 37 OECD Countries | 2010–2021 | Fixed-Effects Panel Regression

H_1	High-tech export intensity → GDP per capita ✓ Supported
H_2	Export productivity → GDP per capita ✓ Supported
H_3	Import penetration → GDP per capita ✓ Significant (negative)

Appendix C

Table C1. Data used for Modeling

Country	Year	GDP (USD bn)	Pop (M)	Labor (M)	Pat. Res.	Pat. Non-Res.	MVA (USD bn)	Med HTMVA%	HTEX (USD M)	Imports (USD bn)	Exports (USD bn)	RDEXP %GDP	RDPop /M
Australia	2010	1,148.84	22.03	14.85	2,409	22,478	917,672.24	27.82	4,556.15	238.10	227.43	2.37	4,568.57
Australia	2011	1,398.61	22.34	15.02	2,383	23,143	103,004.84	26.99	5,601.42	285.51	299.82	2.23	..
Australia	2012	1,547.53	22.73	15.22	2,627	23,731	109,044.28	27.95	5,468.10	335.37	332.61	..	..
Australia	2013	1,577.12	23.13	15.43	3,061	26,656	104,152.84	27.81	5,342.74	335.87	315.06	2.18	..
Australia	2014	1,468.27	23.48	15.60	1,988	23,968	934,242.94	27.26	5,459.16	314.16	309.62	..	..
Australia	2015	1,351.30	23.82	15.77	2,291	26,314	850,273.43	26.12	5,091.39	291.85	271.15	1.92	..
Australia	2016	1,206.84	24.19	15.96	2,620	25,774	733,544.53	26.70	5,379.74	261.83	232.66	..	..
Australia	2017	1,325.58	24.59	16.17	2,503	26,403	767,917.92	26.55	4,902.49	277.67	281.77	1.88	..
Australia	2018	1,427.81	24.96	16.38	2,757	27,200	823,086.22	25.98	5,241.78	311.86	312.61	..	..
Australia	2019	1,392.72	25.33	16.57	2,637	27,121	778,016.73	26.26	6,336.48	306.79	336.48	1.83	..
Australia	2020	1,328.41	25.65	16.70	2,368	26,926	745,043.91	27.34	5,590.75	274.15	319.17	..	..
Australia	2021	1,556.74	25.69	16.65	2,966	29,443	858,180.91	28.85	6,310.40	284.04	343.36	1.86	..
Austria	2010	389.83	8.36	5.65	2,424	249	663,277.31	43.78	15,801.99	187.35	201.09	2.73	4,379.18
Austria	2011	428.95	8.39	5.67	2,154	276	735,427.77	45.00	17,846.80	220.84	232.89	2.67	4,430.54
Austria	2012	406.75	8.43	5.70	2,258	294	70,190.83	44.64	18,475.53	209.54	220.97	2.91	4,720.81
Austria	2013	426.58	8.48	5.72	2,162	244	729,725.24	44.98	20,793.05	217.96	229.88	2.95	4,782.12
Austria	2014	438.56	8.55	5.75	2,092	271	753,736.54	45.40	21,648.23	222.09	236.19	3.08	5,009.44
Austria	2015	379.55	8.64	5.81	2,205	236	654,448.63	45.12	18,084.61	188.72	202.66	3.05	5,073.43
Austria	2016	393.69	8.74	5.87	2,078	237	695,575.69	45.43	17,333.98	192.55	207.33	3.12	5,400.35
Austria	2017	414.93	8.80	5.89	2,073	232	723,743.43	45.43	17,025.65	212.57	225.41	3.06	5,416.08
Austria	2018	452.58	8.84	5.91	2,039	168	783,923.05	45.26	16,680.04	238.40	252.10	3.09	5,682.44
Austria	2019	442.98	8.88	5.92	2,066	208	757,603.29	45.07	15,967.50	231.73	247.61	3.13	5,958.86
Austria	2020	434.40	8.92	5.92	2,124	173	720,983.88	44.75	15,995.94	209.75	224.42	3.20	5,829.29
Austria	2021	480.47	8.96	5.93	1,872	175	815,741.57	45.86	17,019.36	264.64	268.84	3.26	6,323.00
Belgium	2010	481.56	10.90	7.18	620	140	632,218.37	34.39	18,062.11	357.96	365.16	2.06	3,755.64
Belgium	2011	527.20	11.04	7.26	636	127	682,716.47	36.93	19,710.71	424.36	423.19	2.17	3,880.20
Belgium	2012	498.41	11.11	7.28	755	127	629,077.50	47.38	18,578.04	400.40	398.98	2.28	4,116.62
Belgium	2013	524.10	11.16	7.29	715	161	663,736.29	50.04	23,020.45	412.27	413.57	2.33	4,161.62
Belgium	2014	537.99	11.21	7.29	889	137	676,992.04	49.16	26,767.84	425.95	427.14	2.37	4,544.87
Belgium	2015	461.04	11.27	7.31	949	148	57.94	48.98	22,552.91	355.88	359.05	2.43	4,731.82
Belgium	2016	474.27	11.33	7.33	1,054	119	58.09	49.27	27,175.81	376.76	377.61	2.52	4,798.33
Belgium	2017	500.91	11.38	7.33	1,001	216	62.34	49.23	25,235.41	418.06	417.98	2.67	4,757.31
Belgium	2018	542.64	11.43	7.34	892	218	66.98	50.59	28,165.60	460.13	453.01	2.86	5,040.00
Belgium	2019	536.73	11.49	7.36	876	257	66.63	51.29	33,085.09	442.79	445.31	3.16	5,290.88
Belgium	2020	529.69	11.54	7.38	862	288	63.63	52.81	34,318.94	411.85	419.50	3.39	5,418.91
Belgium	2021	598.52	11.59	7.40	799	415	64.31	50.61	60,841.70	525.24	536.43	3.39	6,596.71
Canada	2010	1,617.34	34.01	23.58	4,550	30,899	162.17	35.73	26,567.98	502.04	471.74	1.83	4,664.08
Canada	2011	1,793.33	34.34	23.75	4,754	30,357	178.53	36.65	27,501.35	570.64	550.16	1.79	4,802.72
Canada	2012	1,828.37	34.71	23.90	4,709	30,533	183.21	37.24	33,113.40	589.51	554.96	1.77	4,652.43
Canada	2013	1,846.60	35.08	24.03	4,567	30,174	177.06	36.05	32,182.26	589.05	560.11	1.71	4,647.61
Canada	2014	1,805.75	35.43	24.14	4,198	31,283	170.21	37.04	28,787.81	589.43	573.08	1.71	4,564.22
Canada	2015	1,556.51	35.70	24.19	4,277	32,687	155.27	38.44	28,242.77	534.12	495.75	1.69	4,545.97
Canada	2016	1,527.99	36.11	24.34	4,078	30,667	151.70	37.15	26,203.27	517.40	481.36	1.73	4,400.49
Canada	2017	1,649.27	36.55	24.51	4,053	30,969	159.72	36.35	26,938.35	554.92	518.77	1.69	4,429.06
Canada	2018	1,725.33	37.07	24.76	4,349	31,812	170.22	36.36	30,480.48	591.35	557.72	1.74	4,715.92
Canada	2019	1,743.73	37.62	25.01	4,238	32,250	162.16	36.95	31,537.32	589.71	564.14	1.76	4,850.94
Canada	2020	1,655.68	38.03	25.15	4,452	30,113	149.27	34.14	25,573.87	524.57	487.99	1.93	5,155.65
Canada	2021	2,022.38	38.24	25.15	4,710	32,445	187.16	31.20	29,240.97	628.39	630.48	1.86	5,411.83
Chile	2010	217.05	17.18	11.76	328	748	21.46	20.32	541.80	69.05	82.27	0.33	318.21
Chile	2011	251.38	17.35	11.91	339	2,453	24.91	15.00	554.62	87.46	94.75	0.35	352.02
Chile	2012	267.02	17.52	12.05	336	2,683	26.24	11.44	578.37	91.62	90.38	0.36	389.88
Chile	2013	277.40	17.69	12.18	340	2,732	27.70	15.96	696.43	91.70	89.01	0.39	334.79
Chile	2014	259.56	17.86	12.30	452	2,653	25.75	17.10	731.68	84.70	85.67	0.38	426.80
Chile	2015	242.45	18.05	12.42	443	2,831	25.21	14.06	580.23	73.12	70.77	0.38	455.29
Chile	2016	249.34	18.27	12.57	386	2,521	24.20	7.96	720.87	70.15	69.62	0.37	495.36
Chile	2017	276.15	18.56	12.78	425	2,469	25.30	20.12	634.87	76.48	78.24	0.36	494.63
Chile	2018	295.86	18.89	13.02	406	2,694	28.41	19.63	667.79	87.72	84.41	0.37	523.65
Chile	2019	278.29	19.20	13.24	438	2,799	24.96	19.63	672.15	82.75	77.57	0.34	507.31
Chile	2020	254.04	19.37	13.36	372	2,433	22.85	20.37	1,368.06	68.22	79.59	0.34	515.32

Country	Year	GDP (USD bn)	Pop (M)	Labor (M)	Pat. Res.	Pat. Non-Res.	MVA (USD bn)	Med HTMVA%	HTEX (USD M)	Imports (USD bn)	Exports (USD bn)	RDEXP %GDP	RDPop /M
Chile	2021	315.33	19.46	13.41	402	2,680	26.90	20.37	1,264.19	103.69	100.88	0.36	638.84
Colombia	2010	286.50	44.78	29.87	133	1,739	40.02	24.95	451.12	51.35	46.82	0.19	..
Colombia	2011	334.97	45.26	30.36	183	1,770	44.27	20.95	433.81	67.64	64.57	0.20	..
Colombia	2012	370.69	45.72	30.83	213	1,848	48.52	21.35	565.87	74.23	69.73	0.22	..
Colombia	2013	382.09	46.15	31.29	251	1,781	48.38	20.76	773.38	76.04	69.11	0.26	58.05
Colombia	2014	381.24	46.57	31.72	260	1,898	46.78	20.72	762.82	79.50	63.42	0.30	59.05
Colombia	2015	293.49	46.97	32.13	321	1,921	36.40	22.54	823.79	66.65	45.93	0.37	70.67
Colombia	2016	282.72	47.44	32.59	545	1,658	34.77	23.33	766.93	60.74	41.61	0.27	91.26
Colombia	2017	311.87	48.13	33.23	595	1,777	35.58	23.27	692.53	62.80	47.23	0.26	90.25
Colombia	2018	334.20	49.02	34.02	415	1,808	37.27	22.61	603.55	68.96	53.14	0.31	..
Colombia	2019	323.03	49.91	34.78	422	1,735	35.30	23.93	753.15	70.07	51.26	0.32	..
Colombia	2020	270.35	50.63	35.38	369	1,752	29.00	24.74	689.65	55.51	36.56	0.29	..
Colombia	2021	318.52	51.19	35.83	432	1,855	35.67	24.09	735.07	76.01	51.59	..	..
Costa Rica	2010	37.66	4.56	3.09	8	606	5.49	19.28	2,203.40	13.06	12.36	0.48	385.92
Costa Rica	2011	42.76	4.61	3.14	13	610	6.11	21.08	2,521.52	15.47	13.89	0.46	410.69
Costa Rica	2012	47.23	4.66	3.18	9	578	6.55	21.87	2,732.05	16.72	14.95	0.55	340.94
Costa Rica	2013	50.95	4.72	3.23	18	580	6.54	22.17	3,074.13	17.06	15.59	0.54	359.00
Costa Rica	2014	52.02	4.77	3.27	15	529	6.39	19.92	2,418.53	17.60	16.31	0.56	574.92
Costa Rica	2015	56.44	4.82	3.31	17	582	6.55	16.59	869.82	17.36	16.93	0.44	500.97
Costa Rica	2016	58.85	4.87	3.35	9	497	6.83	15.22	951.90	18.19	18.40	0.44	531.45
Costa Rica	2017	60.52	4.91	3.38	19	504	7.10	14.62	1,106.66	19.55	19.83	0.43	385.02
Costa Rica	2018	62.42	4.96	3.41	8	490	7.44	14.42	1,148.83	20.73	21.06	0.37	349.49
Costa Rica	2019	64.42	5.00	3.44	16	483	7.76	14.36	1,154.57	20.25	22.11	..	..
Costa Rica	2020	62.40	5.03	3.46	12	524	7.98	14.16	1,038.67	17.62	19.91	0.33	371.97
Costa Rica	2021	64.96	5.06	3.48	15	566	8.92	14.16	1,457.23	22.48	23.50	0.27	406.58
Czechia	2010	211.17	10.47	7.35	868	114	43.71	44.99	20,408.15	130.64	137.03	1.33	2,799.13
Czechia	2011	231.43	10.50	7.30	783	97	50.60	45.72	26,864.95	153.90	162.58	1.54	2,925.88
Czechia	2012	210.36	10.51	7.23	867	150	45.68	46.72	25,452.86	148.04	157.99	1.77	3,162.11
Czechia	2013	213.02	10.51	7.15	984	97	46.36	47.34	24,683.56	148.95	161.00	1.88	3,260.83
Czechia	2014	210.91	10.53	7.08	910	62	49.50	48.27	26,896.99	158.32	171.58	1.96	3,428.31
Czechia	2015	189.11	10.55	7.03	880	72	44.88	51.83	24,765.84	139.95	151.31	1.92	3,619.16
Czechia	2016	198.16	10.57	6.97	792	47	47.09	51.22	24,696.48	139.58	154.78	1.67	3,546.93
Czechia	2017	221.56	10.59	6.92	794	66	51.36	52.24	29,455.30	155.24	172.12	1.77	3,718.35
Czechia	2018	251.99	10.63	6.88	678	54	56.21	51.53	36,029.73	175.22	190.53	1.90	3,906.68
Czechia	2019	256.79	10.67	6.84	765	48	56.82	51.77	37,662.94	169.56	185.21	1.93	4,027.59
Czechia	2020	251.11	10.70	6.81	673	56	52.22	50.63	39,608.92	153.23	170.15	1.99	4,187.18
Czechia	2021	290.97	10.51	6.66	541	59	57.15	49.25	41,423.40	194.45	205.28	2.00	4,560.45
Denmark	2010	322.35	5.55	3.63	1,626	142	35.06	48.82	9,012.04	141.18	163.08	2.92	6,763.58
Denmark	2011	344.32	5.57	3.63	1,574	197	37.77	50.92	10,408.23	165.03	186.22	2.94	7,045.77
Denmark	2012	326.79	5.59	3.63	1,406	229	36.63	51.98	9,742.01	161.05	180.00	2.98	7,181.85
Denmark	2013	344.63	5.61	3.63	1,341	193	40.67	52.73	10,196.24	167.81	189.27	2.97	7,115.42
Denmark	2014	352.83	5.64	3.64	1,377	206	41.74	53.31	10,640.79	170.69	193.47	2.91	7,358.19
Denmark	2015	301.76	5.68	3.66	1,462	270	37.67	52.65	10,082.75	148.82	168.42	3.05	7,566.46
Denmark	2016	312.18	5.73	3.68	1,552	298	40.40	53.83	10,051.34	148.94	167.78	3.09	7,852.06
Denmark	2017	331.61	5.76	3.70	1,490	282	43.16	54.39	8,888.11	162.06	183.53	2.93	7,647.62
Denmark	2018	355.29	5.79	3.71	1,262	239	46.20	57.14	9,553.98	184.15	202.46	2.97	7,597.53
Denmark	2019	345.40	5.81	3.71	1,351	228	46.29	55.23	9,580.04	182.86	203.62	2.90	7,693.55
Denmark	2020	355.63	5.83	3.71	1,261	217	45.65	55.69	10,456.16	175.67	195.73	2.97	7,651.28
Denmark	2021	408.38	5.86	3.72	1,090	186	52.03	53.67	12,558.48	214.76	240.23	2.76	7,708.24
Estonia	2010	19.52	1.33	0.90	84	13	2.68	26.75	1,253.05	13.40	14.65	1.58	3,057.78
Estonia	2011	23.30	1.33	0.89	62	15	3.39	27.76	2,547.55	18.75	20.10	2.31	3,392.81
Estonia	2012	23.24	1.32	0.88	20	5	3.22	26.94	2,356.41	19.46	19.84	2.12	3,457.76
Estonia	2013	25.45	1.32	0.87	25	17	3.45	28.14	2,546.33	20.65	21.24	1.72	3,338.40
Estonia	2014	27.06	1.31	0.86	44	6	3.78	28.39	2,709.03	20.93	21.80	1.43	3,285.59
Estonia	2015	23.31	1.32	0.86	30	6	3.18	25.72	2,053.39	16.89	17.73	1.47	3,188.49
Estonia	2016	24.56	1.32	0.85	29	1	3.34	26.64	2,107.83	17.75	18.54	1.24	3,296.37
Estonia	2017	27.47	1.32	0.85	37	4	3.66	27.43	1,846.64	19.32	20.41	1.28	3,552.47
Estonia	2018	31.22	1.32	0.85	24	6	4.13	25.78	1,911.07	22.04	22.76	1.41	3,766.08
Estonia	2019	31.87	1.33	0.84	31	1	4.11	29.43	1,737.78	21.78	22.96	1.62	3,770.42
Estonia	2020	31.82	1.33	0.84	21	2	4.01	28.69	2,169.49	22.03	21.69	1.75	3,836.38
Estonia	2021	37.20	1.33	0.84	25	1	4.85	25.97	2,681.06	30.08	29.59	1.77	4,036.13
Finland	2010	249.43	5.36	3.55	1,731	102	42.50	44.59	7,052.41	92.41	95.81	3.71	7,740.92
Finland	2011	275.13	5.39	3.54	1,650	124	45.00	36.98	6,421.63	109.36	107.21	3.62	7,441.98
Finland	2012	257.44	5.41	3.52	1,698	129	37.57	33.85	5,472.64	104.25	100.24	3.40	7,492.46
Finland	2013	270.27	5.44	3.51	1,596	141	39.58	36.50	4,695.48	106.04	103.38	3.27	7,222.81

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Finland	2014	273.48	5.46	3.49	1,419	126	39.93	42.66	5,102.77	103.43	100.66	3.15	7,022.19
Finland	2015	233.21	5.48	3.48	1,289	127	34.61	42.16	4,250.91	84.38	83.45	2.87	6,855.97
Finland	2016	238.78	5.50	3.46	1,260	108	35.21	44.11	3,966.87	86.89	84.37	2.72	6,543.70
Finland	2017	253.85	5.51	3.45	1,390	139	39.03	44.53	4,405.32	95.88	96.88	2.73	6,731.45
Finland	2018	273.87	5.52	3.44	1,387	100	40.51	43.80	4,515.45	109.53	107.22	2.76	6,872.69
Finland	2019	267.01	5.52	3.43	1,321	75	39.28	39.95	4,563.29	106.66	108.46	2.80	7,245.86
Finland	2020	270.00	5.53	3.42	1,588	97	39.21	42.27	4,419.55	97.12	97.78	2.91	7,548.33
Finland	2021	294.22	5.54	3.42	1,557	105	43.96	43.21	5,489.73	117.30	117.46	2.99	7,870.43
France	2010	2,646.23	65.03	42.06	14,748	1,832	271.97	45.45	105,353.02	761.08	736.35	2.18	3,849.59
France	2011	2,870.41	65.34	42.08	14,655	2,099	296.03	45.41	112,010.15	891.74	847.89	2.19	3,920.66
France	2012	2,683.01	65.66	42.01	14,540	2,092	274.93	46.14	114,180.53	839.87	814.45	2.23	4,052.17
France	2013	2,816.08	66.00	41.99	14,690	2,196	288.74	46.58	118,820.85	879.40	861.47	2.24	4,133.56
France	2014	2,861.24	66.31	41.95	14,500	2,033	291.32	47.04	120,716.76	900.07	879.60	2.28	4,210.04
France	2015	2,442.48	66.55	41.86	14,306	1,994	252.06	47.98	109,712.84	776.98	774.11	2.23	4,310.71
France	2016	2,470.41	66.72	41.79	14,206	2,012	251.12	49.89	108,836.53	786.69	778.13	2.22	4,392.05
France	2017	2,588.87	66.92	41.76	14,415	1,832	259.15	48.38	108,867.69	856.50	836.51	2.20	4,546.76
France	2018	2,781.58	67.16	41.76	14,303	1,919	274.99	50.20	117,655.41	942.52	915.64	2.20	4,670.53
France	2019	2,722.79	67.38	41.76	14,103	1,766	269.85	50.94	120,602.17	913.21	895.51	2.19	4,773.97
France	2020	2,647.93	67.60	41.79	12,771	1,542	244.82	50.41	87,513.99	801.84	756.17	2.27	4,883.22
France	2021	2,966.43	67.84	41.86	13,386	1,373	269.82	48.91	97,729.57	964.10	928.14	2.22	5,058.05
Germany	2010	3,468.15	81.78	53.76	47,047	12,198	673.96	59.08	179,620.90	1,192.33	1,369.34	2.73	4,056.42
Germany	2011	3,824.83	80.27	52.90	46,986	12,458	770.10	59.18	202,328.06	1,406.67	1,592.28	2.81	4,191.52
Germany	2012	3,597.90	80.43	53.06	46,620	14,720	721.63	59.18	202,541.99	1,317.97	1,535.18	2.88	4,355.57
Germany	2013	3,808.09	80.65	53.19	47,353	15,814	752.77	59.29	209,304.93	1,366.94	1,586.78	2.84	4,380.39
Germany	2014	3,965.80	80.98	53.34	48,154	17,811	797.80	60.59	215,660.93	1,393.84	1,653.10	2.88	4,336.02
Germany	2015	3,423.57	81.69	53.70	47,384	19,509	689.88	60.70	199,465.12	1,204.65	1,458.59	2.93	4,755.30
Germany	2016	3,537.78	82.35	53.97	48,480	19,419	722.93	60.91	205,114.89	1,215.61	1,472.56	2.94	4,839.85
Germany	2017	3,763.09	82.66	53.93	47,785	19,927	760.20	60.87	195,727.09	1,326.14	1,592.77	3.05	5,058.40
Germany	2018	4,052.01	82.91	53.88	46,617	21,281	804.96	59.86	209,762.24	1,483.86	1,726.46	3.11	5,209.21
Germany	2019	3,957.21	83.09	53.77	46,632	20,802	766.47	60.38	208,177.44	1,454.17	1,677.31	3.17	5,398.63
Germany	2020	3,940.14	83.16	53.55	42,260	19,845	734.34	60.04	182,393.71	1,332.72	1,543.32	3.13	5,390.05
Germany	2021	4,348.30	83.20	53.25	39,822	18,747	812.99	57.33	211,942.36	1,630.57	1,854.88	3.13	5,520.58
Greece	2010	296.42	11.12	7.36	728	16	23.78	14.71	1,293.89	86.54	64.41	0.60	..
Greece	2011	283.23	11.10	7.31	722	19	23.08	18.31	1,402.49	87.76	71.68	0.68	2,217.33
Greece	2012	238.84	11.05	7.23	628	28	19.43	18.76	1,138.03	79.84	69.18	0.71	2,236.29
Greece	2013	236.56	10.97	7.13	698	19	18.41	22.18	951.30	77.18	71.74	0.81	2,655.47
Greece	2014	233.91	10.89	7.05	651	19	18.23	20.73	1,338.28	79.34	76.06	0.84	2,733.92
Greece	2015	194.57	10.82	6.97	550	23	16.14	19.05	1,330.16	64.37	62.62	0.97	3,196.52
Greece	2016	193.10	10.78	6.91	606	40	15.66	18.31	1,416.99	62.71	60.21	1.01	2,726.60
Greece	2017	200.38	10.75	6.87	498	91	16.07	18.26	1,384.67	72.34	69.69	1.15	3,250.62
Greece	2018	213.30	10.73	6.84	430	149	17.45	18.07	1,755.54	86.61	82.40	1.21	3,416.08
Greece	2019	207.31	10.72	6.82	356	238	16.25	19.98	1,779.26	85.06	82.01	1.27	3,644.60
Greece	2020	191.36	10.70	6.79	400	363	16.13	24.33	2,006.74	74.33	60.18	1.51	4,008.18
Greece	2021	218.30	10.57	6.67	394	532	18.92	21.68	2,250.61	104.24	87.89	1.46	4,244.37
Hungary	2010	131.90	10.00	6.88	649	47	23.92	54.63	20,245.84	101.02	107.17	1.13	2,135.63
Hungary	2011	141.71	9.97	6.85	662	36	26.16	56.39	22,847.62	114.17	122.15	1.18	2,309.99
Hungary	2012	128.47	9.92	6.80	692	66	23.76	55.65	17,575.55	102.89	110.60	1.25	2,400.89
Hungary	2013	135.65	9.87	6.74	642	66	25.38	56.52	17,134.19	107.27	115.83	1.38	2,527.85
Hungary	2014	141.13	9.83	6.67	546	73	27.41	58.50	15,681.73	114.78	122.82	1.34	2,654.80
Hungary	2015	125.24	9.80	6.61	569	64	25.31	59.45	14,668.19	100.01	109.51	1.34	2,569.67
Hungary	2016	128.98	9.76	6.54	616	49	25.23	56.30	15,894.22	100.43	111.07	1.18	2,625.94
Hungary	2017	143.34	9.73	6.49	496	36	27.38	56.97	17,001.57	113.65	122.90	1.32	2,902.24
Hungary	2018	161.18	9.71	6.44	407	36	29.66	51.54	18,051.19	127.83	134.42	1.51	3,846.49
Hungary	2019	164.94	9.69	6.38	427	23	28.58	51.70	18,443.34	130.07	133.71	1.47	4,021.15
Hungary	2020	158.47	9.67	6.32	428	28	27.37	53.21	18,196.44	120.77	123.74	1.59	4,309.62
Hungary	2021	183.28	9.63	6.26	433	13	30.29	50.43	19,838.94	145.36	145.56	1.64	4,452.49
Iceland	2010	13.75	0.32	0.21	57	19	1.78	16.57	142.43	5.76	7.12	..	..
Iceland	2011	15.22	0.32	0.21	50	21	2.01	14.51	157.71	7.13	8.30	2.40	7,088.82
Iceland	2012	14.75	0.32	0.21	37	7	1.72	14.51	111.30	7.25	8.10	..	..
Iceland	2013	16.13	0.32	0.21	33	13	1.87	14.51	94.98	7.34	8.59	1.69	5,739.72
Iceland	2014	17.87	0.33	0.22	51	13	2.04	14.51	119.69	8.07	9.21	1.94	..
Iceland	2015	17.52	0.33	0.22	40	6	1.88	11.27	111.54	7.74	9.05	2.18	..
Iceland	2016	20.79	0.34	0.22	35	3	1.93	11.52	136.82	8.49	9.87	2.11	..
Iceland	2017	24.73	0.34	0.23	36	8	2.19	2.29	167.56	10.20	11.31	2.08	5,979.73
Iceland	2018	26.26	0.35	0.24	56	10	2.24	12.62	143.43	11.18	12.09	2.00	5,645.67

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Iceland	2019	24.68	0.36	0.24	47	15	2.18	17.46	305.00	9.66	10.79	2.34	..
Iceland	2020	21.63	0.37	0.24	44	3	2.00	11.78	144.58	7.50	7.20	2.49	..
Iceland	2021	25.77	0.37	0.25	34	2	2.70	13.90	239.16	10.13	9.63	2.78	6,941.30
Ireland	2010	221.99	4.56	3.08	..	..	43.26	61.69	23,325.04	192.36	229.11	1.59	3,116.62
Ireland	2011	241.34	4.58	3.07	..	..	49.64	60.13	27,791.42	202.16	246.79	1.55	3,341.40
Ireland	2012	227.27	4.60	3.06	..	..	44.49	61.06	26,188.45	196.36	235.59	1.56	4,873.66
Ireland	2013	243.30	4.62	3.05	..	..	46.34	55.93	25,028.07	206.93	254.90	1.57	5,141.74
Ireland	2014	266.79	4.66	3.06	..	..	52.20	55.21	25,849.51	243.66	294.33	1.52	5,290.95
Ireland	2015	302.39	4.70	3.08	..	..	105.41	54.35	30,612.42	282.07	371.71	1.18	5,242.07
Ireland	2016	305.73	4.76	3.12	79	70	99.65	54.31	37,992.13	322.75	379.71	1.18	5,143.86
Ireland	2017	348.53	4.83	3.16	62	75	112.03	54.51	35,187.49	352.50	435.02	1.25	5,114.19
Ireland	2018	395.46	4.90	3.20	76	32	129.74	54.51	36,589.47	369.59	486.28	1.11	4,656.78
Ireland	2019	407.12	4.98	3.25	58	35	130.02	54.79	39,354.39	498.62	518.15	1.16	4,961.44
Ireland	2020	436.56	5.04	3.29	75	54	147.63	54.37	43,116.91	500.49	591.37	1.15	5,196.11
Ireland	2021	531.31	5.11	3.34	75	42	183.71	54.26	49,840.18	485.42	706.93	1.11	5,213.43
Israel	2010	239.68	7.62	4.74	1,450	5,856	33.62	61.97	10,584.55	76.95	80.36	3.86	..
Israel	2011	268.09	7.77	4.80	1,360	5,526	36.38	37.44	11,681.10	93.10	91.64	3.94	..
Israel	2012	263.59	7.91	4.87	1,319	5,473	36.23	42.70	11,566.23	92.91	91.51	4.08	..
Israel	2013	298.48	8.06	4.93	1,201	4,984	38.69	40.52	11,744.66	93.51	97.92	4.12	..
Israel	2014	315.02	8.22	5.01	1,125	5,148	39.31	42.12	12,408.56	95.21	99.20	4.21	..
Israel	2015	303.47	8.38	5.09	1,285	5,623	39.13	39.23	14,350.47	85.08	93.66	4.30	..
Israel	2016	321.94	8.55	5.17	1,300	5,119	38.76	39.00	12,192.74	90.48	95.60	4.59	..
Israel	2017	358.45	8.71	5.25	1,436	5,377	41.01	39.76	12,057.52	97.87	103.87	4.74	..
Israel	2018	376.30	8.88	5.34	1,506	5,857	43.80	40.66	12,968.81	109.32	111.73	4.91	..
Israel	2019	400.65	9.05	5.43	1,368	6,370	46.40	46.55	12,564.14	109.55	116.11	5.33	..
Israel	2020	410.77	9.22	5.52	1,642	6,481	50.08	48.75	12,960.85	98.85	111.62	5.80	..
Israel	2021	489.85	9.37	5.61	1,592	8,017	57.60	49.07	16,092.02	127.74	141.19	5.78	..
Italy	2010	2,144.94	59.82	39.18	8,877	846	303.79	41.11	29,268.48	573.63	534.49	1.22	1,722.67
Italy	2011	2,306.97	60.03	39.20	8,794	927	326.02	40.69	33,756.01	643.86	612.80	1.20	1,760.64
Italy	2012	2,097.93	60.19	39.14	8,439	871	288.64	41.64	29,554.73	564.33	587.11	1.26	1,831.34
Italy	2013	2,153.23	60.31	39.05	8,307	905	296.27	42.22	31,881.51	556.75	606.72	1.30	1,916.30
Italy	2014	2,173.26	60.32	38.89	8,601	781	300.70	41.93	32,867.36	561.40	622.04	1.34	1,947.34
Italy	2015	1,845.43	60.23	38.71	..	..	263.20	42.09	30,092.63	485.19	537.83	1.34	2,075.81
Italy	2016	1,887.11	60.12	38.56	8,848	973	276.72	41.98	31,090.35	483.73	543.46	1.37	2,209.59
Italy	2017	1,970.72	60.00	38.43	8,643	1,031	292.00	42.69	31,877.71	540.83	593.97	1.37	2,322.42
Italy	2018	2,099.44	60.15	38.48	8,921	900	313.83	42.01	32,468.39	597.53	646.91	1.42	2,524.31
Italy	2019	2,019.61	59.73	38.15	9,229	898	298.97	41.63	32,553.69	560.33	625.07	1.46	2,670.65
Italy	2020	1,907.48	59.44	37.92	10,061	947	276.15	42.35	32,899.22	480.95	547.94	1.51	2,614.68
Italy	2021	2,179.21	59.13	37.68	10,281	797	334.84	40.61	36,849.79	634.81	679.29	1.43	2,658.89
Japan	2010	5,759.07	128.07	81.36	290,081	54,517	1,195.94	53.70	130,091.25	782.08	859.17	3.10	5,117.18
Japan	2011	6,233.15	127.83	80.61	287,580	55,030	1,217.68	54.48	133,397.95	961.18	920.91	3.21	5,123.33
Japan	2012	6,272.36	127.63	79.50	287,013	55,783	1,233.57	55.01	129,463.81	1,007.10	904.15	3.17	5,048.63
Japan	2013	5,212.33	127.44	78.38	271,731	56,705	1,007.49	55.34	111,257.96	948.36	822.72	3.28	5,166.83
Japan	2014	4,896.99	127.28	77.39	265,959	60,030	959.50	56.21	107,543.98	980.02	852.99	3.37	5,351.53
Japan	2015	4,444.93	127.14	76.58	258,839	59,882	909.54	56.77	98,394.43	799.67	775.05	3.24	5,198.49
Japan	2016	5,003.68	127.08	75.95	260,244	58,137	1,015.15	56.57	99,260.70	763.17	803.49	3.11	5,232.71
Japan	2017	4,930.84	126.97	75.44	260,292	58,189	1,007.66	56.91	106,357.71	829.95	867.41	3.17	5,319.83
Japan	2018	5,040.88	126.81	75.03	253,630	59,937	1,039.51	57.22	110,919.85	922.23	923.23	3.22	5,339.52
Japan	2019	5,117.99	126.63	74.72	245,372	62,597	1,035.07	56.06	104,081.42	908.59	893.78	3.22	5,376.12
Japan	2020	5,054.07	126.26	74.32	227,348	61,124	1,013.42	54.67	102,929.60	798.65	785.06	3.26	5,450.44
Japan	2021	5,039.15	125.68	73.82	222,452	66,748	1,047.12	54.67	116,695.46	938.76	912.55	3.28	5,589.77
Korea, Rep,	2010	1,143.67	49.55	35.99	131,805	38,296	313.81	61.23	132,322.49	506.60	538.71	3.32	5,429.44
Korea, Rep,	2011	1,253.29	49.94	36.38	138,034	40,890	353.89	58.39	133,617.30	654.58	668.48	3.59	5,908.82
Korea, Rep,	2012	1,278.05	50.20	36.70	148,136	40,779	355.70	59.38	130,675.95	656.46	691.35	3.85	6,393.34
Korea, Rep,	2013	1,370.63	50.43	36.93	159,978	44,611	380.84	60.06	143,485.22	639.54	703.03	3.95	6,457.89
Korea, Rep,	2014	1,484.49	50.75	37.18	164,073	46,219	401.44	59.78	149,062.38	635.11	710.05	4.08	6,867.33
Korea, Rep,	2015	1,466.04	51.01	37.38	167,275	46,419	390.05	60.35	147,046.54	529.87	630.24	3.98	7,020.44
Korea, Rep,	2016	1,499.68	51.22	37.53	163,424	45,406	395.28	59.84	135,906.06	501.96	601.86	3.99	7,057.16
Korea, Rep,	2017	1,623.07	51.36	37.55	159,084	45,691	437.35	61.47	166,656.27	587.34	664.39	4.29	7,453.22
Korea, Rep,	2018	1,725.37	51.59	37.57	162,561	47,431	459.61	62.13	192,788.49	643.14	719.71	4.52	7,918.47
Korea, Rep,	2019	1,651.42	51.76	37.54	171,603	47,372	416.53	61.47	153,549.66	602.46	648.61	4.63	8,328.97
Korea, Rep,	2020	1,644.31	51.84	37.35	180,477	46,282	407.47	62.92	163,074.03	537.27	597.87	4.80	8,620.01
Korea, Rep,	2021	1,818.43	51.77	37.04	186,245	51,753	463.45	64.14	203,986.07	696.86	761.51	4.91	9,071.45
Latvia	2010	23.47	2.10	1.42	178	7	2.85	16.61	430.96	13.26	12.76	0.61	3,923.76
Latvia	2011	26.58	2.06	1.39	173	10	3.23	16.90	633.82	18.05	16.42	0.72	4,166.15

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Latvia	2012	27.12	2.03	1.36	193	12	3.20	16.59	777.10	18.81	17.31	0.66	4,032.34
Latvia	2013	29.15	2.01	1.34	225	8	3.35	16.74	1,082.13	19.56	18.24	0.61	3,605.76
Latvia	2014	30.28	1.99	1.31	103	4	3.30	17.73	1,329.55	20.13	19.25	0.69	3,658.36
Latvia	2015	26.34	1.98	1.29	136	1	2.86	18.87	1,291.46	16.80	16.49	0.62	3,724.60
Latvia	2016	27.12	1.96	1.27	95	18	2.86	18.53	1,203.27	16.65	16.79	0.44	3,992.11
Latvia	2017	29.39	1.94	1.25	90	7	3.20	19.01	1,413.24	18.98	18.85	0.51	4,348.11
Latvia	2018	33.25	1.93	1.23	86	24	3.64	20.22	1,879.36	21.40	21.24	0.64	4,614.40
Latvia	2019	33.10	1.91	1.22	82	5	3.65	19.74	1,469.87	20.77	20.70	0.64	4,859.22
Latvia	2020	33.38	1.90	1.20	93	1	3.83	24.31	1,837.83	20.44	20.92	0.73	5,131.44
Latvia	2021	38.18	1.88	1.19	104	4	4.91	18.87	1,946.95	26.72	25.51	0.75	5,430.94
Lithuania	2010	36.64	3.10	2.09	108	6	6.19	22.81	1,263.21	24.48	23.74	0.78	2,736.49
Lithuania	2011	43.19	3.03	2.03	93	15	7.90	23.29	1,582.17	32.90	31.73	0.90	2,748.03
Lithuania	2012	42.71	2.99	2.01	109	15	7.97	19.46	1,712.70	33.32	33.58	0.89	2,670.61
Lithuania	2013	46.30	2.96	1.98	117	20	8.18	17.84	1,907.24	35.92	36.60	0.95	2,878.75
Lithuania	2014	48.31	2.94	1.96	123	42	8.41	20.84	2,092.95	34.14	35.08	1.03	3,082.28
Lithuania	2015	41.54	2.91	1.94	101	18	7.16	23.62	1,919.84	28.91	28.50	1.04	2,794.92
Lithuania	2016	42.97	2.88	1.90	95	58	7.22	20.16	1,962.94	28.78	29.10	0.84	2,950.40
Lithuania	2017	47.76	2.84	1.86	81	46	7.98	22.15	2,317.78	34.06	35.15	0.90	3,068.33
Lithuania	2018	54.26	2.82	1.84	81	24	8.96	23.01	2,516.32	39.43	40.44	0.94	3,180.95
Lithuania	2019	55.12	2.81	1.83	90	33	8.82	24.68	2,527.08	39.44	42.34	0.99	3,445.31
Lithuania	2020	57.41	2.81	1.83	95	18	8.88	29.06	2,619.28	36.38	41.59	1.13	3,643.25
Lithuania	2021	67.04	2.81	1.82	81	9	10.67	30.91	3,107.24	50.40	53.38	1.10	3,946.51
Luxembourg	2010	56.21	0.51	0.35	79	21	2.79	7.58	963.10	73.62	91.47	1.42	5,200.14
Luxembourg	2011	61.70	0.52	0.36	85	43	3.00	7.13	1,136.38	85.90	104.57	1.42	5,525.73
Luxembourg	2012	59.78	0.53	0.37	109	52	2.72	22.41	925.51	84.03	102.34	1.21	4,401.98
Luxembourg	2013	65.20	0.54	0.38	113	56	2.97	23.95	743.29	93.99	115.01	1.23	4,661.60
Luxembourg	2014	68.80	0.56	0.38	128	90	3.25	22.46	679.51	103.65	125.76	1.22	4,782.94
Luxembourg	2015	60.07	0.57	0.39	128	119	3.29	23.67	702.21	95.69	115.24	1.25	4,636.38
Luxembourg	2016	62.22	0.58	0.40	143	301	3.61	20.41	811.60	97.89	118.90	1.27	4,802.18
Luxembourg	2017	65.71	0.60	0.41	156	512	3.14	21.77	623.10	105.83	126.66	1.24	4,969.23
Luxembourg	2018	71.09	0.61	0.42	152	243	3.53	21.13	858.27	117.04	140.19	1.17	4,754.92
Luxembourg	2019	69.87	0.62	0.43	117	359	3.52	22.07	791.00	123.03	144.34	1.18	5,090.82
Luxembourg	2020	73.67	0.63	0.44	129	679	3.63	23.63	618.38	126.08	150.61	1.10	4,688.80
Luxembourg	2021	86.39	0.64	0.44	112	1,441	4.01	24.09	806.65	157.13	186.27	1.04	4,939.09
Mexico	2010	1,105.42	113.62	72.87	951	13,625	197.19	36.89	49,221.67	334.46	320.77	0.47	341.24
Mexico	2011	1,229.01	115.24	74.26	1,065	12,990	217.79	38.65	52,264.96	389.84	374.15	0.45	348.03
Mexico	2012	1,255.11	116.82	75.58	1,294	14,020	235.96	40.15	57,555.13	410.65	395.85	0.40	250.69
Mexico	2013	1,327.44	118.34	76.83	1,210	14,234	240.30	40.79	58,518.03	423.93	408.26	0.41	254.47
Mexico	2014	1,364.51	119.78	78.03	1,246	14,889	248.74	41.52	61,566.55	445.31	429.31	0.42	262.92
Mexico	2015	1,213.29	121.07	79.15	1,364	16,707	240.52	41.02	60,222.70	439.82	414.50	0.41	284.58
Mexico	2016	1,112.23	122.25	80.22	1,310	16,103	220.99	43.24	62,489.60	432.38	409.48	0.38	319.55
Mexico	2017	1,190.72	123.40	81.28	1,334	15,850	240.07	44.44	69,637.77	469.60	446.69	0.32	273.57
Mexico	2018	1,256.30	124.57	82.35	1,555	14,869	253.64	44.87	74,827.26	517.20	490.51	0.30	278.44
Mexico	2019	1,304.11	125.76	83.40	1,305	14,636	259.36	45.60	75,166.40	507.72	502.47	0.28	282.19
Mexico	2020	1,121.06	126.80	84.36	1,132	13,180	226.28	45.79	71,010.21	421.74	439.85	0.29	281.59
Mexico	2021	1,316.57	127.65	85.21	1,117	15,044	274.32	42.53	74,981.57	559.54	534.12	0.27	273.15
Netherlands	2010	852.46	16.62	11.16	2,527	240	83.30	45.62	77,595.34	518.25	590.65	1.70	3,210.02
Netherlands	2011	913.14	16.69	11.16	2,585	310	92.24	47.05	86,024.48	607.99	686.46	1.88	3,648.13
Netherlands	2012	845.69	16.75	11.12	2,375	338	85.08	47.44	81,513.74	590.44	672.96	1.92	4,336.15
Netherlands	2013	883.95	16.80	11.10	2,315	449	85.29	45.80	80,936.82	621.25	710.44	2.16	4,887.39
Netherlands	2014	901.56	16.87	11.09	2,294	288	87.11	46.79	84,488.37	634.10	733.71	2.17	4,926.09
Netherlands	2015	775.74	16.94	11.11	2,207	287	78.73	47.28	69,778.89	592.16	651.56	2.15	4,891.78
Netherlands	2016	797.16	17.03	11.15	2,290	314	81.84	47.44	71,059.99	558.46	644.98	2.15	5,109.21
Netherlands	2017	848.23	17.13	11.20	2,241	365	89.49	48.57	77,793.88	631.65	722.52	2.18	5,273.18
Netherlands	2018	929.73	17.23	11.24	2,111	394	99.10	50.51	85,620.09	711.27	809.39	2.14	5,496.79
Netherlands	2019	928.90	17.34	11.30	2,228	449	96.80	50.21	87,057.98	699.98	792.55	2.18	5,592.12
Netherlands	2020	932.56	17.44	11.34	2,198	825	97.77	51.35	87,232.91	662.04	757.63	2.32	5,799.28
Netherlands	2021	1,054.47	17.53	11.37	2,080	1,390	113.41	51.73	101,413.26	808.83	911.59	2.27	6,003.87
New Zealand	2010	146.89	4.35	2.88	1,585	5,051	15.89	21.51	616.51	40.99	44.36	..	..
New Zealand	2011	168.48	4.38	2.89	1,501	4,708	18.42	21.51	729.70	48.54	51.15	1.23	3,731.34
New Zealand	2012	176.56	4.41	2.90	1,425	5,674	18.69	20.03	762.20	49.63	50.85	..	..
New Zealand	2013	191.01	4.44	2.91	1,614	5,167	19.54	20.72	775.32	51.93	55.01	1.15	4,043.96
New Zealand	2014	201.52	4.52	2.96	1,636	6,092	22.29	21.45	699.12	54.70	56.33	..	..
New Zealand	2015	178.22	4.61	3.02	1,184	5,317	20.09	19.53	657.06	47.97	49.78	1.23	5,476.90
New Zealand	2016	189.10	4.71	3.10	1,075	5,311	19.06	19.68	640.14	48.78	50.32	..	..

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New Zealand	2017	206.75	4.81	3.16	1,014	5,146	21.06	22.15	564.06	54.95	56.99	1.35	5,035.56
New Zealand	2018	211.99	4.90	3.22	1,017	5,221	21.08	21.40	605.40	59.15	59.07	..	..
New Zealand	2019	213.01	4.98	3.26	324	5,690	20.60	21.61	572.76	57.78	58.21	1.41	5,665.38
New Zealand	2020	213.03	5.09	3.33	348	5,417	20.00	22.12	589.05	47.46	46.22	..	..
New Zealand	2021	253.98	5.11	3.33	330	6,522	22.06	23.74	673.72	66.62	56.66	1.47	5,094.76
Norway	2010	431.05	4.89	3.24	1,117	696	31.37	32.54	4,943.82	121.08	173.07	1.64	5,444.70
Norway	2011	501.36	4.95	3.28	1,122	654	34.25	32.45	5,379.99	140.07	208.87	1.62	5,533.92
Norway	2012	512.78	5.02	3.31	1,009	555	34.39	32.14	5,157.08	139.53	211.71	1.61	5,584.13
Norway	2013	526.01	5.08	3.35	1,101	648	35.29	35.22	5,372.32	147.87	209.68	1.64	5,605.02
Norway	2014	501.74	5.14	3.38	1,106	457	34.56	31.80	5,698.28	147.56	199.33	1.70	5,722.57
Norway	2015	388.16	5.19	3.41	1,153	652	27.02	32.26	5,035.66	123.65	149.73	1.92	5,929.86
Norway	2016	370.96	5.23	3.43	1,227	833	24.74	29.78	4,312.06	123.08	133.31	2.03	6,120.85
Norway	2017	401.75	5.28	3.45	1,152	908	26.02	28.94	4,265.64	130.07	147.57	2.08	6,396.17
Norway	2018	439.79	5.31	3.47	1,082	592	26.84	30.12	4,318.61	139.77	168.09	2.03	6,484.24
Norway	2019	408.74	5.35	3.49	957	582	25.65	35.03	4,610.75	140.01	149.77	2.14	6,737.55
Norway	2020	367.63	5.38	3.50	880	564	24.22	30.81	4,069.22	121.85	118.42	2.24	6,766.01
Norway	2021	503.37	5.41	3.51	946	634	28.27	30.16	4,733.48	141.88	216.63	1.89	7,228.72
Poland	2010	478.11	38.04	27.15	3,203	227	77.31	35.88	9,591.09	202.40	190.73	0.73	1,697.03
Poland	2011	527.85	38.06	27.07	3,879	244	85.19	35.84	9,700.76	235.45	223.38	0.75	1,679.90
Poland	2012	498.15	38.06	26.96	4,410	247	82.48	34.31	10,728.84	223.21	219.30	0.89	1,752.08
Poland	2013	518.18	38.04	26.80	4,237	174	80.62	35.21	13,662.05	231.37	237.16	0.88	1,868.31
Poland	2014	542.13	38.01	26.62	3,941	155	91.66	33.94	16,757.32	248.37	250.58	0.95	2,054.75
Poland	2015	480.05	37.99	26.41	4,676	139	84.82	33.98	16,502.31	216.47	226.40	1.00	2,158.38
Poland	2016	473.26	37.97	26.18	4,261	135	86.49	32.99	16,888.12	222.24	236.19	0.97	2,304.01
Poland	2017	528.36	37.97	25.95	3,924	117	89.22	32.74	18,837.38	258.23	273.12	1.04	2,994.86
Poland	2018	594.62	37.97	25.70	4,207	115	98.02	32.23	21,763.15	298.56	310.58	1.21	3,079.49
Poland	2019	602.68	37.97	25.44	3,887	112	100.42	32.28	19,830.98	295.12	317.09	1.32	3,159.07
Poland	2020	605.91	37.52	24.92	4,010	88	98.21	32.28	19,970.87	283.72	317.70	1.39	3,260.15
Poland	2021	689.17	36.98	24.41	3,377	111	119.23	32.15	23,837.27	371.02	393.09	1.43	3,558.20
Portugal	2010	238.44	10.57	7.01	499	46	27.62	23.96	1,498.65	89.80	71.60	1.54	3,923.76
Portugal	2011	245.43	10.56	6.98	571	75	27.80	24.65	1,830.31	94.73	84.46	1.46	4,166.15
Portugal	2012	216.54	10.51	6.92	621	26	24.63	24.72	1,919.38	82.76	81.69	1.38	4,032.34
Portugal	2013	226.68	10.46	6.86	647	22	26.22	24.51	2,135.84	87.19	89.68	1.32	3,605.76
Portugal	2014	230.08	10.40	6.80	722	18	27.24	24.62	2,314.87	92.11	92.46	1.29	3,658.36
Portugal	2015	199.04	10.36	6.75	925	20	24.38	24.41	2,095.89	79.51	80.98	1.24	3,724.60
Portugal	2016	206.31	10.33	6.70	724	27	25.26	23.90	2,457.36	80.80	83.25	1.28	3,992.11
Portugal	2017	220.86	10.30	6.65	644	36	27.46	24.42	2,816.26	92.46	94.84	1.32	4,348.11
Portugal	2018	242.09	10.28	6.62	661	29	29.83	24.47	2,978.02	104.24	105.74	1.35	4,614.40
Portugal	2019	240.12	10.29	6.59	703	104	28.66	25.96	3,594.84	103.53	104.82	1.40	4,859.22
Portugal	2020	229.62	10.30	6.57	695	263	27.76	26.53	3,366.11	90.13	85.55	1.61	5,131.44
Portugal	2021	256.06	10.36	6.59	711	42	31.86	25.55	3,561.40	113.50	106.39	1.67	5,430.94
Slovak Republic	2010	91.11	5.39	3.89	234	48	15.61	45.75	4,117.59	71.09	70.23	0.61	2,817.37
Slovak Republic	2011	99.71	5.40	3.88	224	33	17.36	44.52	5,015.33	84.30	84.38	0.65	2,840.92
Slovak Republic	2012	94.72	5.41	3.88	168	35	16.36	47.82	6,429.71	81.52	86.11	0.79	2,827.15
Slovak Republic	2013	99.13	5.41	3.86	184	26	16.52	49.43	8,084.07	87.26	92.07	0.82	2,723.15
Slovak Republic	2014	101.71	5.42	3.85	211	23	18.85	47.48	8,401.87	87.79	91.99	0.88	2,723.67
Slovak Republic	2015	89.18	5.42	3.82	228	28	16.79	49.39	7,465.24	79.13	80.88	1.16	2,658.90
Slovak Republic	2016	90.35	5.43	3.80	220	15	16.05	47.77	7,474.40	81.72	83.40	0.79	2,609.15
Slovak Republic	2017	95.98	5.44	3.77	183	23	16.43	47.93	8,807.97	88.96	90.17	0.88	2,803.05
Slovak Republic	2018	106.61	5.45	3.74	217	14	19.30	49.51	8,973.44	100.01	101.20	0.84	3,003.39
Slovak Republic	2019	105.84	5.45	3.71	206	28	19.79	51.17	8,049.16	97.25	97.21	0.82	3,116.82
Slovak Republic	2020	107.73	5.46	3.68	206	15	18.90	48.86	7,860.14	89.67	90.82	0.90	3,167.43
Slovak Republic	2021	120.56	5.45	3.64	146	13	20.04	44.08	8,421.90	109.93	109.38	0.92	3,211.13
Slovenia	2010	47.79	2.05	1.42	442	11	8.36	46.76	1,259.93	30.48	30.99	2.05	3,770.38
Slovenia	2011	51.20	2.05	1.42	470	11	9.32	47.78	1,499.88	35.61	36.24	2.41	4,287.96
Slovenia	2012	46.17	2.06	1.42	..	..	8.64	49.66	1,492.66	32.37	33.95	2.56	4,330.67

Country	Year	GDP (USD bn)	Pop (M)	Labor (M)	Pat. Res.	Pat. Non-Res.	MVA (USD bn)	Med HTMVA%	HTEX (USD M)	Imports (USD bn)	Exports (USD bn)	RDEXP %GDP	RDPop /M
Slovenia	2013	47.87	2.06	1.41	..	..	9.18	48.03	1,586.85	33.61	35.88	2.56	4,236.51
Slovenia	2014	49.51	2.06	1.40	..	..	9.73	47.33	1,622.65	34.57	37.95	2.37	4,166.88
Slovenia	2015	42.71	2.06	1.39	..	..	8.53	37.41	1,552.49	29.71	33.17	2.20	3,836.54
Slovenia	2016	44.29	2.07	1.37	..	..	8.96	37.09	1,672.14	30.92	34.74	2.01	3,939.83
Slovenia	2017	48.15	2.07	1.36	..	..	9.94	37.98	1,755.02	36.03	40.39	1.87	4,510.81
Slovenia	2018	53.69	2.07	1.36	255	23	10.96	37.13	2,091.05	41.36	45.94	1.95	4,880.49
Slovenia	2019	53.91	2.09	1.36	..	..	11.09	36.88	2,333.30	40.80	45.48	2.04	5,060.57
Slovenia	2020	53.38	2.10	1.36	232	12	10.95	37.24	2,542.92	36.98	41.79	2.14	5,187.16
Slovenia	2021	61.53	2.11	1.36	222	17	12.28	37.11	2,476.35	48.06	51.67	2.13	5,235.18
Spain	2010	1,427.99	46.58	31.76	3,566	213	158.71	34.73	11,505.35	383.64	368.82	1.36	2,880.00
Spain	2011	1,487.57	46.74	31.70	3,430	196	166.63	34.88	13,556.13	433.23	437.17	1.33	2,775.30
Spain	2012	1,330.99	46.77	31.56	3,266	209	143.91	35.95	13,769.97	389.37	416.60	1.30	2,693.40
Spain	2013	1,362.19	46.60	31.26	3,026	218	148.02	37.39	14,480.41	393.85	446.19	1.27	2,623.65
Spain	2014	1,380.24	46.46	30.99	2,953	225	151.79	38.68	14,666.99	416.56	458.51	1.24	2,614.45
Spain	2015	1,206.16	46.42	30.85	2,799	221	132.84	38.52	13,111.34	365.27	401.64	1.22	2,622.53
Spain	2016	1,243.02	46.46	30.81	2,745	177	136.14	38.77	15,601.71	368.08	417.37	1.19	2,712.75
Spain	2017	1,321.75	46.57	30.82	2,167	176	146.91	38.70	14,946.96	413.27	461.30	1.21	2,847.37
Spain	2018	1,431.64	46.78	30.94	1,525	149	154.45	37.80	15,484.72	460.05	499.28	1.24	2,984.30
Spain	2019	1,403.50	47.12	31.18	1,288	159	150.13	37.79	15,037.42	444.78	486.67	1.25	3,048.15
Spain	2020	1,289.78	47.36	31.37	1,431	124	139.98	39.28	15,726.20	373.59	393.14	1.41	3,051.65
Spain	2021	1,461.24	47.44	31.44	1,308	126	164.48	36.46	23,457.23	478.81	493.27	1.41	3,229.65
Sweden	2010	492.75	9.38	6.11	2,196	353	73.52	42.96	22,906.42	198.16	221.13	3.17	5,279.17
Sweden	2011	570.54	9.45	6.11	2,004	337	84.25	45.90	25,829.99	233.79	259.10	3.19	5,172.47
Sweden	2012	549.74	9.52	6.11	2,288	148	77.38	52.53	22,170.26	222.45	246.46	3.23	5,196.72
Sweden	2013	584.13	9.60	6.12	2,332	163	78.05	53.15	21,883.05	227.25	249.63	3.26	6,717.62
Sweden	2014	577.73	9.70	6.14	1,984	441	75.36	44.65	21,308.04	233.20	251.07	3.10	6,909.46
Sweden	2015	501.70	9.80	6.17	2,038	390	69.05	51.61	18,916.11	202.93	219.28	3.22	6,846.13
Sweden	2016	514.22	9.92	6.22	2,032	352	69.31	50.94	18,699.49	207.50	220.65	3.25	7,143.44
Sweden	2017	536.08	10.06	6.29	1,992	305	71.49	51.11	17,107.88	225.30	235.31	3.36	7,316.65
Sweden	2018	551.31	10.18	6.35	1,838	442	73.72	51.62	17,178.54	243.89	253.00	3.32	7,425.67
Sweden	2019	532.17	10.28	6.40	1,802	388	69.91	51.89	17,422.27	235.64	256.72	3.39	7,685.84
Sweden	2020	545.15	10.35	6.44	1,764	432	68.91	50.31	17,731.15	218.45	241.28	3.49	7,754.64
Sweden	2021	637.19	10.42	6.47	1,771	425	85.85	49.09	18,973.18	273.85	301.39	3.40	8,159.80
Switzerland	2010	598.85	7.82	5.32	1,622	533	107.17	61.32	43,609.99	330.73	388.28	..	..
Switzerland	2011	715.89	7.91	5.37	1,597	446	130.25	60.82	51,150.14	423.14	475.71	..	..
Switzerland	2012	686.42	8.00	5.42	1,480	1,508	121.71	62.71	51,281.75	402.77	465.24	2.87	4,497.49
Switzerland	2013	706.23	8.09	5.47	1,525	631	125.71	62.57	54,348.47	435.22	508.97	..	..
Switzerland	2014	726.54	8.19	5.52	1,480	568	127.50	63.00	57,103.94	398.70	474.70	..	..
Switzerland	2015	694.12	8.28	5.57	1,477	446	118.82	63.56	53,797.22	370.03	443.05	3.08	5,308.73
Switzerland	2016	687.90	8.37	5.62	1,462	309	119.23	63.32	55,537.84	390.37	459.64	..	..
Switzerland	2017	695.20	8.45	5.65	1,337	291	121.60	65.66	29,834.78	398.16	461.26	3.08	5,257.33
Switzerland	2018	725.57	8.51	5.67	1,283	332	128.18	63.86	30,116.13	413.21	489.60	..	..
Switzerland	2019	721.37	8.58	5.69	1,369	348	129.29	68.17	29,874.36	412.35	483.07	3.20	5,561.73
Switzerland	2020	742.00	8.64	5.72	1,384	301	131.98	64.43	29,246.59	428.06	475.59	..	..
Switzerland	2021	815.31	8.70	5.74	1,288	267	156.84	68.48	38,186.91	480.19	579.84	3.31	6,021.40
Turkiye	2010	776.97	73.14	48.94	3,180	177	116.96	30.23	1,943.40	198.13	164.67	0.79	882.50
Turkiye	2011	838.79	74.22	49.91	3,885	228	137.97	29.95	2,202.18	254.24	192.87	0.79	977.26
Turkiye	2012	880.56	75.18	50.78	4,434	232	139.43	29.89	2,327.32	250.69	214.51	0.83	1,100.19
Turkiye	2013	957.80	76.15	51.68	4,392	269	155.93	29.93	3,782.78	275.22	227.89	0.81	1,175.90
Turkiye	2014	938.93	77.18	52.64	4,766	331	157.48	29.24	4,293.03	268.17	236.66	0.86	1,163.54
Turkiye	2015	864.31	78.22	53.59	5,352	489	144.31	29.78	3,872.19	229.54	212.03	0.97	1,136.18
Turkiye	2016	869.68	79.28	54.45	6,230	618	144.33	29.94	3,421.69	219.55	200.75	1.12	1,263.79
Turkiye	2017	858.99	80.31	55.20	8,175	380	151.11	30.82	4,069.12	255.31	223.68	1.18	1,420.60
Turkiye	2018	778.97	81.41	55.93	7,156	310	148.66	33.66	3,735.74	244.45	243.29	1.27	1,585.08
Turkiye	2019	761.01	82.58	56.58	7,871	217	139.85	35.83	4,280.26	229.21	251.70	1.32	1,737.58
Turkiye	2020	720.34	83.38	56.92	7,920	238	137.87	36.74	4,172.76	232.11	209.77	1.37	1,964.66
Turkiye	2021	819.87	84.15	57.31	8,234	242	182.32	34.34	5,715.37	289.73	293.05	1.40	2,209.70
United Kingdom	2010	2,485.48	62.77	41.36	15,490	6,439	234.76	45.19	66,763.51	755.99	717.23	1.64	4,089.21
United Kingdom	2011	2,663.81	63.26	41.54	15,343	6,916	248.52	45.00	75,837.99	857.15	836.10	1.65	3,973.02
United Kingdom	2012	2,707.09	63.71	41.58	15,370	7,865	253.01	43.47	73,189.56	855.64	828.87	1.58	4,017.98
United Kingdom	2013	2,784.85	64.14	41.61	14,972	7,966	267.17	44.97	74,738.91	883.07	843.52	1.62	4,171.81

Country	Year	GDP (USD bn)	Pop (M)	Labor (M)	Pat. Res.	Pat. Non-Res.	MVA (USD bn)	Med HTMVA%	HTEX (USD M)	Imports (USD bn)	Exports (USD bn)	RDEXP %GDP	RDPop /M
United Kingdom	2014	3,064.71	64.62	41.73	15,196	7,844	288.41	43.85	76,056.35	935.31	875.78	2.26	4,280.80
United Kingdom	2015	2,927.91	65.09	41.88	14,867	7,934	272.14	41.89	74,614.81	859.56	815.30	2.28	4,389.56
United Kingdom	2016	2,689.11	65.61	42.08	13,876	8,183	244.97	43.22	73,847.89	828.19	777.21	2.32	4,434.66
United Kingdom	2017	2,680.15	65.97	42.16	13,301	8,771	242.67	44.58	73,731.45	863.85	827.91	2.33	4,472.69
United Kingdom	2018	2,871.34	66.29	42.22	12,865	8,076	257.05	43.50	75,625.93	946.47	906.77	2.71	..
United Kingdom	2019	2,851.41	66.63	42.32	12,061	7,189	250.83	43.06	76,930.49	941.83	902.04	2.67	..
United Kingdom	2020	2,696.78	66.74	42.33	11,990	8,659	242.64	42.78	58,169.58	785.07	800.13	2.94	..
United Kingdom	2021	3,143.32	66.98	42.50	11,592	7,263	271.78	40.57	66,732.69	924.27	917.23	2.90	..
United States	2010	15,048.97	309.38	208.69	241,977	248,249	1,788.32	46.48	166,133.84	2,389.55	1,857.25	2.71	3,550.09
United States	2011	15,599.73	311.84	209.98	247,750	255,832	1,863.08	46.00	166,559.64	2,695.48	2,115.86	2.74	3,659.10
United States	2012	16,253.97	314.34	210.97	268,782	274,033	1,921.31	46.05	169,157.79	2,769.32	2,217.70	2.67	3,642.58
United States	2013	16,880.68	316.73	211.86	287,831	283,781	1,970.53	45.78	169,309.01	2,766.38	2,287.92	2.70	3,725.76
United States	2014	17,608.14	319.26	212.90	285,096	293,706	2,009.69	47.04	176,029.44	2,887.45	2,378.55	2.71	3,824.01
United States	2015	18,295.02	321.82	214.00	288,335	301,075	2,071.14	46.60	175,321.67	2,794.94	2,270.62	2.77	3,858.04
United States	2016	18,804.91	324.35	215.10	295,327	310,244	2,035.23	46.25	173,983.45	2,738.83	2,235.56	2.84	3,809.12
United States	2017	19,612.10	326.61	215.96	293,904	313,052	2,109.72	46.57	154,602.42	2,931.59	2,388.26	2.88	3,931.60
United States	2018	20,656.52	328.53	216.57	285,095	312,046	2,261.82	45.54	153,893.52	3,131.17	2,538.09	2.99	4,226.79
United States	2019	21,539.98	330.23	217.02	285,113	336,340	2,268.79	45.23	154,028.53	3,116.68	2,539.39	3.15	4,265.92
United States	2020	21,354.10	331.58	217.30	269,586	327,586	2,149.53	45.09	141,612.13	2,777.34	2,151.14	3.42	4,464.09
United States	2021	23,681.17	332.10	217.13	262,244	329,229	2,408.76	43.27	169,281.83	3,415.46	2,555.43	3.48	4,825.18

Source: prepared by auther based on Worldbank Data

Appendix D

Code Used for Accuracy Graphs

```
Regression Model (estimated in EViews 10):
   LS(CX=F) GDP/POP C HTEX/LABOR IMP/POP MVA*MEDHTMVA/100
   LOG(PATENTRE) EXP01/LABOR LOG(PATENTNONRE) RDEXP*GDP/100
   RDPOP/LABOR

   Sample: 37 OECD countries, 2010-2021, 386 observations (unbalanced)
   R-squared: 0.9897 | F-statistic: 741.85 | DW: 0.881
"""

import pandas as pd
import numpy as np
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
import os

print("Loading data...")
DATA_PATH = '/Users/nushabehuseynova/Desktop/Data EXCELSIOR.xlsx'
OUTPUT_DIR = '/Users/nushabehuseynova/Desktop/thesis_graphs'

df = pd.read_excel(DATA_PATH, sheet_name='Sheet2')
df = df.replace('..', np.nan).replace(',,', np.nan)

numeric_cols = ['GDP ', 'Pop', 'Labor ', 'PatentRe', 'PatentNonRe',
                'MVA', 'MedHTMVA', 'HTEX', 'IMP', 'EXP', 'RDEXP', 'RDPop']
for col in numeric_cols:
   df[col] = pd.to_numeric(df[col], errors='coerce')

# Full dataset 2010-2024 for actual GDP per capita line
df_full = df[df['Year'].between(2010, 2024)].copy()
df_full['GDP_POP'] = df_full['GDP '] / df_full['Pop']

df_full['HTEX_LABOR']      = df_full['HTEX'] / df_full['Labor ']
df_full['IMP_POP']         = df_full['IMP'] / df_full['Pop']
df_full['MVA_MEDHT']       = df_full['MVA'] * df_full['MedHTMVA'] / 100
df_full['EXP_LABOR']       = df_full['EXP'] / df_full['Labor ']
df_full['RDEXP_GDP']       = df_full['RDEXP'] * df_full['GDP '] / 100
df_full['RDPOP_LABOR']     = df_full['RDPop'] / df_full['Labor ']
df_full['LOG_PATENTRE']    = np.log(df_full['PatentRe'].replace(0, np.nan))
df_full['LOG_PATENTNONRE'] = np.log(df_full['PatentNonRe'].replace(0, np.nan))
```

```
innovation_vars = ['LOG_PATENTRE', 'LOG_PATENTNONRE', 'RDEXP_GDP',
'RDPOP_LABOR']

for country in df_full['Country'].unique():
    mask = df_full['Country'] == country
    for col in innovation_vars:
        last_known = df_full.loc[mask & (df_full['Year'] <= 2021), col].dropna()
        if len(last_known) > 0:
            last_val = last_known.iloc[-1]
            fill_mask = mask & (df_full['Year'] > 2021) & df_full[col].isna()
            df_full.loc[fill_mask, col] = last_val

COEFFICIENTS = {
    'const':           46422.750,
    'HTEX_LABOR':      1.241924,
    'IMP_POP':         -0.497622,
    'MVA_MEDHT':       6.26e-8,
    'LOG_PATENTRE':    -2172.586,
    'EXP_LABOR':       0.572588,
    'LOG_PATENTNONRE': -1757.572,
    'RDEXP_GDP':       2.81e-8,
    'RDPOP_LABOR':     -997902.5,
}

df_full['fitted_pooled'] = (
    COEFFICIENTS['const']
    + COEFFICIENTS['HTEX_LABOR']       * df_full['HTEX_LABOR'].fillna(0)
    + COEFFICIENTS['IMP_POP']          * df_full['IMP_POP'].fillna(0)
    + COEFFICIENTS['MVA_MEDHT']        * df_full['MVA_MEDHT'].fillna(0)
    + COEFFICIENTS['LOG_PATENTRE']     * df_full['LOG_PATENTRE'].fillna(0)
    + COEFFICIENTS['EXP_LABOR']        * df_full['EXP_LABOR'].fillna(0)
    + COEFFICIENTS['LOG_PATENTNONRE']  * df_full['LOG_PATENTNONRE'].fillna(0)
    + COEFFICIENTS['RDEXP_GDP']        * df_full['RDEXP_GDP'].fillna(0)
    + COEFFICIENTS['RDPOP_LABOR']      * df_full['RDPOP_LABOR'].fillna(0)
)

df_reg = df_full[df_full['Year'].between(2010, 2021)].copy()
df_reg['resid_pooled'] = df_reg['GDP_POP'] - df_reg['fitted_pooled']
fixed_effects = df_reg.groupby('Country')['resid_pooled'].mean().rename('fixed_effect')

df_full = df_full.merge(fixed_effects, on='Country')
df_full['fitted'] = df_full['fitted_pooled'] + df_full['fixed_effect']

HIGH_ACCURACY = [
    'Germany',
    'Korea, Rep,',
```

```
    'Japan',
    'Hungary',
    'Poland',
    'United States',
    'Ireland',
    'Switzerland',
]

MEDIUM_ACCURACY = [
    'Luxembourg',
    'Denmark',
    'Estonia',
    'Netherlands',
    'Norway',
    'Lithuania',
    'Sweden',
    'Czechia',
    'United Kingdom',
    'Finland',
    'France',
    'Canada',
    'Latvia',
    'Slovak Republic',
    'Greece',
    'New Zealand',
    'Colombia',
    'Israel',
]

LOW_ACCURACY = [
    'Costa Rica',
    'Turkiye',
    'Spain',
    'Mexico',
    'Portugal',
    'Chile',
    'Iceland',
    'Italy',
    'Slovenia',
    'Australia',
    'Austria',
    'Belgium',
]

os.makedirs(OUTPUT_DIR, exist_ok=True)
```

```
def generate_graph(country, group, fig_num):
    cdf = df_full[df_full['Country'] == country].sort_values('Year')
    if len(cdf) < 2:
        print(f'  Skipping {country} — insufficient data')
        return

    fig, ax = plt.subplots(figsize=(7, 4))

    # Actual GDP per capita — blue solid line
    ax.plot(cdf['Year'], cdf['GDP_POP'],
            'b-o', linewidth=2, markersize=5, label='Actual GDP per Capita')

    # Fitted GDP per capita — orange dashed line
    ax.plot(cdf['Year'], cdf['fitted'],
            color='orange', linewidth=2, linestyle='--',
            marker='s', markersize=5, label='Fitted GDP per Capita')



    display_name = country.replace('Korea, Rep,', 'Korea')
    ax.set_title(f'Figure {fig_num}. {display_name} — '
                 f'Actual vs Fitted GDP per Capita (2010–2024)',
                 fontsize=10, pad=10)
    ax.set_xlabel('Year', fontsize=9)
    ax.set_ylabel('GDP per Capita (USD)', fontsize=9)
    ax.legend(fontsize=8, loc='upper left')
    ax.grid(True, alpha=0.3)
    ax.set_xlim(2009.5, 2024.5)
    ax.xaxis.set_major_locator(plt.MaxNLocator(integer=True))
    ax.yaxis.set_major_formatter(
        plt.FuncFormatter(lambda x, p: f'${x:,.0f}'))

    plt.tight_layout()
    safe = country.replace(', ', '_').replace(' ', '_').replace(',', '')
    path = os.path.join(OUTPUT_DIR, f'fig{fig_num:02d}_{group}_{safe}.png')
    plt.savefig(path, dpi=150, bbox_inches='tight')
    plt.close()
    print(f'  Figure {fig_num:02d}: {country}')

print('\nGenerating HIGH accuracy graphs...')
for i, c in enumerate(HIGH_ACCURACY):
    generate_graph(c, 'high', i + 1)

print('\nGenerating MEDIUM accuracy graphs...')
for i, c in enumerate(MEDIUM_ACCURACY):
    generate_graph(c, 'medium', i + 9)
```

```
print('\nGenerating LOW accuracy graphs...')
for i, c in enumerate(LOW_ACCURACY):
    generate_graph(c, 'low', i + 27)

total = len(os.listdir(OUTPUT_DIR))
print(f'\nComplete. {total} graphs saved to {OUTPUT_DIR}')
```